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Essays on environmental change and inequality

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Abstract

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This dissertation is comprised of three essays which may be loosely described as empirically studying the implications of environmental change for distinct concepts of inequality.

Income inequality. Chapter 1 characterizes the relationship between anthropogenic climate change and global inequality, a subject intersecting two of the defining challenges of the 21st century but which remains virtually unstudied at subnational levels. To make overdue progress addressing this research gap, I improve upon frontier methods for estimating climate impacts to overcome outstanding methodological limitations and apply these refinements to fine-scaled data on globally representative income distributions. Next, I document new evidence that temperature shocks significantly and persistently impact distributions of income within countries, an effect driven by concentrations of harm onto the lowest income-earners in warm climates as well as a surprising vulnerability of the top 1% in these countries to environmental shocks. Integrating these inequality effects over observed distributions of income and the spatial incidence of global warming, I find that climate change between 1981 and 2016 regressively redistributed global shares in a reduced stock of income both between and within countries largely by depriving the world's poorest of economic opportunity that would otherwise have been available. Altogether, these results constitute the most comprehensive evidence yet of the regressive impact of climate change on global inequality.

Institutional inequality. Chapter 2 builds on recent political economy work which observes that democratizations tend to be preceded by negative transitory shocks. A compelling theoretical explanation for this phenomenon refers to a "democratic window of opportunity" wherein the shock decreases the opportunity cost of contesting political power; the worst impacted have nothing to lose, so to speak.

Gendered inequality. Chapter 3 seeks to merge open questions regarding the determinants and evolution of gendered norms with an emerging literature on cultural persistence and breaks from tradition. When unexpected and severe droughts induce negative income shocks, the resultant shock to intra-household bargaining power should tend to manifest in gendered effects. To the extent that these environmental factors can affect the intergenerational persistence of gendered attitudes and behaviors, I refer to this as the environmental disruption of gender norms. Taking this hypothesis to the rural India, I find that gender norms are highly persistent across generations and that norms related to beliefs about the role of women are relatively impervious to disruption. In contrast, gender norms related to intra-household distributions of power and the

violent exercises of such power are more sensitive. Estimating a survival model on a uniquely rich survey module on domestic violence, I find that droughts that a drought that occurs in the early years of a marriage reduce the hazard of becoming domestically violent by between 1.8 and 2.0 percentage points in the year of the drought's occurrence, equivalent to a 48-53% reduction in the overall hazard of spousal violence.

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Acknowledgements

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Dedication

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Chapter 1: Climate inequalities

> "There is no progress trying to do welfare economics without [going] beyond the Pareto principle and taking on the real problems of distribution, inequality, and poverty. This could not be done, then or now."
>
> Amartya Sen (Sen et al., 2020)

1.1 Introduction

Mitigating anthropogenic climate change and narrowing economic inequalities are widely considered among the most pressing global challenges of the 21st century. For example, the Sustainable Development Goals adopted by all United Nations members explicitly resolve to "take urgent action to combat climate change and its impacts" and to "reduce inequality within and among countries" (United Nations, 2015). These global priorities are also understood to be intimately related, often framed as twinned crises (Pande, 2023): the Intergovernmental Panel on Climate Change identifies the "widening opportunities between and within countries" (IPCC, 2018) as first-order considerations in any sufficient policy response to climate change. Likewise, the World Inequality Report remarks that "climate mitigation is largely a distributional issue, not only between countries but also within them" (Chancel et al., 2022). When surveyed in 2021, an overwhelming majority of published environmental economists indicated a belief that climate change will "likely" or "very likely" exacerbate inequalities between (89%) and within (71%) countries (Howard & Sylvan, 2021).

These frames exemplify how the distributional implications of climate change are increasingly operationalized along its international and intranational dimensions. The conventional wisdom that the social consequences of climate change tend to concentrate in less developed countries with tropical geographies is supported by a robust empirical literature and reinforced by the visibility of calamitous extreme weather events. In contrast, within-country 'climate inequality' remains remarkably undercharacterized and largely inactive as a subject of academic inquiry even as public and institutional demand for these equity considerations has surged (Office of Management and Budget, 2023) and intranational disparities increasingly dominate total global inequality[1]. As a result, our understanding of how climate change affects the economic well-being of the Earth's eight billion human inhabitants is largely limited to aggregate comparisons across fewer than 200 polities, necessarily agnostic to urgent considerations such as social stratification and mass immiseration. Addressing this ignorance is necessary for adequately specifying the problem of climate change, let alone formulating an efficient and equitable response to it. This chapter aims to initiate and stimulate long overdue progress in this area by providing the most comprehensive characterization

yet of the dynamic and distributional impact of climate change on economic inequality.

Credible progress in this area first requires overcoming long-standing challenges in the tradition of estimating the causal effect of weather variation on economic outcomes of interest (Pindyck, 2013). In Section 1.2, I briefly summarize specific technical issues in this niche literature to contextualize the methodological choices made in the present analysis. First, estimates of cumulative impacts implied by these "damage functions" are known to diverge by orders of magnitude depending on the modeler's assumptions about the persistence of these effects. While 76% of recently surveyed climate economists indicated a belief in highly persistent or permanent effects of weather on macroeconomic aggregates like GDP, statistical tests of their presence have often been inconclusive (M. Burke et al., 2015a; Newell et al., 2021). Second, recent work has challenged the validity of raw weather variables conventionally used for as explanatory variables in these regressions on the grounds of identificaiton issues. Finally, I discuss the use of these reduced-form damage functions to make inferences about systematic climate change.

My implementation is briefly described in Section 1.3 with more detailed exposition provided in Section A.2. I adapt recent innovations in empirical macroeconomics to construct more plausibly exogenous shocks and estimate dynamic impulse response functions by the method of local projections. To credibly measure global income distributions, the analysis uses new data from the most recent update to the World Inequality Database, which provides well-balanced percentile-year panels of pre-tax income for a set of countries now representing almost the entire global population. By applying a refined econometric apparatus to unprecedentedly rich distributional data, my empirical analysis accomplishes two objectives. First, it substantially improves the frontier of an analytic tradeoff between subnational disaggregation and global representativeness which has historically hindered the joint study of global inequality and climate change. Second, the dynamic estimates which result are entirely invariant to hotly contested assumptions about the persistence structure of these economic impacts, thus obviating the dominant source of extreme estimation instability observed in the climate impacts literature.

The results of this empirical analysis are documented in Section 1.3.2. Here, we present dynamic distributional income response surfaces which depict how the impact of a temperature shock may vary across income groups and climates and accumulate over time. We find evidence of substantial, significant, and persistent impacts on within-country inequality, particularly in warmer countries. This distributional effect is driven by substantial transfers of national income shares among countries in warm climates. More specifically, the distributional impacts in these settings are marked by severe reductions in income among the lowest income-earning decile and to a lesser extent, a perhaps surprising precipitous fall in income among the top 1%. These losses are shown to compound for at least three years following a shock even when accounting for persistence in the shock itself. As impacts to warmer countries are shown to be significantly negative across the entire income distribution, national income in sufficiently warm countries are expected to uniformly decrease in response to a transitory temperature shock. Thus, positive temperature shocks in already-hot countries result in regressive redistribution of shares in a persistently diminished national income.

In Section 1.4, we present a counterfactual analysis which produces the most disaggregated evidence yet of the impact of historically observed climate change on the global distribution of

[1]Within-country income inequality now accounts for almost 70% of total income inequality as measured by the entropy-based Theil measure of inequality, an increase from just over 40% in 1980. (Chancel et al., 2022)

income. This entails integrating the distributional response surfaces estimated in the previous section over the observed histories of country-percentile-year income and the incidence of temperature shocks between 1981 and 2016. My results suggest that systematic changes to the distribution of temperature shocks over this 36 years has unambiguously worsened total global inequality, both between and within countries. This effect mainly stems from foregone income growth for the poorest 20% of global income-earners. For the lowest positive income-earning global percentile in our data, the analysis implies that cumulative income growth over this 36-year period would be 29% higher than historically observed had the climate remained stable at 1980 levels. The evidence is also consistent with the standard finding in the literature on aggregate impacts that climate change is extremely likely to have reduced total global income over this period though we cannot reject a null of no impact for segments of the global middle class, income groups largely comprised of individuals resident in cool climates where impacts are expected to be relatively mild.

Section 1.5 concludes by summarizing the contributions and limitations of the present analysis and suggesting promising extensions of this work. In particular, I briefly consider how the subject of climate inequalities poses urgent challenges the economic analysis of climate change and the design of climate policy.

1.1.1 Related literature

Spurred by methodological advances, accessibility of computing power, and political urgency, the past decade has seen a proliferation of economic research dedicated to estimating the social impacts of climate change. To contextualize the relative sparsity of within-country distributional impact studies in climate economics, it is instructive to familiarize the reader with the subfield's taxonomy of empirical designs into complementary "top-down" and "bottom-up" approaches each premised on an underlying utilitarian cost-benefit logic.

Top-down approaches restrict attention to a widely available outcome of interest. This outcome is taken as a comprehensive enough proxy for social welfare that climatically attributable impacts to the stock of this outcome may represent the socially relevant climate "damage". The pioneering works in this category include Dell et al. (2012) and M. Burke et al. (2015a), which relate annual observations of country-level temperature to GDP growth. Alternatively, bottom-up approaches consider impacts to distinct "sectors" of an economy; mortality (T. Carleton et al., 2022), energy consumption (Rode et al., 2021), and labor (Graff Zivin & Neidell, 2014) are common areas of focus. Results using this approach can be presented as partial costs or combined with other sectoral damages in an integrated model which accounts for their interdependencies. The advantage of the former approach is in its convenience and interpretability: the wide availability and cross-country comparability of economic aggregates such as those provided by national accounts substantially simplifies the task of calculating global climate impacts. The benefit of the latter approach is in its ability to account for disaggregated and non-market impacts to which single-dimensional aggregates are largely indifferent.

Conversely, the simplicity of the top-down approach and comprehesiveness of the bottom-up approach pose a critical tradeoff between geographic disaggregation and global coverage. In perhaps the most prominent existing work directly concerned with characterizing global 'climate inequality', Diffenbaugh and Burke (2019) estimate that the ratio between the top and bottom deciles of the global income distribution was 25% larger in 2010 than it would have been had anthropogenic climate forcings been constrained to 1960 levels, all else held equal. But this is necessarily calculated

by assigning all individuals their country's mean GDP and crudely defining global deciles accordingly, an unavoidable restriction when limited to 160 or so distinct economic units. The authors remark that "documenting the impact of global warming on within-country inequality remains an important challenge."

S. Hsiang et al. (2017) is a relatively rare study which makes the opposite compromise between specificity and breadth in order to make progress in this area. Integrating fine-scaled agricultural, crime, energy, mortality, and labor data, the authors determine that climate risks in the continental United States are disproportionately borne by counties in the US South and Midwest, regions generally already poorer than their coastal counterparts; climate change then is likely to imply worsening inequality within the United States. Yet to make more generalizable claims about the implications of global warming for global inequality, analyses must extend to settings which lack comparably rich or granular data.

The distributional focus of this paper complements a much more active area of research measuring unequal impacts across countries. Like Diffenbaugh and Burke (2019), a subset of these studies explore the implications of this type of inequality directly, for example by highlighting the inverse relationship between country-level contributions to cumulative global greenhouse gas emissions and exposure to their social consequences (Chancel & Piketty, 2015; Kotz et al., 2024; Ricke et al., 2018). There is by now general agreement that non-linearities in the response of aggregate variables to marginal warming concentrate damages and risk in relatively warm and poor countries which have contributed proportionally little to anthropogenic warming (Howard & Sylvan, 2021). However, estimated magnitudes of these impacts are notoriously divergent, largely because of unsettled disagreement over the persistence of these non-linear effects (Newell et al., 2021). We draw from recent work by Nath et al. (2024) and Bilal and Känzig (2024) which use less restrictive time series methods to circumvent these statistical bugbears.

Similar non-linearities emerge at the micro level across a range of outcomes as varied as labor supply (Graff Zivin & Neidell, 2014), agricultural yields (Schlenker & Roberts, 2009), and standardized test performance (J. Goodman et al., 2020). T. A. Carleton and Hsiang (2016) overviews the proliferation of multidisciplinary studies in this category whose results describe patterns of setting-specific inequality suggestive of a broader micro-level environmental inequality similar to those observed in the aggregate. The present study bridges this micro evidence with the aforementioned macro evidence on inequalities between countries, estimating distributional responses consistent with the general features of both categories of impacts. We also document evidence of negative impacts to an elite minority, which I have not seen examined in existing work.

The literature on climate impacts on economic inequality *within* countries has remained comparatively sparse in large part due to an absence of adequate research infrastructure. Estimation of distributional equivalents to the aggregate damage functions standard in the literature requires data comparable in credibility and coverage to domestic production data whose construction across countries follows rigorous standards set forth by the international System of National Accounts (European Commission et al., 2008). Historically, distributional measurement of welfare has lacked and arguably been crowded out by the wider interest and investment in measuring aggregate production (Jorgenson, 2018). While secondary inequality data products compiling country-year inequality variables from a wide array of official and survey sources date at least as far back as Deininger and Squire (1996), the validity of these products for cross-country causal analysis is limited by inconsistencies across primary sources in construction and availability (Atkinson & Brandolini, 2001). For example, the most recent rigorous evaluation I could find of these secondary datasets

concluded that results derived from the conveniently packaged Standardized World Income Inequality Database should be regarded as "not sufficiently credible" and that analyses using the United Nations' long-running World Income Inequality Database should at best be accompanied by justifications for their sample and variable selection criteria (Jenkins, 2015). Nonetheless, these inequality products have been used in at least two recent studies (Cevik & Jalles, 2023; Gilli et al., 2024) relating the inequality measures provided in these products to variations in weather.

In the absence of credible global inequality measures, a handful of works have managed to contribute to the study of within-country inequality impacts by using idiosyncratic data available for individual countries. S. Hsiang et al. (2017), described earlier, is a particularly impressive example. Elsewhere, Marx (2024) uses rich tax data from France to infer that years with marginally more days above 30°C are associated with wider income disparities between the richest and poorest cantons (administrative units described as roughly equivalent to 10 municipalities). Dasgupta et al. (2023) uses data from perhaps the most unequal country in the world to project a 3-6 Gini point increase in South Africa by 2100 under a moderate warming scenario.

The research programs of Thomas Piketty, Emmanuel Saez, and their co-authors since the turn of the century are widely credited with revitalizing public and academic interest in the measurement of within-country inequality. Beginning with work tracing the long-run evolution of income and wealth inequality in France (Piketty, 2003) and the United States (Piketty & Saez, 2003), their mobilization of extensive tax returns data has enabled credible measurement of top incomes over long time horizons, stimulating new programs of research and equipping the general public with new political language around which to mobilize (Jones, 2015). This paper takes advantage of the cross-country research apparatus that has rapidly developed and been made accessible in recent years with the intention of maturing the economic analysis of climate change in a similar manner. Palagi et al. (2022) is the only other climate-economic study I could locate which makes use of the same "distributional national accounts" (World Inequality Lab, 2024) used in this study. The authors find that extreme precipitation negatively impacts low-income individuals much more severely in countries more dependent on agriculture.

Finally, I locate this paper in the context of a growing demand for distributional considerations in economic analyses of climate change and the design of climate policy. For example, Chancel et al. (2023) estimates that within-country inequality of carbon emissions have recently exceeded the equivalent "carbon inequality" between countries and advocates for targeted carbon taxation programs based on individual footprints to more efficiently internalize the climate change externality. The modern SCC framework accommodates equity considerations by concavifying the underlying utility function so that the welfare weights assigned to individuals are inversely related to their relative levels of consumption; a logarithmic specification is commonly chosen, for example. Variations of this more progressive utilitarianism have been institutionalized to some degree in a handful of countries including recently in the United States (Office of Management and Budget, 2023). Theoretically, the way equity weighting is implemented should result in substantially higher SCC estimates depending on the extent to which the distributive effect of climate change is modeled as regressive. This paper provides the first comprehensive estimate of the global regressivity of climate change at subnational resolution.

1.2 Data

1.2.1 Distributional income data

We make use of the recently published grouped income series from the World Inequality Database (World Inequality Lab, 2024). Its predecessor, the World Top Income Database launched in 2011, drew from comprehensive tax records from over 30 mostly Western countries to capture the distribution of income among their top income-earners with unprecedented precision. A critical innovation used to develop the new product is the systematic integration of this tax data with national accounts and household survey data in order to impute the full income distribution within those countries (Piketty et al., 2019) in a manner patterned after the System of National Accounts (European Commission et al., 2008). The newest generation of WID data extends coverage within those countries—now claiming to capture "100% of income" in the United States (Piketty et al., 2018)—and to countries with relatively limited available data. By 2019, historical data for over 120 countries had been incorporated into the database. As of this writing, their equal-split adult pre-tax income series was available for 215 countries and territories collectively representing almost the entire global population with distributional data imputed every year since 1980. All income estimates are converted to 2023 US dollars using the provided PPP-adjusted exchange rates. Accompanying metadata scores the availability and reliability of each category of input source by country, unsurprisingly indicating that the input mix of these sources varies substantially by country with less developed countries being more reliant on household surveys and wealthier countries on tax records.

Notably for the percentile-level panels, the income data exhibits substantial bottom-coding with almost every country-year reporting exactly zero income for at least the bottom four percentiles of the pre-tax income distribution. Even if an accurate measure of adult unemployment, this evokes well-known deficiencies of income as an identifier and measure of poverty compared to alternatives like consumption expenditures (Meyer & Sullivan, 2003). For example, while the unemployed subpopulation of a given economy may well be composed of its most impoverished, it is likely to also include unemployed individuals of all levels of precarity and social status and does not account for security provided by communal and social safety nets. It also positively biases estimated changes in welfare among the poor since zero incomes mechanically cannot decrease.

Given these inherent disadvantages of income measures as a proxy for relevant welfare, we intentionally do not make precise claims about the poverty implications of our findings. Instead, we consider the incomes of the bottom 10 percentiles as a group in all analyses and consider estimated impacts for this quantile to be a conservative bound on true impacts on the very poorest because it implicitly assumes perfect equality among its constituent population (Cowell, 2011).

1.2.2 Historical weather data

We merge all income data with an annual country-level panel of population-weighted near-surface air temperature data derived from the Global Meteorological Forcing Dataset for Land Surface Modeling (GMFD). which corrects model-based biases in NCEP/NCAR reanalysis product using observational meteorological station data (Department of Civil and Environmental Engineering/Princeton University, 2006). The GMFD reports historical weather data for the entire global

surface at 0.25-degree resolution[2] for every three hours from the start of January 1, 1948 to the end of December 31, 2015. We use the product which reports daily averages of near-surface air temperature.

While other temperature series of similarly high resolution can provide a larger estimation sample by beginning before 1948 or updating more recently than 2015, results reported in the broad climate impacts literature have been generally consistent across data products. We favor the GMFD product as it has the unique feature of being the only reference product used to bias-correct and downscale the CMIP6 projection data series produced by the NASA Earth Exchange Global Daily Downscaled Projections; this allows the results of the present analysis to seamlessly integrate with future work exploring its implications under different future warming scenarios or attribution studies which attempt to decompose estimated climate change impacts into their anthropogenic and natural contributions, as introduced in Section A.4.

1.2.3 Temperature as an explanatory variable

It has been customary in the climate impacts literature to use functions of observed weather variables such as temperature or precipitation as explanatory variables in causal analyses. Historically, their use in panel settings has been justified by an intuition that annual fluctuations in these geophysical conditions are plausibly exogenous with respect to most economic activity. Among several challenges to this intuition, Nath et al. (2024) objects to their validiity on the grounds that because levels of temperature do not satisfy conditions of an identified shock[3] and exhibit substantial autocorrelation, they are ill-suited for dynamic causal inference.

I follow Kahn et al. (2021) in defining temperature shocks $\widehat{\tau}_{it}$ as deviations in temperature relative to a climate 'state' variable $\overline{T}_{it}$ in turn defined as the M-period moving average of local temperature $\overline{T}_{it}$:

$$\overline{T}_{it} := \frac{1}{M}\sum_{m=1}^{M} T_{i,t-m} - 18 \tag{1.1a}$$

$$\widehat{\tau}_{it} := T_{it} - \overline{T}_{it} \tag{1.1b}$$

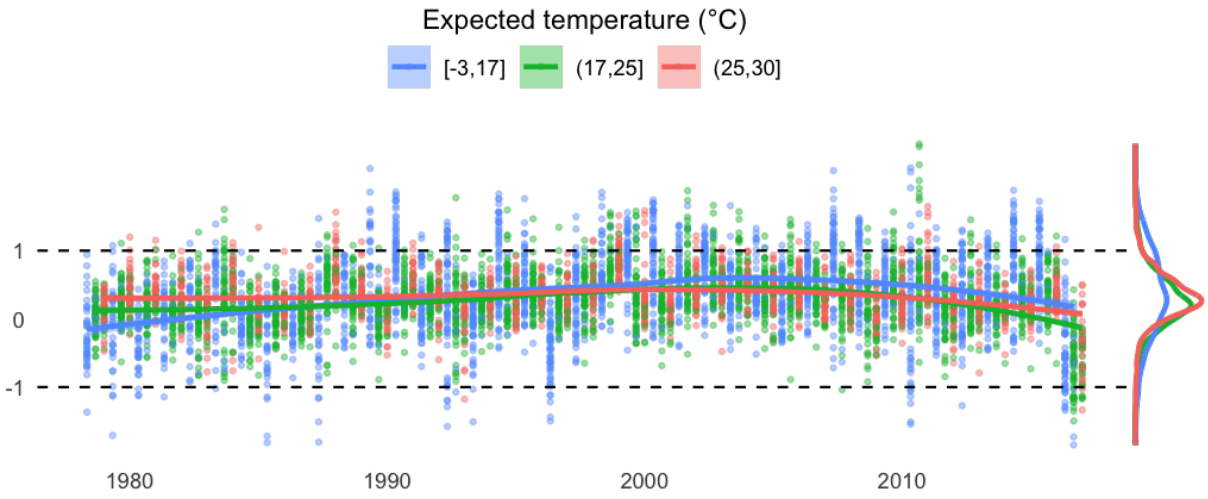
We subtract 18 in Equation (1.1a) so that our main effects in state-dependent regressions correspond to an intermediate representative climate of 18°C. For sufficiently large M, this definition of the state variable $\overline{T}_{it}$ has an appealing coherence with the convention of using long-run moving averages of weather realizations to define the slow-moving state of the climate at a fixed period[4]Most commonly, these "climate normals" are defined as 30-year averages to accommodate natural variation which would occur even in a stable climate so we set $M = 30$ in our main results. The most economically relevant example of this natural variability is the El Niño Southern Oscillation, an ocean-warming phenomenon which can vary average global surface temperatures by as much as

[2]At this resolution, cells near the equator are approximately 27 km^2 (17 mi^2).

[3]To crudely summarize the conditions as outlined in the handbook chapter of Ramey (2016), a valid shock in a time series setting must i) measure unanticipated movements in relevant exogenous variables, ii) be exogenous with respect to current and lagged endogenous variables, and iii) be uncorrelated with other exogenous shocks included in the model. Temperature specified in levels do not satisfy the first condition and the third condition is violated in commonly used non-linear specifications where temperature enters as a polynomial.

0.4°C across its three phase cycles which may span 2-7 years. Other minor oscillations can span multiple decades.

Figure 1.1: Country-level time series of identified temperature shocks



Points correspond to shocks observed in individual country-years. The three colors correspond to an equal-sized binning of country-years by their moving-average temperature. Flexible local regression curves describe time trends split by this binning. Marginal distribution plots on the right show that colder country-years observe greater variance in shocks.

Figure 1.1 depicts the implied history of temperature shocks $\widehat{\tau}_{it}$ constructed in this manner with individual points corresponding to distinct country-year observations. These observations are divided into three equally sized groups according to their corresponding climate states $\overline{T}_{it}$; points and local regression curves are colored accordingly. The marginal distribution plots on the right demonstrate that colder seasonal countries experience more positive and more volatile temperature shocks than their warmer counterparts, a reflection of both greater natural weather variability in these settings and the fact that absolute anthropogenic warming is greater in regions closer to the Earth's poles[5].

The positive values of the curves in Figure 1.1 over almost the entire observation period captures the idea that systematic warming is characterized by the increased frequency and greater magnitude of positive (hot) shocks relative to negative (cold) shocks. Intervals where the curves are closer to 0 thus reflect relative balance in the incidence of hot and cold shocks, corresponding to periods of relative climate stability and slower warming.

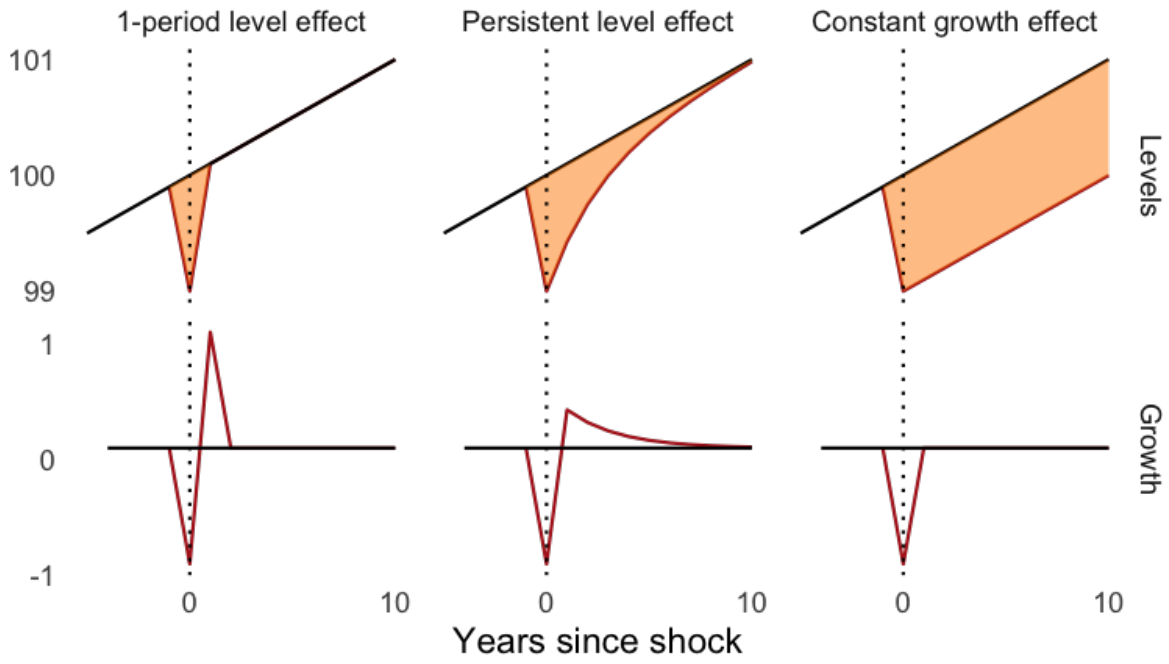
[4]Nath et al., 2024 use a state variable that is constant for a given country. They suggest this is justified by the moderate warming over their observation period but I find that observed warming in my sample is non-trivial: the median country observes, on average, a 0.26°C (IQR: 0.18-0.36) increase in these 30-year temperature normals every 15 years.

[5]Figure A.6 provides a more direct illustration of this phenomenon for different global climate models

1.2.4 Modeling the persistence of social impacts: levels vs. growth effects

Figure 1.2 illustrates the conceptual distinction between transitory level effects and permanent growth effects. Each column corresponds to a different persistence structure for the dynamic effect of an isolated temperature shock in period 0 on an outcome measured in log levels (top panels) and first differences in these log levels (bottom), a convenient approximation of a growth rate. Black and red lines depict the evolution of the outcome variables in the absence and presence of a unit shock in period 0 respectively under the different described dynamics. Filled areas in orange measure the cumulative loss (or gain) in the outcome which may be attributed to the shock.

Figure 1.2: Levels and growth under different persistence structures



Under a one-period level effect, the level of, say, GDP falls by an estimated level $\widehat{\beta}_0$ in the year of the shock but fully recovers to its pre-shock trend (black) in the next period. In this scenario, the production lost in the year of the shock is exactly offset by an above-trend rate of growth which exactly offsets the negative growth effect in the year of the shock. The shock has no additional effect on future production and the area in orange represents the one-off impact from the shock.

Under the constant growth effect depicted in the rightmost panels, the same shock incurs the same effect on the levels but this effect is permanent so that GDP in all future periods is reduced by the same amount relative to the no-shock counterfactual. The difference between the two trajectories thus compound over time as observed by the orange area which grows linearly with the projection horizon. The corresponding growth effect depicted in the bottom panel is entirely transitory with no compensatory positive growth relative to trend.

The middle column describes an intermediate case where the economy requires several periods

to return to its pre-shock trend. Cumulative differences attributable to the one-off shock continue to rise over time but to a diminishing degree until eventually compensatory positive growth exactly offsets the initial negative growth, i.e., the integral of the growth curve equals zero in the long run.

Now consider the following two models, which we may refer to as the levels and growth models respectively.

$$\log Y_{it} = \beta^L X_{it} + v_{it} \tag{1.2a}$$

$$\Delta \log Y_{it} = \beta^G X_{it} + u_{it} \tag{1.2b}$$

For simplicity, imagine the outcome variable Y_{it} represents a conventional economic aggregate like GDP and X_{it} a perfectly identified temperature shock. In the levels model, β^L represents an estimate of the change in log GDP attributable to a unit shock. Graphically, this impact is represented by the vertical distance between the black and red lines in the top-left plot. Since this is an entirely static model, there is no mechansim for X_{it} to affect future values of Y_{it} and so the model assumes a perfect return to trend in all ensuing periods. In the growth plot directly below, this is represented by a $\beta^L - 0 = \beta^L$ effect in period 0, an exactly offsetting $0 - \beta^L = -\beta^L$ effect on growth in period 1, and no effect on growth in all ensuing periods. In models specified in levels, level effects which persist for p periods can be accommodated by including p additional lags of X in the regression. A permanent growth effect can only be accommodated with the inclusion of infinite lags.

In the growth model, $\beta^G \neq 0$ represents a permanent change in the difference between consecutive values of Y_{it}. This is represented by the gap between the black pre-shock trend line and the observed GDP series in the top-right plot staying constant at β^G for all periods following the shock and is never offset. Level effects which persist for p periods can be accommodated by including p lags of X and finding that the sum of the $p + 1$ coefficients is exactly 0 so that the initial negative growth effect is eventually offset by sufficient compensatory growth.

The differing restrictions imposed by these stylized regression models is a simplification of perhaps the most influential unsettled question in climate economics: the persistence of climate impacts. For example, two studies projecting the proportional loss in cumulative global production by 2100 arrive at point estimates of -20% (M. Burke et al., 2015a) and -3.4% (Casey et al., 2023) under the same worst-case warming scenario. Methodologically, the two studies use similar methods and data except for their choice of outcome variable.

Intermediate persistence structures where level effects are diminishingly persistent can be accommodated by either model by including sufficient lags of the explanatory variables. Of course, identifying variation is reduced with the addition of additional explanatory variables, a problem exacerbated in specifications where shocks enter non-linearly and settings where shocks exhibit significant autocorrelation, both general features of modern climate models using absolute temperature as their explanatory variables. In addition, models specified in levels are notoriously vulnerable to unit root concerns, further undermining credible identificaiton.

Indeed, the appendix to M. Burke et al. (2015a) concludes that the identifying variation which remains after lag augmentation to test for persistence "cannot reject the hypothesis that this effect is a true growth effect nor the hypothesis that it is a temporary level effect", rendering these extremely influential assumptions effectively unfalsifiable. Absent substantially improved data, resolving this issue may require richer data or an alternative methodological approach relatively robust to misspecification.

1.3 Empirical analysis

1.3.1 Methods: cumulative temperature multipliers

To characterize the dynamic effects of an identified temperature shock on national income distributions, we adapt the method of local projections first introduced in Jordà (2005). As Jordà (2023) notes, this semi-parametric method of estimating impulse response functions has found growing appeal outside of applied macroeconomics in recent years as recent econometric literature has demonstrated its flexibility in incorporating non-linearities (Cloyne et al., 2023), its coherence with the potential outcomes framework (Dube et al., 2024), and its robustness to misspecification and non-stationarity (Montiel Olea & Plagborg-Møller, 2021; Montiel Olea et al., 2024; Piger & Stockwell, 2023). Berg et al. (2024), Bilal and Känzig (2024), and Nath et al. (2024) are the first works I am aware of to apply this method to estimate dynamic economic responses to climate shocks, demonstrating how these features can help address the identification issues described in Section 1.2.

Our application of this method to our inequality setting entails the following steps:

1. Estimating the state-dependent impulse response of income to an identified temperature shock
2. Estimating the state-dependent impulse response of temperature shocks to an identified temperature shock
3. Using both to estimate a state-dependent multiplier to isolate the dynamic income response net of the shock's effect on future shocks
4. Statistically evaluating the significance and persistence of this multiplier
5. Repeating the above steps for each income quantile of interest

We briefly describe this process below and provide additional exposition in Section A.2 using an application to country-level GDP data.

Estimating state-dependent impulse responses

We use country-year panel datasets to estimate dynamic responses to temperature for an outcome of interest represented by the outcome variable y_{it}. For example, y may represent the average logarithm of income of the bottom 10% of individuals in country i in year t and we may repeat the estimation process for each income decile and use the results to infer a total effect on national income distributions.

The method of local projections entails estimating $H + 1$ iterations of the following single-equation model each corresponding to different horizons $h \in \{0, 1, ..., H\}$ periods after an identified temperature shock:

$$\begin{aligned} \Delta_h y_{i,t+h} &:= y_{i,t+h} - y_{i,t-1} \\ &= \beta_{1h}\widehat{\tau}_{it} + \beta_{2h}\widehat{\tau}_{it} \cdot \overline{T}_{it} + \lambda_h \overline{T}_{it} + \boldsymbol{\gamma}_h^\top \mathbf{Z}_{it} + \mu_i + \eta_t + u_{i,t+h} \end{aligned} \tag{1.3}$$

In our main specification, the matrix $\mathbf{Z}_{it}$ includes lagged controls $\{\widehat{\tau}_{i,t-j}, \widehat{\tau}_{i,t-j} \cdot \overline{T}_{it}, \Delta y_{i,t-j}\}_{j=1}^{p}$. The inclusion of p lags of the temperature shock $\widehat{\tau}$ is included to address potential autocorrelation in the explanatory variable. Interactions of these lagged shocks with the state variable capture

potential state-dependencies in these autocovariance structures motivated partially by the finding in Figure 1.1 that shocks in cooler climates tend to be more positive on average and feature greater year-to-year variability. Lags of the outcome are included as a guard against trend non-stationarity. Finally, the sets μ_i and η_t of fixed effects are included to absorb time-invariant unit-specific effects and common year-specific effects. In all, this model differs from the state-dependent model of Nath et al. (2024) only in how temperature shocks are constructed and in permitting the state variable to vary over time as described in Section 1.2.3.

For each projection horizon h, the main regression coefficients of interest attribute to a unit temperature shock a main effect β_{1h} representing a change in the "long difference" between y observed h periods after the shock and y observed the period before the shock. Equivalently, this can be interpreted as the change in y h periods after the shock relative to a counterfactual absent the shock. In most of our results, y represents the logarithm of income for some quantile of the national income distribution; in these cases, coefficients are interpreted as percentage-point impacts on income.

An interaction term β_{2h} allows the effect at each period to scale with an observation's time-varying expected temperature $\overline{T}_{it}$, defined in Section 1.2.3. This 'state dependence' is motivated by the standard result in the climate impacts literature that the marginal effect of a given temperature shock on aggregate economic outcomes is increasingly negative for warmer climates.

The inclusion of a main effect λ_h associated with the time-varying state variable allows for the possibility that expected impacts may depend on changes in expected temperatures, net of the impact of climate change on the distribution of temperature shocks. For example, if a hot country stabilizes at a climate that is 1°C warmer than some reference period, the productive potential of the country is altered even in the absence of temperature shocks as the new climate will be differently conducive to some economic activity. Neglecting this possibility amounts to assuming perfect climate adaptation in the long run and that climate change is only socially impactful by biasing the trend in temperature shocks.

It is worthwhile to take stock of the assumptions underlying the sequence of regressions specified in Equation (1.3). First, because coefficients for each horizon are estimated through distinct regressions using differently constructed variables from different subsamples of the data, there are in principle no restrictions imposed by the model on the shape of the resulting impulse response function for a given value of the state variable; it is in this sense that the method is partially non-parametric.

This flexibility is particularly appealing in the context of estimating responses to weather shocks. As summarized in Section 1.2.4, fully parametric methods have necessarily required the modeler to assume either of the extreme persistence patterns depicted in Figure 1.2 with limited ability to accommodate intermediate persistence dynamics. By instead allowing the modeler to remain mostly agnostic to the data generating process, this method circumvents a notorious source of extreme estimation instability and can accommodate any persistence structure in principle without restriction[6]. Specifically, the sequence of coefficient estimates $\{\beta_{1h}\}_{h=0}^{H}$ in Equation (1.3) maps directly to the logged differences between the black and red time series in the first "levels" row or equivalently the sequence of its first differences correspond to the red time series depicted in the second "growth" row. Thus, a one-period level effect would be represented by a sequence where $\beta_{1h} = 0$ for all horizons $h > 0$ while a constant growth effect would be reflected in a sequence converging to a

non-zero constant[7].

Another relevant benefit of the method is that local projections specified in long differences have recently been shown to be particularly robust to biases associated with highly persistent and even non-stationary processes Jordà (2024) and Piger and Stockwell (2023). This is a long-standing concern with regressions specified in levels such as those which use logged GDP as an outcome (Campbell & Mankiw, 1987; Nelson & Plosser, 1982). Altogether, the method is relatively robust to different forms of misspecification

These minimal restrictions naturally come with costs in estimation efficiency; IRFs estimated by local projection regressions are fundamentally noisier than their parametric and structural counterparts such as vector autoregressions[8]. For this reason, we expect resulting estimates to be less precise than more restrictive methods, particularly for increasingly distant projection horizons.

At the same time, the model is not entirely free of restrictions. Most critically, the model specified in Equation (1.3) imposes within-horizon linearity assumptions on the coefficients of interest. Since the shock enters additively into Equation (1.3), responses to the shock are assumed to scale proportionally so that a 1°C shock has exactly twice the impact of a 0.5° shock and the exact negative of the effect of a -1°C shock.

By the same principle, the linear interaction term imposes a symmetry in the state dependence such that holding all other factors fixed, the impact of a shock is assumed to be $k\beta_{2h}$ percentage points larger in a climate some quantity k degrees warmer than baseline and $-k\beta_{2h}$ in a climate k degrees cooler than baseline. While appreciating that unpredictable non-linearities abound in the interactions between social and climate systems (Dietz et al., 2021), I consider these necessary and palatable restrictions.

Collecting estimates of the coefficients of interest $\{\widehat{\beta}_{1h}, \widehat{\beta}_{2h}\}_{h=0}^{H}$ from Equation (1.3) and adopting the notation of Jordà (2023), our set of local projection estimates imply a family $\mathcal{R}_{\tau \to y}(h; \overline{T})$ of state-dependent impulse responses as a function of the state variable $\overline{T}$:

$$\mathcal{R}_{\tau \to y}(h; \overline{T}) := \widehat{\beta}_{1h} + \widehat{\beta}_{2h} \cdot \overline{T} \tag{1.4}$$

Estimating state-dependent cumulative multipliers

To make progress accounting for the persistence of temperature shocks themselves (see: Section 1.2.3), Figure 1.3 depicts impulse responses which result from using temperature shocks themselves as an outcome variable y as constructed by two commonly used datasets for gridded historically observed weather. We plot separate impulse responses for three representative states corresponding to the 20th and 80th percentiles of country-year expected temperatures in our sample (10°C colored blue and 26°C colored red) and their average (18°C, roughly the 40th percentile and colored green). This coloring scheme and these reference temperatures will be used throughout the

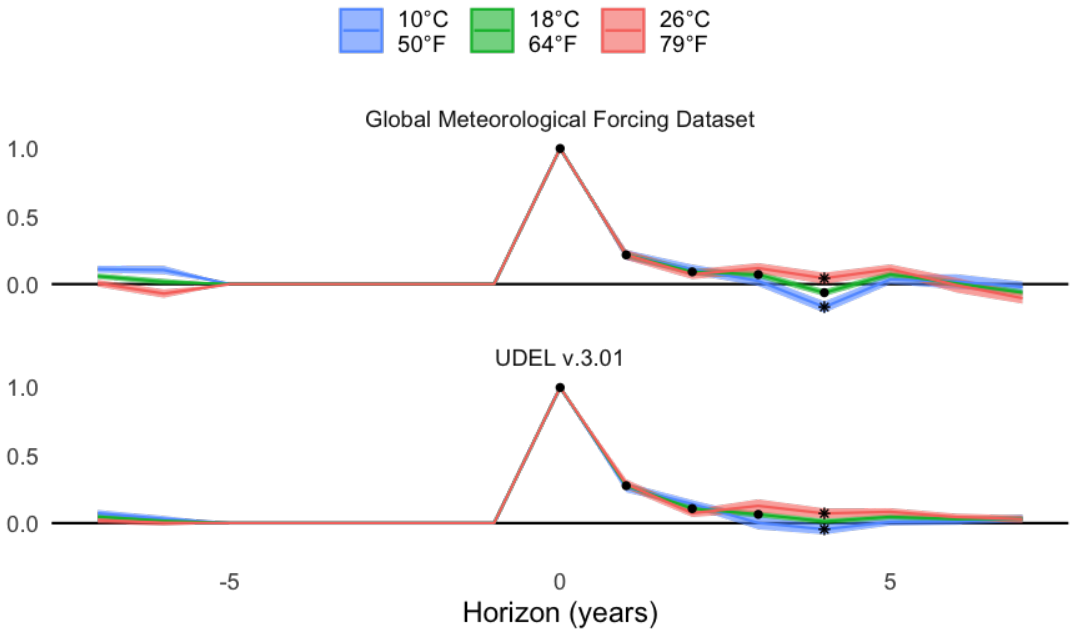
[7]Residuals across the projection horizon are generally autocorrelated but do not affect consistency of point estimates.

[7]Of course, under a conservative null hypothesis of non-persistence at all horizons, a permanent growth effect cannot be affirmatively evidenced and the modeler can only produce evidence of a lower bound on persistence

[8]Different compromises in this bias-variance tradeoff can be obtained by imposing additional restrictive parametric assumptions such as the functional approximation approach of Barnichon and Matthes (2018) which can substantially diminish estimation error.

rest of this paper. [9]

Figure 1.3: State-dependent impulse response of temperature shocks to a unit temperature shock



State-dependent impulse responses of identified temperature shocks estimated using two widely used temperature datasets. The points in black denote significant evidence of autocorrelation. In both cases, persistence is large in magnitude for the first 1-2 periods following a shock before flattening. Negative horizons are included as a check for significant pre-shock trends.

Points along the IRFs depicted in black correspond to point estimates which are deemed significantly persistent by a joint hypothesis test with $\alpha = 0.05$. The underlying test is based on the significance-band test recently introduced in Inoue et al. (2024) and adapted for our state-dependent setting as described in Section A.2.2. In short, we have adopted a convention where filled circles correspond to significant main effects and asterisks correspond to significant interaction effects under a joint null hypothesis of non-persistence at each horizon. The presence of a significant effect up to horizon h is interpreted as evidence of a lower bound of h-period persistence. Thus, the IRFs presented in Figure 1.3 constitute significant evidence that temperature shocks positively influence successive temperature shocks for at least four periods although the magnitude of this persistence is mostly diminished within 1-2 years.

[9]Note that for expositional convenience, we have transformed our state variable to center on the intermediate reference temperature of 18°C so that the responses in green correspond to the sequence of estimated main effects $\widehat{\beta}_{1h}$ while the responses in blue and red incorporate equivalent but opposite interaction effects corresponding to ±8°C deviations from this reference state

As emphasized by Nath et al., we will want to directly account for this persistence when attributing social impacts to a unit temperature shock. An analogy can be drawn to the estimation of multipliers in empirical macroeconomics: when estimating the impulse response of, for example, GDP to a fiscal policy shock, one must account for the tendency of stimulus programs to be implemented in stages or to be followed by additional, if typically smaller, rounds of stimulus. Otherwise, the effect sizes implied by simple impulse response functions misattribute cumulative changes in the outcome only to the initial unit shock; the calculation of multipliers scale these effects at each horizon by the corresponding accumulation of shocks. As such, we will want to present our results in the form of these cumulative temperature multipliers rather than as raw impulse response functions.

In Section A.2.3, I describe how we modify methods for efficiently estimating cumulative temperature multipliers in order to implement this transformation while maintaining the ability to statistically test for persistence. Ultimately, the process of estimating cumulative temperature multipliers for some outcome y amounts to estimating a cumulative variant of the original single-equation model in Equation (1.3). That is, for every discrete income quantile q of interest and projection horizon $h \in \{0, 1, ..., H\}$, we estimate the following state-dependent model:

$$\Delta_h^c y_{i,t+h}^q = m_{1,h}^q \widehat{\tau}_{i,t+h}^c + m_{2,h}^q \widehat{\tau}_{i,t+h}^c \overline{T}_{it} + \lambda_h \overline{T}_{it} + \mathbf{Z}_{it}^q \boldsymbol{\gamma}_h^q + \mu_i + \eta_t + \varepsilon_{i,t+h}^q \tag{1.5}$$

Here, variables denoted by a superscript c are cumulative versions of the original variables constructed using the one-step process summarized by Algorithm 1 in Section A.2.3. The quantile-specific matrix of controls $\mathbf{Z}_{it}^q$ include lags $\ell \in \{1, ..., L\}$ of the shocks, the interaction of the shock with the state variable, the outcome $\Delta y_{i,t-\ell}^q$, and country-level aggregate growth $\Delta y_{i,t-\ell}$.

State-dependent cumulative temperature multiplier estimates $\{\widehat{m}_{1,h} + \widehat{m}_{2,h}\overline{T}_{it}\}_{h=0}^H$ describe the effect of an identified temperature shock on the cumulative stock of outcome y^q after h periods as a function of the expected temperature.

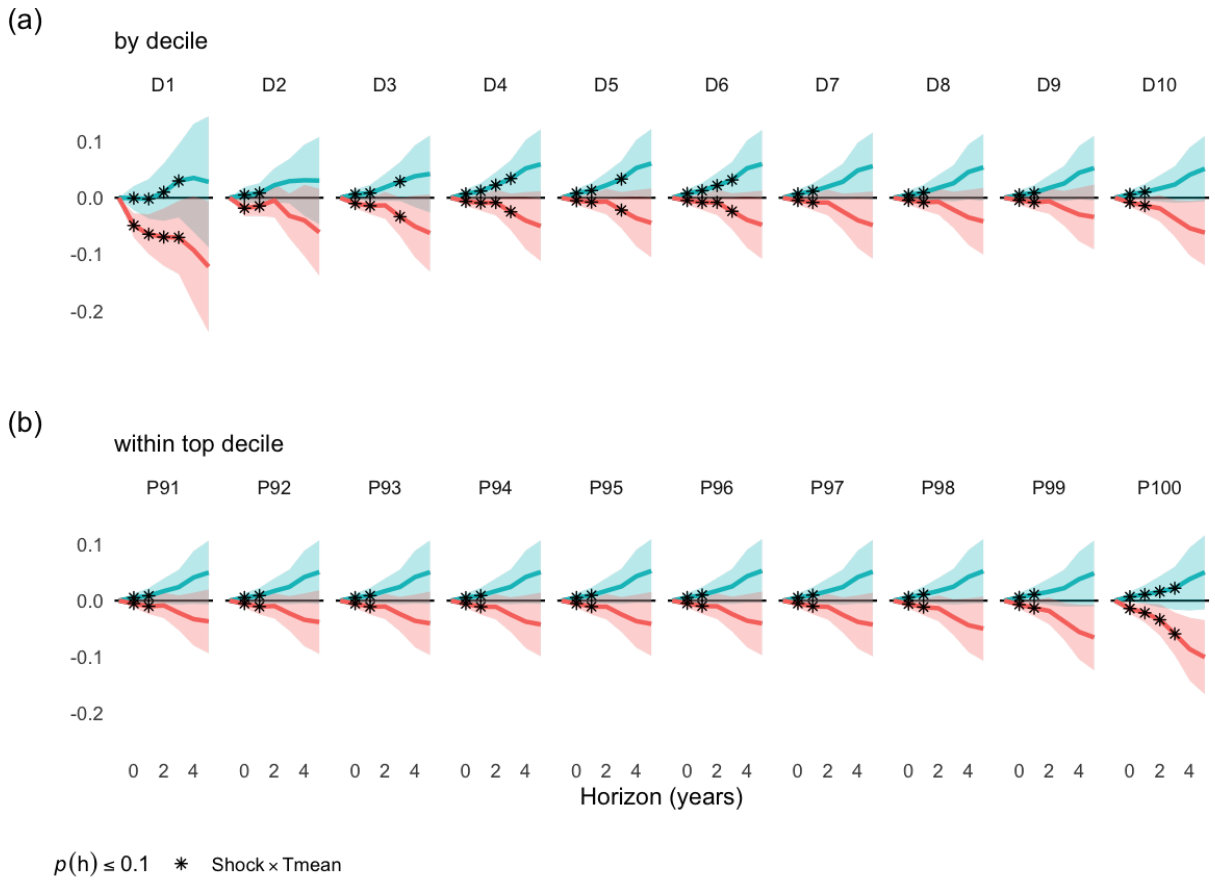
1.3.2 Empirical results: evidence for a temperature effect on within-country inequality

Panel (a) of Figure 1.4 summarizes the results of estimating separate sets of state-dependent umulative temperature multipliers for all 10 within-country income deciles. Here, color-coding follows the same intuitive scheme as previous plots with blue and red corresponding to representative expected temperatures of 10 and 26°C respectively[10]. The points in asterisks indicate significant interactions which impose differentiated effects based on expected climate. These effects persist for between 1-3 years, concentrating negative impacts in hot countries while keeping cooler countries relatively insulated, consistent with the aggregate results standard in the climate literature and the dynamic GDP results reported in Section A.2.3. Estimation error is larger than when estimating GDP impacts in Figure A.2b, unsurprising given the difference in temporal coverage and data quality.

Importantly, we are now able to compare impacts across quantiles within countries. Focusing on hot-climate countries, the estimates reported in Panel (a) imply an incidence pattern wherein negative effects of a transitory temperature shock to hot countries are found at all quantiles but are

[10]We omit the green 18°C main effect to avoid over-busy plots, but it is of course implied as the average of the two depicted curves

Figure 1.4: Cumulative income response to a 1°C shock



particularly dire at the bottom. For 26°C countries, the 90% confidence intervals for the impact of a temperature shock on cumulative income a year after the shock is $[-10.0, -2.9]$ percent for the bottom decile compared to $[-3.2, 0.3]$% for the second decile. This generally improves as one moves up the income distribution, stabilizing at approximately $[-2.4, 0.6]$ percent by the fourth decile through to the 99th percentile. Interestingly, the estimated impacts then discontinuously drops to $[-3.9, -0.5]$ for the richest percentile. Panel (b) shows this discontinuity within the top decile and also indicates this effect is significantly persistent for two additional years. In contrast, cumulative growth incidence is much flatter across the income distribution with point estimates suggesting positive impacts are somewhat larger for the middle classes between the third and sixth deciles.

Figure 1.5a and the bottom panel of Figure 1.5b capture these discontinuities at the extremes more clearly by plotting point estimates of percentile-level impacts over time for our three representative climates. These figures imply that negative expected impacts to hot countries are indeed concentrated on both the bottom 10% and the top 1% of the distribution[11]. To the best of my knowledge, this particular vulnerability of the highest incomes in hot and generally poor countries to transitory environmental shocks has not been previously observed or theorized although it is consistent with recent work outside environmental economics demonstrating this group's disproportionate sensitivity to aggregate conditions even compared to other members of the top income decile (Alvaredo et al., 2018; Roine et al., 2009). In ongoing work, I investigate the extent to which this may be driven by weather-induced capital depreciation, capital shares in agricultural income, and the relative unavailability of weather derivatives in these settings.

This first set of new results describe inequalities in the growth incidence of impacts across a national income distribution as a function of the local climate. The disproportionate impacts to the bottom decile strongly imply a regressive effect: In the aftermath of an isolated temperature shock, a representative member of the the bottom decile in a 26°C country can expect to earn 7% less in total over the next three years than they would had the shock never occurred. This compares to losses of 2.2% for the median adult in their country and a *gain* of 3.0% for their counterpart in a country with a 10°C climate. At the same time, a representative member of the 100th percentile in the same 26°C country, represented by the thinnest sliver at the top of the rightmost distributional contour plot in (b), can expect to earn 5.9% less over those three years than they would otherwise compared to 3.6% for the 99th percentile.

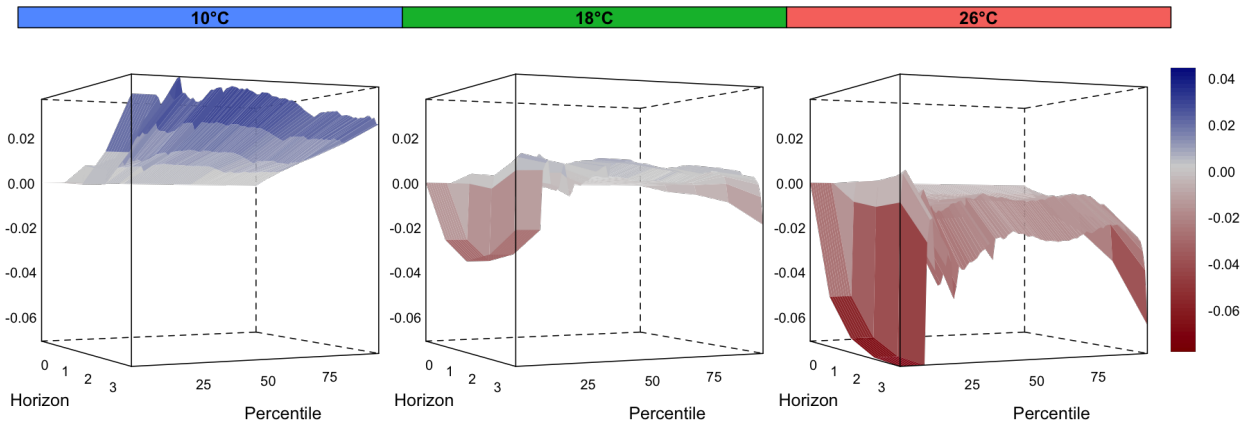
While these results strongly imply substantial redistributive effects from a temperature shock, their implied effect on inequality depends both on the initial distribution of income shares across quantiles and the measure of choice, especially given that growth effects are non-monotonic in income quantiles. Figure 1.6 depicts the annualized effect of a temperature shock on various measures of inequality. Units are not directly comparable across measures but all measures are increasing in (their respective conception of) inequality. Measures are provided in order from the most sensitive to redistribution of top incomes (the Theil index) to the most sensitive to redistribution of bottom incomes (the mean logarithmic deviation index).

The filled circles in the middle column drawn with green error bands correspond to significant main effects of the temperature shock on the given measure of inequality. The asterisks on the plots

[11]As mentioned in Section 1.2, the income data for the bottom 10% is not well suited for decomposition into percentile effects but one would expect an ideal decomposition would reveal a similarly steep within-decile gradient in which negative impacts are driven by the poorest percentiles within the bottom 10%.

Figure 1.5: Point estimates of cumulative income growth incidence responses to a 1°C shock

(a) Incidence patterns for three representative climates



(b) Aggregate vs. distributional responses

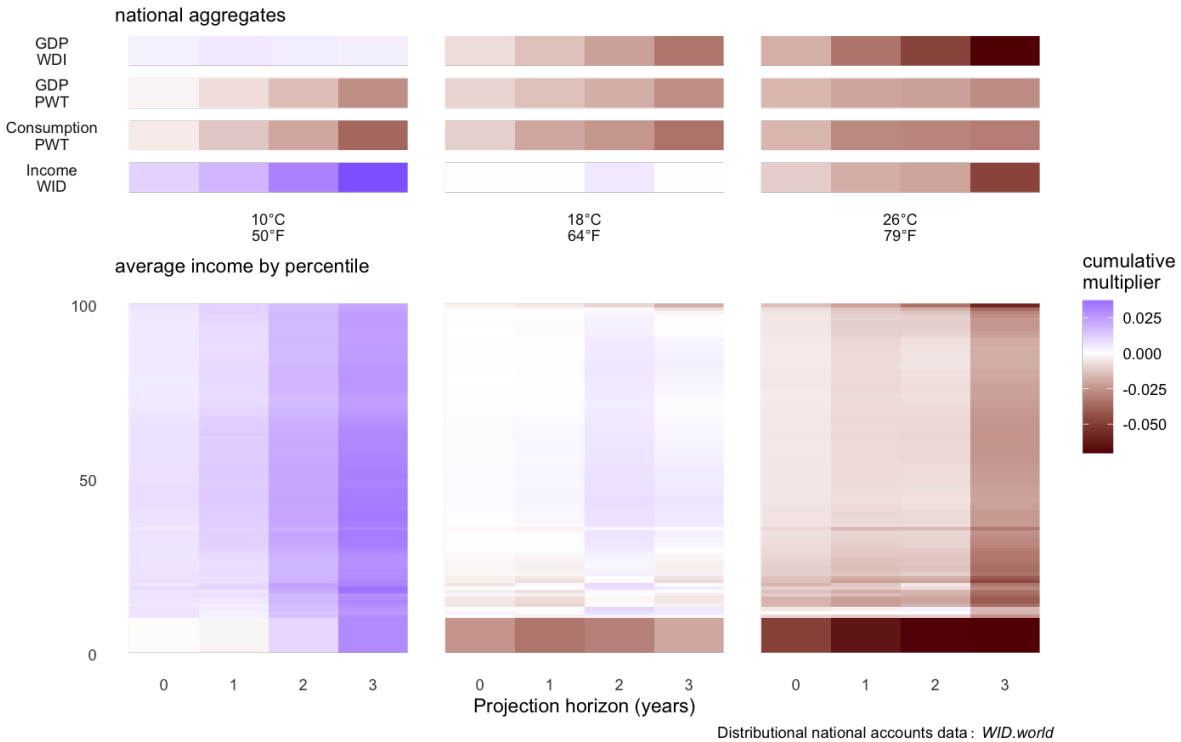
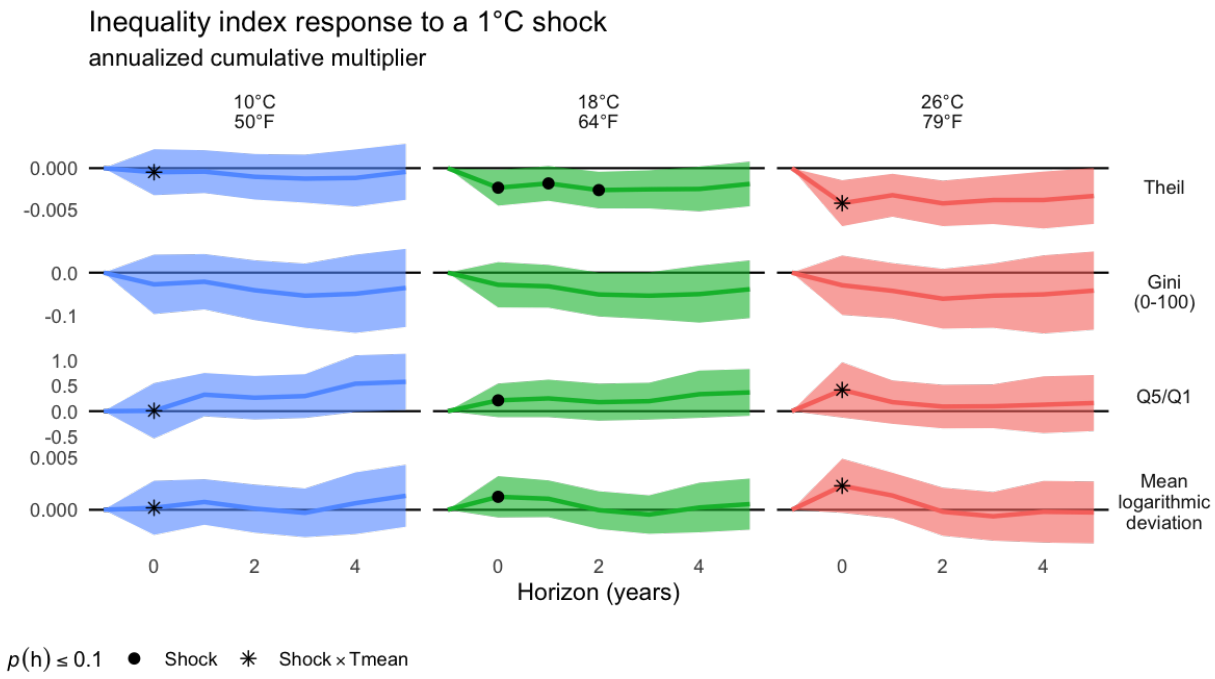


Figure 1.6: Inequality index response to a 1°C shock



on either side correspond to significant interactions, measuring linear heterogeneity by expected temperature. In light of the growth incidence results just discussed, the near-zero impacts to cold-country inequality and substantial effects to hot-country inequality are unsurprising. Null effects on the Gini index, known to be relatively insensitive to changes at either end of the income distribution, are consistent with the disproportionate impacts to the bottom 10% and top 1%.

By the Theil index[12], a transitory temperature shock appears to actually *improve* inequality while inequality as measured by the ratio of incomes of the top 20% to the bottom 20% and mean logarithmic deviation[13] both suggest the opposite. From the previous analysis, we can surmise that the progressive effect on the Theil index is being driven by the transfer away from top 1% while for the other two measures, the regressive transfer of income shares away from the bottom 10% outweighs this 'improvement' in inequality. While the impacts to the bottom 10% are substantially larger proportionally and represent a population segment 10 times as large, for the Theil measure, the disproportionately large share of income originally held by the top 1% constitutes a large enough share of total national income that these measures disagree on the net effect on inequality. Of course, since the overall effect on income is negative at all percentiles for hot countries, these inconsistencies

amount to differently equitable shares of a strictly smaller national income.

1.4 Counterfactual analysis: global inequality absent climate change

1.4.1 From identified temperature shocks to systematic climate change

The results presented in Section 1.3.2 are expressed as responses to a 1°C shock, which notably is much larger in magnitude than shocks we would expect to observe in a typical year in any country. Referring back to the time series of shocks illustrated in Figure 1.1, the median magnitude of a shock varies by the rate of climate change which will in turn vary across geographies and time. For the warmest 33% of country-years in our data, the median shock was approximately 0.19°C in absolute value in the 1970s and then increased to between 0.30-0.37° in the ensuing decades. Thus, hot-country effects can be understood as responses to a shock 3-5 times larger than is typical. For the coldest 33% of country-years, the median magnitude of a shock was approximately 0.35°C in the 1970s and 1980s, 0.51-0.53°C in the 1990s and 2000s, and 0.59°C in the 2010s. Their effect sizes then can be interpreted as the response to a shock just 2-3 times larger in magnitude than the median.

So while those results consistently demonstrated much larger distributional responses to the same shock for residents in warmer geographies, this disparity is at least partially offset by the relatively mild incidence of shocks in these settings. The counterfactual analysis we document in this section is valuable as a means of interpreting these reduced-form results practically, accounting for the spatial incidence of temperature shocks as well as other factors such as the absolute distribution of global incomes across country-quantiles and the spatial distribution of the global population across different geographies as observed in the historical record.

In the absence of a structural climate-economy model, it is standard practice to extrapolate reduced-form effects estimated using short-term weather variation to infer long-term impacts of sustained climate change. Broadly, this involves applying the estimated impact model to a counterfactual climate scenario to produce a counterfactual trajectory for the social outcome of interest. The difference between the social outcome of interest under this counterfactual relative to its trajectory under a reference scenario (e.g., for historical attribution studies like the present analysis, the actual historically observed trajectory of this outcome) are then attributed to the differences in the climate under the two scenarios.

In our case, we are estimating the effect of changes to the global climate system between 1980 and 2016 on the global distribution of income. Using the observed history of country-year temperatures, we calculate the implied history of country-year termperature shocks as defined in Equations (1.1a) and (1.1b). For the bottom national income decile and each of the 11th to 100th national income percentiles, we collect $B = 500$ bootstrap estimates of the state-dependent cumulative tem-

[13]One interpretation of the Theil measure is as a geometric measure of inequality. For any given income distribution, an individual with income $100,000 giving $10 to someone with income $10,000 would improve the Theil index measure of inequality by the exact same amount as someone with a $1,000 income giving the same $10 to someone with income $100 because the ratio of incomes between the donor and recipient are the same (Cowell, 2011).
By the same principle, a $100 income earner regressively transferring 10% of their income to a $1,000 income earner would offset the progressivity of a $100,000 income-earner transferring a ten-thousandth of their income to a $10,000 income earner. If that is still not intuitive, then you are in good company (Sen, 1973).

[13]One interpretation of the mean logarithmic deviation measure is as the average disutility of a given income distribution compared to an equal distribution of the same total income, assuming logarithmic utility.

perature multipliers of income (Equation (1.5)). Then integrating these percentile-bootstrap specific estimates to the observed histories of temperature shocks implies a counterfactual history of country-percentile-year income absent any temperature shocks between 1980 and 2016. The differences between this simulated history of global income distributions and the observed history of global income distributions are then attributed to changes in the climate over this 36-year period, a long-enough horizon to dominantly attribute to anthropogenic causes by the conventional definition of a climate normal.

This process (and its analogous implementation for projecting future climate change impacts) offers several significant advantages over prior approaches to this type of counterfactual attribution exercise. First, since estimates of cumulative damages result from local projection regressions, dynamic impacts are identified directly from medium-term long differences in income and disciplined by a statistical test of the persistence of these impacts rather than imposed by modeling choices, circumventing the contentious levels vs. growth criticism described in Section 1.2.4.

Second, models which use levels of temperature as an explanatory variable have been challenged on the grounds that these counterfactual exercises require them to extrapolate effects of treatments that have never been observed in the estimation sample. For example, the model of M. Burke et al. (2015a) specifies a globally generalized damage function which attributes to a given level of temperature an expected impact on GDP growth. This means that studying scenarios where annual country temperature levels exceed any observed in the estimation sample, much less the history of the human species, requires increasingly heroic extrapolations. This is much less of a concern in our model where the treatment variable is an identified temperature shock which is dynamically anchored by a time-varying state variable defined as a moving average of temperature levels, which can accommodate a capacity for adaptation in the sense ofS. M. Hsiang (2016) wherein the "direct" harm of exposure to a given temperature level can be partially offset by Bayesian "belief effects" wherein adaptive expectations about the distribution from which local weather realizations are drawn.[14]

Finally, I argue that our approach to attribution is more intuitive than those deployed by comparable studies which use identified temperature shocks as their treatment variable. Bilal and Känzig (2024) and Nath et al. (2024) are other recent papers which use local projection regressions to attribute economic impacts to identified temperature shocks. Curiously, both define their shocks as residuals from slightly different autoregressive distributed lag models of temperature levels. As I understand it, doing so seems to assume a model of nearly immediate and costless climate adaptability such that if a country is exposed to, for example, 40°C weather five years in a row, the fifth year will impose no economic impact because it will register as a 0°C shock. Because of this hyper-adaptability, the time series of historically observed temperature shocks is largely homoskedastic around a constant mean of zero, a stark contrast to the positively biased time series we observed in Figure 1.1[15]. Because of this absolute symmetry in positive and negative shocks and the within-horizon linearity assumption discussed in Section 1.3.1, were they to adopt the approach I had outlined for attributing damages to long-term climate change, the cumulative impacts of historically observed temperature shocks would net out to essentially nil[16].

[14]More extreme adaptation can be accommodated by lower values of M, the number of lags used to construct the moving average defining the state variable. However, a review of the empirical literature on climate adaptation, however, concludes "limited evidence" of adaptive behavior moderating the direct impacts of weather shocks in the context of climate change. (M. Burke & Emerick, 2016)

[16]In order to project the trajectory of GDP under different warming scenarios, Nath et al. instead seem to modify a

This lag between true and expected climate can be interpreted as a bound on the ability for belief or anticipation effects to offset the direct effects of weather shocks, corresponding to climate adaptation in the sense of Kahn et al. (2021) and S. M. Hsiang (2016). Alternatively, these lagged expectations may also be intuitively rationalized in a model of limited or costly adaptive capacity; even a farmer internalizing climate change trends will find it inconvenient or impractical to transform their land and re-optimize their crop mix every period in response to a gradual long-term trend. This imperfect adaptability gives rise to the positive trends we observed in Figure 1.1 during periods of more rapid climate change. To the extent that local expectations about the climate are anchored by recent experience, shocks defined this way may be interpreted as deviations relative to adaptive expectations. It is intuitive that one of the main channels through which systematic climate change imposes economic harms is through the increased frequency and magnitudes of positive "hot" shocks relative to negative "cold" shocks and the implausibility of perfect adaptation.

1.4.2 Results

Upon constructing these counterfactual series, each bootstrap b assigns each country-quantile-year to a global percentile corresponding to their population-weighted income ranking across the global income distribution for that year. Finally, we calculate the long-term growth between 1981 and 2016 implied by each series for each bootstrap series. The resulting long-term growth incidence curve is depicted in Figure 1.7a.

These results may be interpreted as impacts of observed climate change insofar as it the influence of climate change is sufficiently well-captured by asymmetries in the incidence of temperature shocks and changes to expected temperatures over the 36-year period of analysis. Section A.4 presents additional results decomposing these effects into their anthropogenic and natural contributions by applying methods from climate attribution science to output from state-of-the-art climate models, arriving at largely the same conclusions.

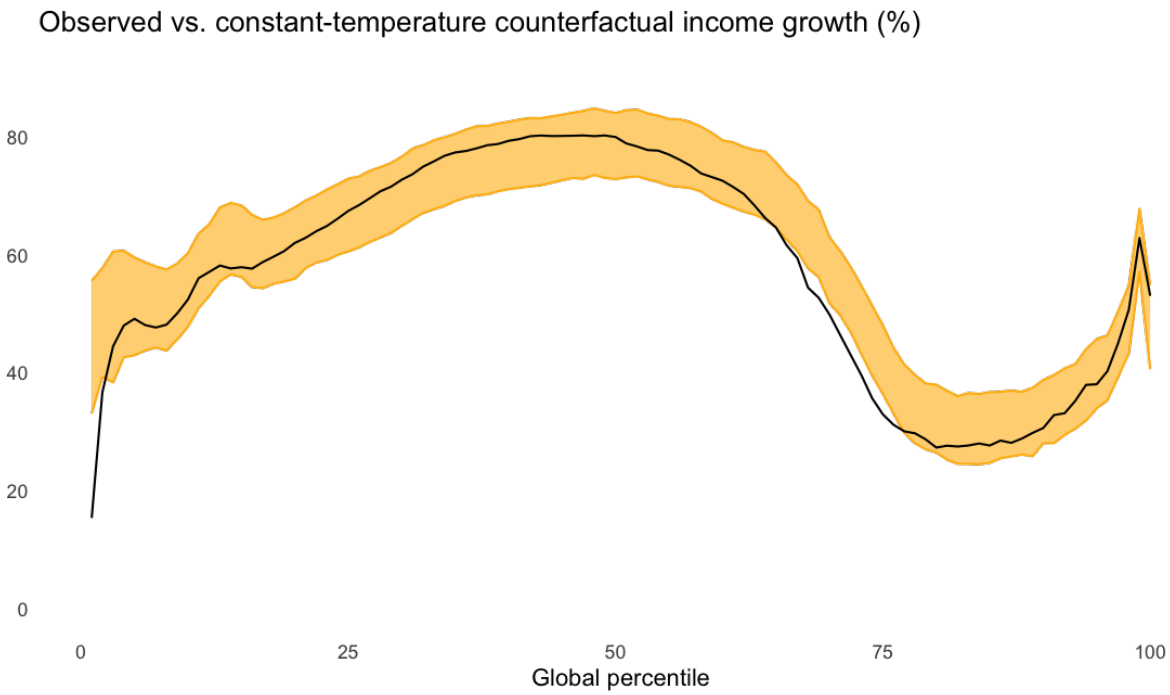
Here, the historically observed level of growth over this period is represented by a black filled line, reproducing a version of the "elephant curve" famously documented in Lakner and Milanovic (2016). The blue band depict the 90% confidence interval for the counterfactual removing the influence of temperature shocks on income. Regions where this band is above the observed incidence curve imply deprivations of income attributable to the incidence of temperature shocks over this period. Point estimates imply probable net harm for around 65% of the global sample.

Despite proportionally milder absolute warming in the poorest countries (see Figure A.6), this negative incidence still concentrates on the world's poorest 20%, representing 1.2 billion people in our 2016 sample and mostly located in warm developing countries. Importantly, as in Alvaredo et al. (2018), calculations underpinning this figure omit the bottom 10% of each country's income distribution because of the inclusion of unemployed adults and compositional instability effects which

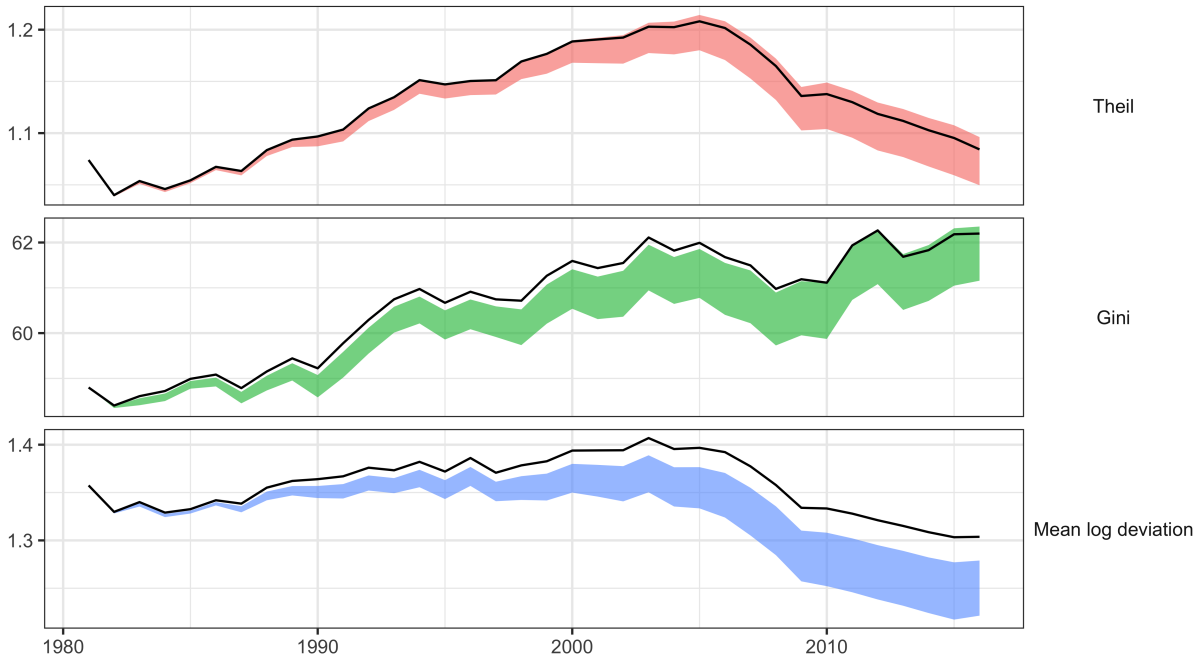
delta approach commonly used in studies which use damage functions specified in temperature levels. This entails linearizing the difference between country-level temperatures at the end of the 21st century and a baseline period ending in 2010. Under the most extreme warming scenario, the population-weighted mean temperature change over this period is 4.3°C with the largest differences in countries furthest from the equator. Nath et al. then multiply these projected long differences in *levels* of temperature by their estimated cumulative responses to an identified temperature *shock*, seemingly interpreting the result as the cumulative loss in GDP attributable to climate change over these 90 years. The motivation for this approach is unclear to me especially in combination with the hyperadaptability implied by the construction of the temperature shock.

Figure 1.7: Global inequality, observed vs. zero-warming counterfactual

(a) Global incidence of growth, 1981-2016



(b) Global inequality indices, 1981-2016



arise from assigning deciles of very large populations to percentiles representing smaller populations. For the poorest percentile in our data, we find that incomes absent these disproportionate temperature shocks would be 29% [18, 41] higher than they are today. As these effects exclude precisely the subpopulation we have already found to be the most economically vulnerable to temperature shocks, this growth incidence curve should be interpreted as a likely very conservative estimate of deprivations to the global poor.

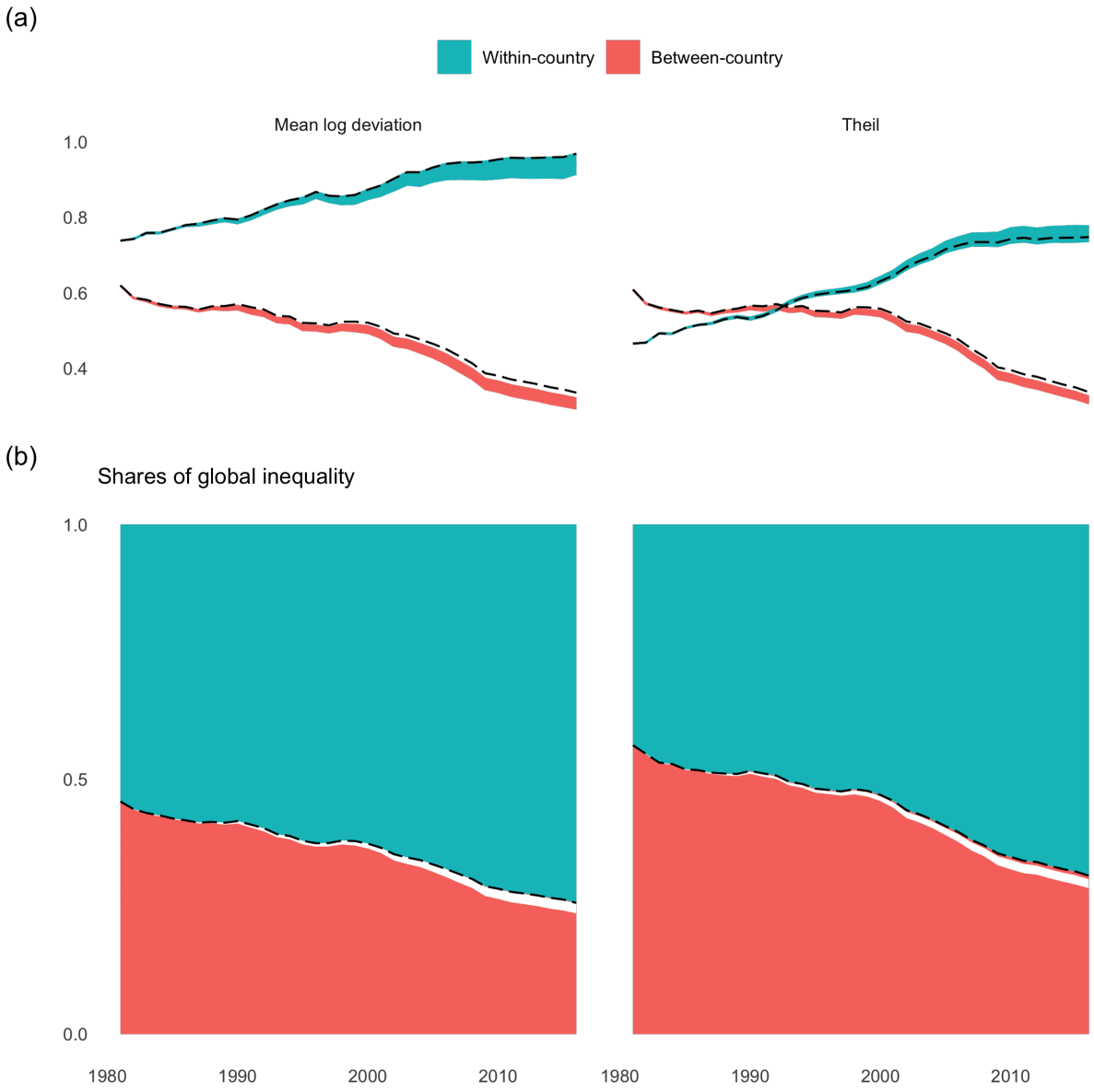
Elsewhere, estimates also imply likely harm from warming for the 52nd-97th percentiles. That the top 3% are relatively unaffected despite the negative effects for the top 1% in Section 1.3 is because these groups are dominated by extreme incomes in relatively cold countries. Effects on the global middle class are relatively tempered with slightly positive point estimates estimated for essentially the entire second quartile, also consistent with the within-country incidence curves which found the middle class experienced the mildest economic impacts across all climates.

Despite these different-signed effects, the counterfactual growth incidence band is noticeably flatter than the elephant curve, implying that historical climate change has been globally regressive in addition to being a likely drag on total growth. Figure 1.7b plots the time series for the global equivalents of the within-country inequality measures introduced earlier. In our within-country analysis, we identified opposing regressive transfers from the bottom decile and progressive transfers from the top decile which opposed one another to produce contradictory effects on within-country inequality depending on the measure of choice. In comparison, the historical effects of these shocks on global inequality are large and regressive across all measures, increasing the global Theil index by 0.9% $[-1.8, 2.8]$, the Gini index by 0.5 points $[-0.2, 1.0]$, and the mean logarithmic deviation index by 3.9% $[1.6, 6.4]$.

Figure 1.8 decomposes these effects on global inequality into their within-country and between-country dimensions. Panel (a) compares the observed (black dashed line) and counterfactual (filled colored bands) evolution of these entropy-based decomposable measures in levels. As measured by mean logarithmic deviation, which is relatively sensitive to the welfare of low-income quantiles, observed temperature shocks have unambiguously exacerbated both varieties of inequality decreasing within-country inequality by 2.6% [0.0, 5.6] and between-country inequality by 8.7% [4.9, 13.3]. For the Theil index, the relative sensitivity to negative impacts to top incomes observed in Section 1.3 re-appear, resulting in an estimated 1.3% [-1.3, 3.8] *improvement* in within-country inequality as a result of temperature shocks. These same shocks still worsen between-country inequality by 6.1% [3.6, 9.0].

Panel (b) adopts the same color scheme to visualize their relative contributions to global inequality. The dashed black lines trace the stark observed trend of within-country inequality (teal) increasingly dominating between-country inequality (red) since 1981. The region in white separating the two colors corresponds to the counterfactual shares had average temperatures remained fixed since 1981. That these white regions are entirely under the black dashed lines imply that warming since 1981 has very likely slowed this trend by reducing the within-country share of global inequality by between 1.1 percentage points $[0.1, 2.0]$ by the MLD measure and 1.6 percentage points $[0.7, 2.5]$ by the Theil index. Thus, the incidence of temperature shocks primarily influence the global distribution income by slowing convergence between countries.

Figure 1.8: Global inequality between and within countries, 1981-2016



1.5 Conclusion

The results presented in this paper constitute the first comprehensive evidence of regressivity in the global incidence of climate change, complementing an expansive and active economic literature which has remained effectively agnostic on global distributional impacts at subnational levels. By importing methods recently developed in empirical macroeconomics to distributional data thus far largely unused in environmental economics, we were able to overcome several long-standing statistical and practical constraints which have long obstructed progress on this urgent research question.

Given the sparsity of other work on this broad topic, several limitations of the present study suggeset avenues for potentially fruitful future research. Firstly, even the impressive coverage of the World Inequality Database is fundamentally limited in its ability to speak to the most important of distributional considerations, namely the impacts of climate change on global poverty. Our evidence affirms and quantifies the conventional understanding that the world's poorest have been and will continue to be by far the most severely impacted by climate change. But due to the limitations of pretax income as a measure of poverty and welfare, even these impacts are likely to be a conservative lower bound on true impacts.

In revisions in progress, the distributional analysis reported here will be supplemented by an analysis of compositional effects identifying which income groups from which countries have been transitioning up and down the global income distribution as a result of climate change. If income effects are as severe and state-dependent as described here, then it becomes plausible that an important distributional consequence of climate change is that inhospitable environments becomes increasingly deterministic of economic development, severely limiting the upward mobility of populations in hotter climates.

To the best of my knowledge, the vulnerability of the highest incomes in hot and generally poor countries to transitory environmental shocks has not been previously documented or theorized. Weather-induced capital depreciation, capital shares in agricultural income, and the relative unavailability of insurance instruments such as weather derivatives in developing countries are example plausible mechanisms that call for empirical testing.

From a policy perspective, the results presented in this paper suggest that incorporating equity considerations as through a concave utility function would indeed increase the social weighting of the most impacted by climate change. But it is may still be worth taking seriously criticisms that a "profound indifference to inequality" (Sen et al., 2020) remains embedded even in these progressive varieties of cost-benefit analyses predominant in the climate economic literature. For example, Prest et al. (2024) show that one proposed implementation would increase the US government's official estimate of the social cost of carbon by a factor of eight. But somewhat paradoxically, these greater cost estimates are primarily used to prescribe an increased price on carbon which may then disproportionately burden these same segments of the population whose increased welfare weight induced the inequality-exacerbating policy. Equity weighting should of course factor into both the estimation of costs and the design of policy internalizing them.

To this end, it is intuitive that democratic pressures and a richer portfolio of policy instruments should be more effectively mobilized domestically than coordinated across a Westphalian international system (Barrett, 2003; Keohane & Victor, 2011; Levin et al., 2009). If so, levers available domestically but not internationally—such as compensatory fiscal transfers—may yet emerge as viable instruments of climate policy.

Chapter 2: Temperature, institutions, and the political climate

> "A change in the weather is sufficient to recreate the world and ourselves."
>
> Marcel Proust, *In Search of Lost Time*

2.1 Introduction

In 2007, a panel of recently retired United States military officers published an assessment of the national security risks presented by climate change, the result of consultations with intelligence officers, climate scientists, and high-level figures in government and business. The report concluded that projected environmental pressures would lead to "widespread political instability and the likelihood of failed states" and "add to tensions even in stable regions of the world." (Sullivan et al., 2007) In a period defined by military-industrial expansion and the mainstreaming of climate advocacy, the report found influence by popularizing the term "threat multiplier" to describe how environmental outcomes associated with climate change may amplify social risks interpreted as falling within the interest of national security (S. Goodman & Baudu, 2023). By the following year, the government of the United Kingdom had revised its national security strategy, naming climate change "potentially the greatest challenge to global stability" (Her Majesty's Government, 2008).

This chapter investigates the degree to which environmental change may influence political preferences and facilitate institutional transformation. In a first step, we measure the impacts of identified temperature shocks on proxies for the demand for democracy. Using nationally representative survey data from over 300,000 respondents from up to 98 countries, we find evidence that exclusively for respondents in warmer climates, experiencing an additional positive temperature shock in the three years prior to being surveyed significantly diminished respondent attitudes towards democracy and increased support for features of government associated with authoritarianism. The same shocks were also associated with more inclusive conceptualizations of democracy in cooler countries with the opposite effect for warm countries.

We complement these findings with an analysis of aggregate effects on institutional change. To measure effects on the propensity for democratic "intensity", we use various data products which score country-year regimes along different aspects of democracy. We find relatively imprecise but suggestive evidence that a marginal temperature shock facilitates democratic backsliding in warm countries but that this effect does not persist beyond the year of the shock and is null for other geographies.

Finally, we look at the effects of these same shocks on the propensity for discrete regime change. Here, we report significant evidence that a transitory temperature shock greatly decreases the likelihood of warm-climate autocracies to transition to democracies while increasing the propensity

for warm-climate democracies to regress to autocracies. For temperate countries, the same shock greatly increases the probability of democratic transition but has null effects on the propensity for democratic reversal. These effects are also largely concentrated in the period of the shock but remain mildly elevated for an additional 1-4 years.

2.1.1 The demand for democracy

Summarizing the institutional risk posed by climate change, the aforementioned Military Advisory Board report concluded that the consequences of climate change "will likely foster political instability where societal demands exceed the capacity of government to cope." While such a direct relationship between the incidence of climate change and the demand for institutional change is undertheorized by my reading, two provisional hypotheses can be constructed from the broader literature on the economic determinants of democratic deepening, transition, and stability (as well as their negations).

Under a "mercury uprising" theory, adverse climate outcomes affect the latent demand for inclusive institutions and in turn the propensity for concessionary reform. This is motivated by invoking the influential economic model of Acemoglu and Robinson (2001) wherein diminutions in the perceived opportunity cost of challenging autocratic rule result in a more democratic political equilibrium; in other words, citizens have less to lose by taking costly political action to demand institutional reform of elites. For example, P. J. Burke and Leigh (2010) reports evidence that economic downturns worldwide are associated with democratic improvements while Acemoglu et al. (2019) highlight that democratic transitions are correlated with temporary reductions in GDP growth rates. Brückner and Ciccone (2011) provides evidence associating negative income shocks with a transitory increase in the propensity for select sub-Saharan autocracies to democratize. Aidt and Leon (2015) reports complementary evidence linking drought-induced rioting to improvements in democratic institutions.

Perhaps notably, the assessment of the Military Advisory Board concluded the opposite effect, suggesting that for particularly vulnerable tropical and subtropical countries, these demands will manifest in "movement toward increased authoritarianism." For the United States and Europe, it alludes to intergroup fracturing and resentments associated with the arrival of environmental refugees and migrants.

These potential consequences are more consistent with an alternative"state of exception" theory wherein citizens respond to threats associated with climate change with an increased willingness to trade features of liberal democratic institutions for security perceived as more credibly ensured by authoritarian and exclusionary forms of governance. The COVID-19 pandemic provides a particularly resonant intuition for this dynamic, wherein emergency restrictions on movement and privacy were generally deployed with wide approval and successful health outcomes. Alsan et al. (2023) reported that experimentally induced increases in perceived health insecurity at the outset of the pandemic led to significant increases in respondents' willingness to trade off civil liberties. Amat et al. (2020) found similar effects on the preference for technocratic and authoritarian government in the Spanish experience of the pandemic. Mittiga (2022) is a political theory paper which directly considers how these pandemic experiences may legitimize authoritarian approaches to responding to climate change in the future. Fritsche et al. (2012) finds that simply reminding respondents of local adverse consequences of climate change increased discriminatory attitudes towards out-groups such as prisoners and accentuated authoritarian preferences.

All in all, our results are highly consistent with this latter "state of exception" interpretation.

2.1.2 Other related literature

In the nearly two decades since the Military Advisory Board report operationalized climate change as a "threat multiplier", a prolific interdisciplinary social scientific literature has emerged dedicated to empirically examining climate change as a phenomenon of social disruption (T. A. Carleton & Hsiang, 2016; Intergovernmental Panel on Climate Change, 2023). The present analysis is closely related to an active subset of this work relating various environmental outcomes to the onset of conflict; Buhaug and von Uexkull (2021) and Koubi (2019) provide reviews of this research, construed broadly enough to include concepts of ranging from psychological aggression to self-harm to crime to warfare. Obradovich (2017) and a tertiary result in Dell et al. (2012) each find that higher temperatures in developing countries are associated with irregular changes in political leadership. Bellemare (2015) links food price increases to increased incidence in various forms of social unrest.

2.2 Data

2.2.1 Surveys of political preferences

Data comes from the Integrated Values Survey, which harmonizes the World Values Survey (Haerpfer et al., 2022) and the European Values Study (European Values Study, 2022). In total, the final dataset summarizes the responses of 663,965 total respondents to 506 nationally representative surveys conducted in 116 countries between 1981 and 2022. We retain data on the respondent's demographic information, including their birth country, immigration status, income range, and political preferences.

Table 2.1: Summary statistics for IVS respondent data

	Unique	Missing Pct.	Mean	SD	Min	Median	Max
Year	36	0	2005.2	10.3	1981.0	2007.0	2022.0
Male	2	1	0.47	0.53	0.0	0.0	1.0
Citizen	2	74	0.98	0.1	0.0	1.0	1.0
Support index	41	55	0.0	0.8	-3.4	0.1	0.8
Skepticism index	65	72	0.0	0.8	-1.7	0.0	2.0
Authoritarian index	65	30	0.0	0.7	-1.3	-0.1	1.9
Inclusiveness index	140902	69	0.0	0.5	-2.0	-0.1	1.2
Democracy	2	38	0.74	0.36	0.0	1.0	1.0
Polity score	19	41	6.3	5.3	-10.0	9.0	10.0

Our primary outcomes of interest are derived from responses to questions identified as indicative of attitudes towards democracy or its negation. We assign these questions to one of four categories as summarized in Table 2.2. For each category, we then construct an inverse-covariance weighted index of responses to the questions using the method of Anderson (2008), which assigns greater weight to more informative component questions.

2.2.2 Ordinal measures of democracy

We use two sources of democracy 'scores' which purport to measure how democratic a given country-year regime is. The first is a measure provided by the Polity Project (Marshall & Gurr, 2020) which ranges discretely between -10 and 10 in increments of 0.5 with more positive values corresponding to higher perceived levels of democracy. The Project recommends a classification where country-years scoring below -5 are labeled autocracies, those above +5 are democracies, and those in between are 'anocracies'.

The second source of democracy scores comes from the Varieties of Democracy (V-Dem) Project (Coppedge et al., 2025). These measures are produced b combining primary sourcing with expert elicitation to assign a panel of country-year observations a score ranging continuously between 0 and 1. There are up to 400 variables by which observations are scored and these are in turn aggregated to produce five indices which operationalize democracy according to its deliberative, egalitarian, electoral, liberal, and participatory connotations respectively. Differences between these concepts of democracy can be pertinent if, for example, the extension of civil liberties is a systematically slower-moving process or is considered a much greater democratic concession than the administration of a nominally competitive election. A contemporary example is Chile's ongoing constitutional reform process which began in 2021 in response to civil uprising in 2019. That the drawn-out process has seen the rejection of multiple proposed constitutions by referendum may be interpreted as an exercise of improved democratic deliberation and participation. But until a replacement is ratified, democracy as measured by formalized rights and protections are still constrained by the 1980 Constitution being challenged.

The distribution of scores are summarized in Section B.1, showing a bimodal distribution with peaks at -6 and +10 for the Polity IV scores and heavily right-skewed distributions for each of the V-Dem scores. Correlations between these indexes are summarized in Table B.1. The wide use of these ambitious intensive measurements of democracy seems likely to reflect their usefulness more so than it does their credibility, especially for capturing intra-regime variation; **data-democracy-critique**; **data-democracy-critique2<empty citation>** provide thoughtful critiques of these data products along these lines.

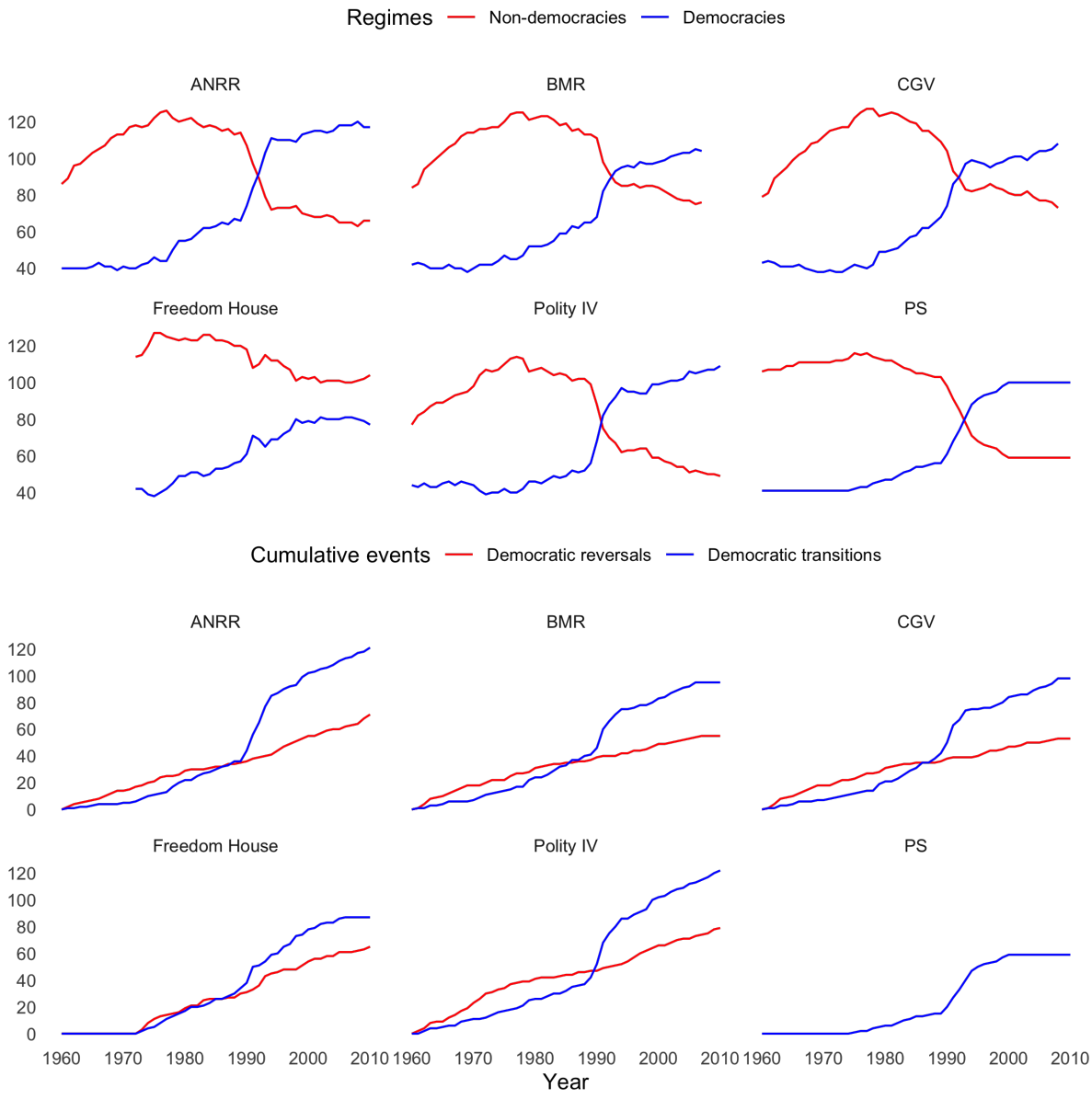
2.2.3 Dichotomous indicators of democracy

To study systematic transitions in and out of democracy, we require data which crudely labels country-year regimes a label of 'free' or 'democratic' or their negation. Table B.2 summarizes differences between notable products which serve this purpose and have been commonly used in the political economy literature. For each data product, Section 2.2.3 plots the time series of the number of democracies and non-democracies worldwide as well as the cumulative number of transition events. In our main results, we will restrict attention to the aforementioned Polity index (made dichotomous by using a score of +5 as a threshold for democratic status) as well as the panel datasets from Acemoglu et al. (2019) and Cheibub et al. (2010).

2.2.4 Weather

We merge all data with population-weighted near-surface air temperature data derived from the Global Meteorological Forcing Dataset for Land Surface Modeling (GMFD). This data corrects

Figure 2.1: Time series of regimes and regime changes by data source



model-based biases in NCEP/NCAR reanalysis product using observational meteorological station data (Department of Civil and Environmental Engineering/Princeton University, 2006). The GMFD reports historical weather data for the entire global surface at 0.25-degree resolution[1] for every three hours from the start of January 1, 1948 to the end of December 31, 2015. For interpretive convenience, we center population-weighted average temperatures $\overline{T}$ at 18°C, roughly in the 40th percentile of country-year temperatures in our dataset.

As in Chapter 1, we define temperature shocks $\widehat{\tau}_{it}$ in relation to an M-period moving average of local temperature $\overline{T}_{it}$:

$$\overline{T}_{it} := \frac{1}{M} \sum_{m=1}^{M} T_{i,t-m} - 18 \tag{2.1a}$$

$$\widehat{\tau}_{it} := T_{it} - \overline{T}_{it} \tag{2.1b}$$

To avoid redundancy, I refer the reader to Section 1.2 in Chapter 1 for a graphical depiction of the resulting history of temperature shocks constructed in this manner (Figure 1.1) as well as additional elaboration on this construction.

We next define a variable τ_{ct}^{N} representing the recent exposure to temperature shocks of a respondent in country c surveyed in year t:

$$\tau_{ct}^{N} := \sum_{j=1}^{N} \mathbb{1}(\widehat{\tau}_{c,t-j} > \underline{\tau}) \tag{2.2}$$

This shock will be used when estimating impacts on the demand for democracy as measured by cross-sectional surveys from the IVS. Because we cannot infer the month of the year when respondents were surveyed, shocks defined using temperature observed only in the calendar year the respondent was surveyed would be strongly influenced by weather only respondents surveyed later in the year would have experienced. Using exposure over a longer period before the calendar year helps mitigate the effect of this non-equivalence and also accounts for persistence in temperature shocks such as that caused by natural periodicity in the climate system (e.g., the El Niño Southern Oscillation which spans between 2-7 years). In our preferred construction, we set $N = 3$ to span the persistence in temperature shocks documented in Chapter 1.

This variable also depends on the value $\underline{\tau}$, the threshold value defining a temperature shock. As a benchmark, we use a common value of 1°C (corresponding to the 75th percentile in our full estimation sample) for all settings though there is also a reasonable case to use relative thresholds, for example based on country-specific variability in temperature.

[1]At this resolution, cells near the equator are approximately 27 km^2 (17 mi^2).

Table 2.2: Categorization of relevant survey questions from the Integrated Values Survey

1. Support for democratic institutions
• Agree: Democracies are good for governance
• Agree: Democracy has its problems but is better than all alternatives
2. Democratic skepticism/ineffectiveness
• Agree: Democracies cause disorder
• Agree: Democracies are too bureaucratic
• Agree: Democracies are bad for the economy
3. Support for authoritarian leadership
• Preference for a strong leader who does not bother with parliament and elections
• Preference for army rule
• Preference for governance by experts over elected officials
4. Inclusiveness of definition of democracy
• Agree: Holds open elections
• Agree: Extends civil rights
• Agree: Promotes equality of women
• Disagree: Army takes over when government is incompetent
• Agree: Redistributes from the rich to the poor
• Agree: Provides unemployment insurance
• Disagree: Government by religious leaders

2.3 Methods

2.3.1 The demand for democracy

We modify the specification of Acemoglu et al. (2024), which uses the same IVS dataset to estimate the effect of exposure to democracy on attitudes towards democracy.

$$Y_{i,w,c,t,a} = \beta_1 \tau^N_{ct} + \beta_2 (\tau^N_{ct} \times \overline{T}_{ct}) + \boldsymbol{\gamma}^\top \mathbf{W}_{i,w,c,t,a} + \varepsilon_{i,w,c,t,a} \tag{2.3}$$

Indexes refer to respondent i, a resident of country c who in year t was surveyed as part of IVS wave w at age a. The matrix $\mathbf{W}_{i,w,c,t,a}$ of controls include the expected temperature $\overline{T}_{ct}$ in country c at year t, respondent gender, respondent town size, and fixed effects for country, year of survey, age of respondent, respondent's birth cohort , and IVS wave. Standard errors are clustered at the country and year levels.

2.3.2 Democratic deepening and backsliding

To estimate dynamic effects of temperature shocks on an outcome y measuring the degree of democracy, we estimate a series of state-dependent local projection regressions on a panel of country-year regime observations for projection horizons $h \in \{0, ...H\}$:

$$y_{c,t+h} = \beta_{1h} \widehat{\tau}_{ct} + \beta_{2h} (\widehat{\tau}_{ct} \times \overline{T}_{ct}) + \lambda_h \overline{T}_{ct} + \boldsymbol{\gamma}_h^\top \mathbf{Z}_{ct} + \mu_c + \eta_t + u_{c,t+h} \tag{2.4}$$

The primary quantity of interest for each projection horizon is $\beta_{1h} + \beta_{2h} \overline{T}_{it}$, the first derivative of the equation with respect to the temperature shock $\widehat{\tau}$. The first term represents the expected effect of a unit temperature shock on the outcome of interest for a country-year with an expected temperature of 18°C. The second term represents the linear dependence of this effect for country-years with different expected temperatuers. For example, the model would imply that a unit shock in a country with an expected annual temperature of 26°C would be expected to increase outcome y for the country by $\widehat{\beta_{1h}} + \widehat{\beta_{2h}} \cdot (26 - 18)$ units h periods after the shock relative to the counterfactual where no shock occurred.

Controls $\mathbf{Z}_{ct}$ include lags $\ell \in \{1, ..., L\}$ lags each of the shock $\widehat{\tau}_{c,t-\ell}$, the shock $\widehat{\tau}_{c,t-\ell}$ interacted with expected temperature $\overline{T}_{c,t}$, logged precipitation, GDP growth, and the outcome variable $y_{c,t-\ell}$.

The reader may refer back to Section 1.3 and Section A.2 for exposition of this method and interpretation of its results.

2.3.3 Discrete regime change

To measure impacts on the propensity for autocracies to transition to democracies and vice versa, we estimate Equation (2.4) using a binary dependent variable indicating the country-year's regime type as the outcome variable. The corresponding coefficient estimates are interpreted as the state-dependent effect of an exogenous 1°C shock on the absolute propensity to transition to a democracy (or regress to an autocracy) h years after the shock. For interpretation as a proportional hazard, the magnitude of this effect can be compared to the unconditional annual propensity to undergo a regime change.

2.4 Results

2.4.1 The demand for democracy

Table 2.3 reports the main results of our demand analysis. To minimize susceptibility to inferential error from multiple testing, we restrict attention here to results for the four survey category indices but provide the equivalent results for their component survey questions in Tables B.3 to B.6. To aid interpretation of our non-linear state-dependent model, Figure 2.2 depicts these coefficients graphically as a function of the expected temperature in the respondent's country of residence in the year of their survey.

Our favored specification, depicted as a green filled line with a corresponding 90% confidence interval, uses an explanatory variable defined as the number of temperature shocks above 1°C the respondent was exposed to in the three years preceding the year of the interview. Additional dashed lines correspond to alternative constructions which vary the number of years considered.

The coefficients on the main effect imply that recent exposure to an additional positive temperature shock for respondents in a represenentative 18°C decreases support for democracy and leads respondents to favor more restrictive definitions of democracy while increasing skepticism of democracy and preferences for authoritarian features of government. These effects are large in magnitude, ranging between 0.12-0.13 standard deviations for the attitudinal indexes and 0.46 standard deviations for the inclusiveness index.

The state-dependency in this effect is significant at the 5% level for all outcomes except the democracy-support index. Among cold countries, effects are positive but nell with the exception of the large and highly significant impacts to the inclusiveness index. That these impacts change sign for all outcomes at some intermediate temperature well within the range of human habitation is consistent with the convention that marginal temperature shocks are increasingly harmful for countries above a economically optimal temperature of approximately 13°C and increasingly beneficial for increasingly cool countries (M. Burke et al., 2015a; Nath et al., 2024). That the impacts are larger in magnitude and much more precisely estimated for the inclusiveness index despite being estimated for by far the smallest sample is most likely a reflection of the diversity and informativeness of questions from which it is constructed compared to the two highly correlated variables which comprise the democratic support index (see Table 2.2).

These estimates are generally consistent as we vary the period of exposure defining the temperature shock with a general pattern that increasingly restricting attention to environmental exposure in the most recent years tends to increase the magnitude of the impact. While the repeated cross-sectional design and other limitations of the survey data limits our ability to capture dynamic impacts, this provides some suggestive evidence that these demand effects are concentrated either contemporaneous to the period of the shock or with a one-period lag and unlikely to be highly persistent.

All in all, these results are strongly consistent with a 'state of exception' demand story.

2.4.2 Institutional change

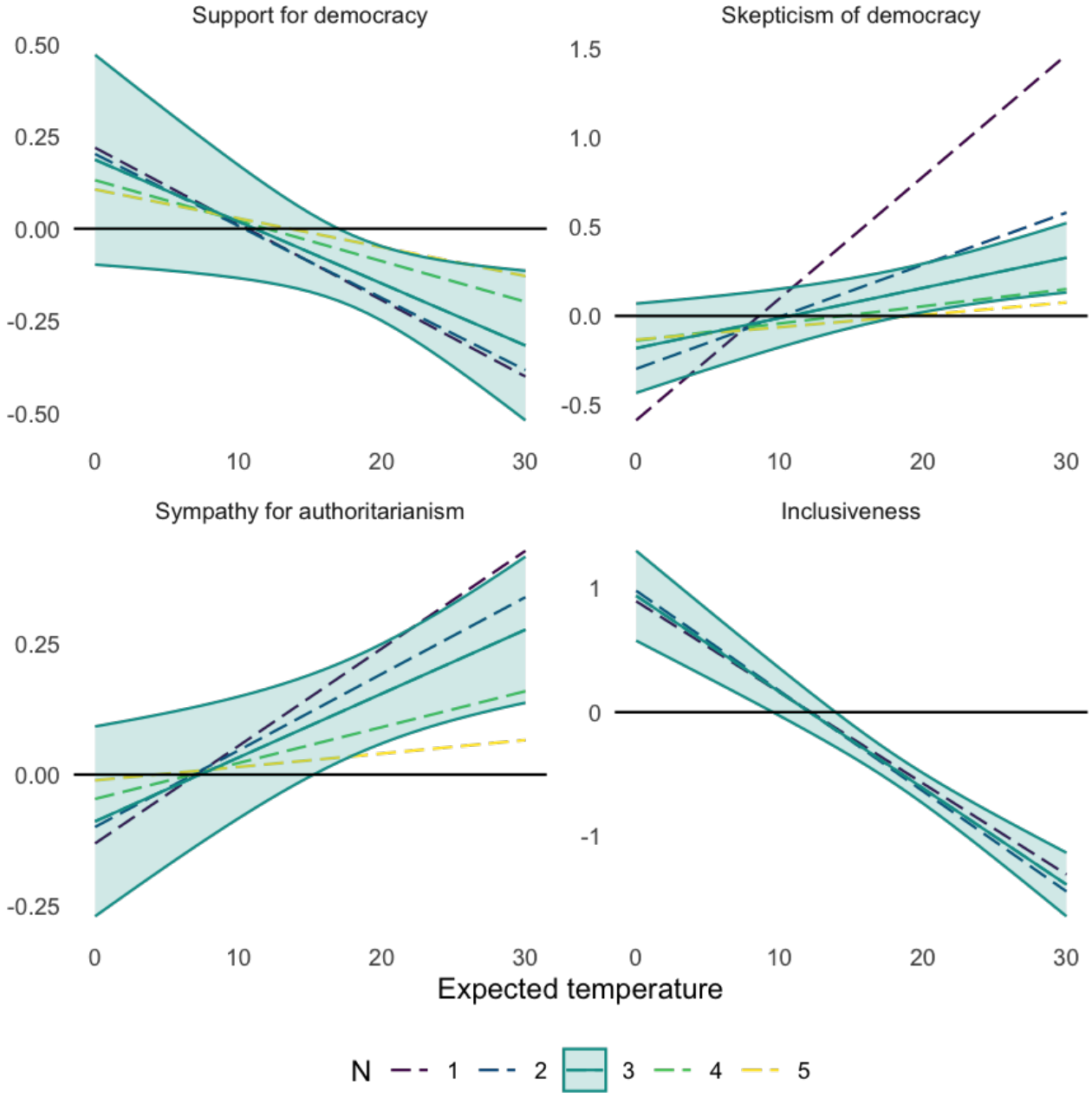
While our prior results provided strong evidence of transitory temperature effects on the latent demand for democracy, they are not directly informative of the mobilization of these new preferences to effect institutional change. To this end, Figures 2.3 and 2.4 depict the dynamic

Table 2.3: Impact of an additional 1°C shock in prior three years on attitudes towards democracy

Index:	Support	Skepticism	Authoritarian	Inclusiveness
Model:	(1)	(2)	(3)	(4)
Variables				
1°C Shock	-0.1152$^{\cdot}$	0.1235	0.1303*	-0.4582**
	(0.0585)	(0.0815)	(0.0568)	(0.0704)
1°C Shock $\times \overline{T}$	-0.0168$^{\cdot}$	0.0169*	0.0122*	-0.0775**
	(0.0090)	(0.0072)	(0.0052)	(0.0116)
$\overline{T}$	-0.6651$^{\cdot}$	-0.2732	0.0832	-0.3258*
	(0.3382)	(0.3732)	(0.1460)	(0.0970)
Gender	Yes	Yes	Yes	Yes
Precipitation	Yes	Yes	Yes	Yes
Fixed effects, # unique values				
Country	75	76	98	58
Year	11	13	21	6
Age	84	87	87	83
Birth year	92	93	101	86
Survey wave	4	4	8	3
Observations	150,292	170,798	317,655	77,392
Outcome mean	0.02250	-0.00387	-0.03017	-0.05107
R^2	0.08924	0.10248	0.18527	0.17156

Clustered (Country & Year) standard-errors in parentheses
*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, .: 0.1*

Figure 2.2: Effect on political preferences of an additional 1°C shock in last *N* years



state-dependent cumulative responses of institutional democracy as measured along intensive and extensive margins respectively.

Figure 2.3 plots the marginal effect of an identified temperature shock on democracy scores for regimes in three representative climates, net of the shock's effect on ensuing temperature shocks. Figure 2.3a depicts impacts measured by the widely used Polity IV score while Figure 2.3b report results for scores from the Varieties of Democracy Project which grade country-year regimes according to distinct aspects of democracy.

The results corresponding to the Polity IV score indicate a significant state dependency contemporaneous with an identified temperature but this effect is imprecisely estimate and only large for warm climates. Point estimates ascribe a transitory 0.18 reduction in the polity score to a unit shock in a representative 26°C country. While not exactly rich in intuitive meaning, this is approximately 1.8% of the difference between a borderline democracy and a borderline autocracy going by the group's suggested classification. The results for the V-Dem scores are qualitatively similar but report null effects across the board.

Finally, Figure 2.4 plots estimates of the impact of a unit shock on the propensity for democratic transition and breakdown, split across rows by data source.

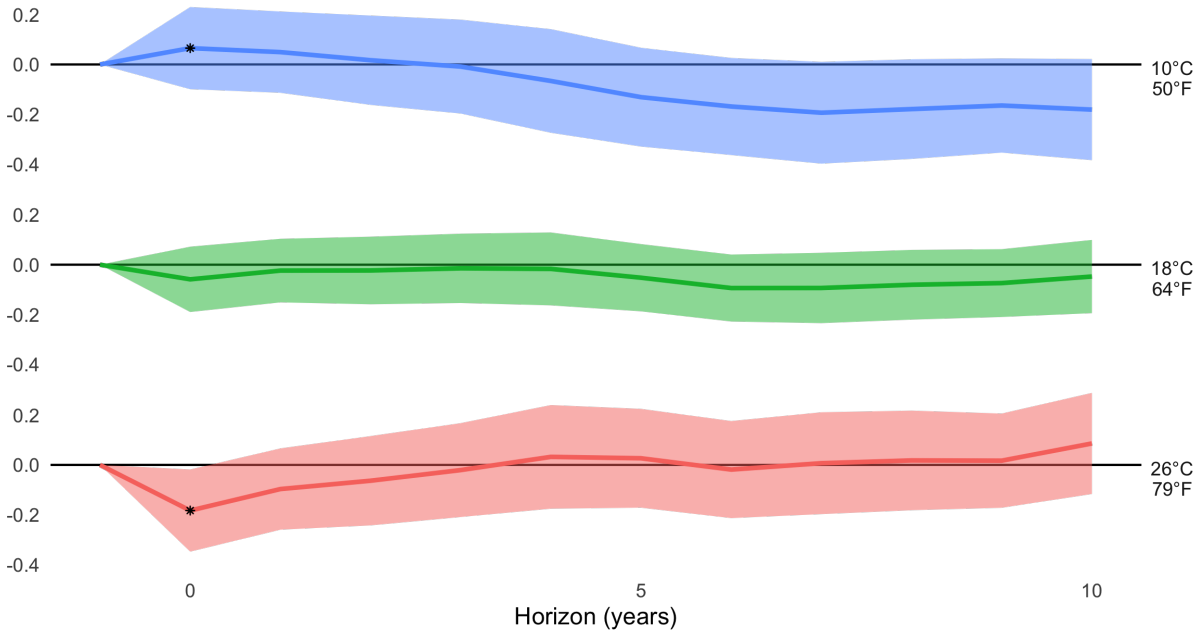
Figure 2.4a indicates that a transitory temperature shock substantially and significantly increases the propensity of temperate autocracies to democratize by between 0.030 and 0.075 percentage depending on the data source. Compared to an unconditional transition probability of approximately 0.03[2], these represent proportional increases to the probability of a 10°C autocracy to democratically transition of 96% by the ANRR measure, 108% by the CGV measure, and 252% by the Polity IV measure. Referring to Table B.2, the relative responsiveness of the Polity IV score may be explained by its relatively limited focus on "competitiveness of political participation and constraints on executive authority". Indeed, the ANRR definition of democracy is strictly more selective since the Polity IV indicator is a necessary but insufficient qualifying condition.

For 26°C autocracies, the propensity to democratize instead decreases as a result of a unit temperature shock by between 0.032-0.048 percentage points corresponding to proportional decreases between 134-236%. Now, this is clearly nonsensical since probabilities cannot take on negative values; this is partially a well-known consequence of linear probability models but more importantly, is worth emphasizing that these impacts are expressed in relation to a 1°C shock, which is much larger in magnitude than shocks we would expect to observe in a typical year in any country. Referring back to the time series of shocks illustrated in Figure 1.1, the median magnitude of a shock varies by the rate of climate change which will in turn vary across geographies and time. For the warmest 33% of country-years in our data, the median shock was approximately 0.19°C in absolute value in the 1970s and then increased to between 0.30-0.37° in the ensuing decades. Thus, effects of a 1°C shock in settings above 25°C can be understood as responses to a shock 3-5 times larger than is typically observed. For the coldest 33% of country-years, the median magnitude of a shock was approximately 0.35°C in the 1970s and 1980s, 0.51-0.53°C in the 1990s and 2000s, and 0.59°C in the 2010s. Their effect sizes then can be interpreted as the response to a shock just 2-3 times larger in magnitude than the median. By the assumption of linearity inherent in the local projections model specified in Equation (2.4), we can scale down these estimated effects accordingly

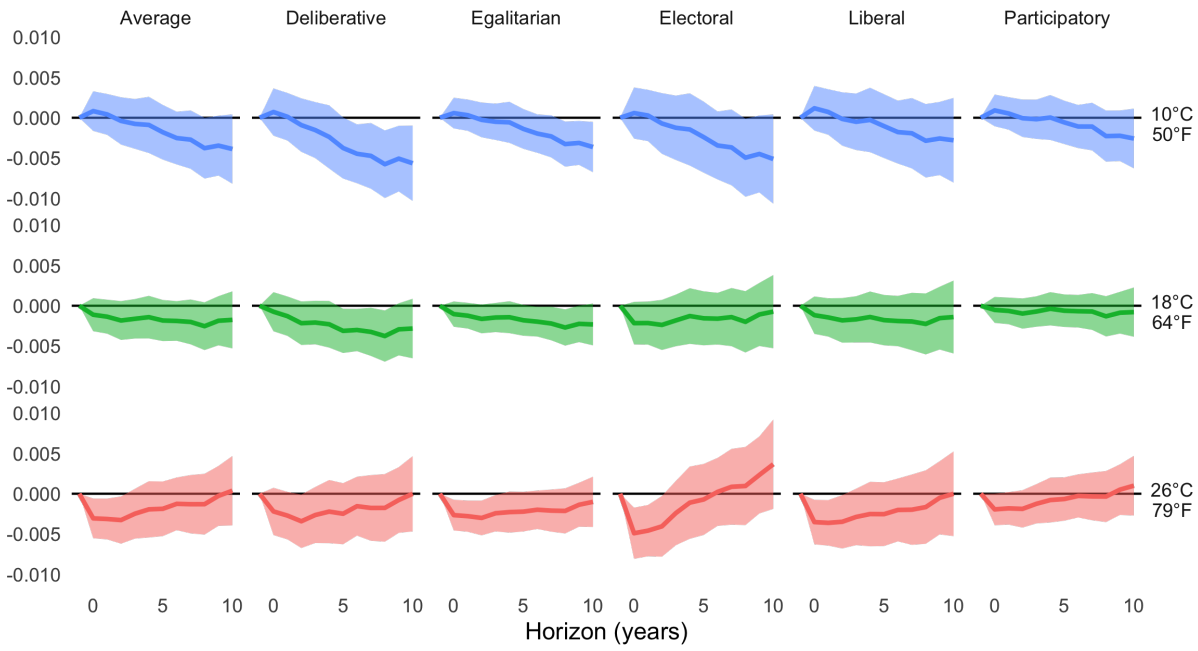
[2]The unconditional probability to democratize in any given period is surprisingly consistent across climates: 0.027 for autocracies cooler than 17°C compared to 0.024 for those warmer than 25°C by the ANRR measure, 0.024 vs. 0.019 by the CGV measure, and 0.030 vs. 0.032 by the Polity IV measure.

Figure 2.3: Intensive margin: Effect of a 1°C shock on democracy scores

(a) Polity IV democracy score (-10 to +10)



(b) V-Dem Scores (0 to 1)

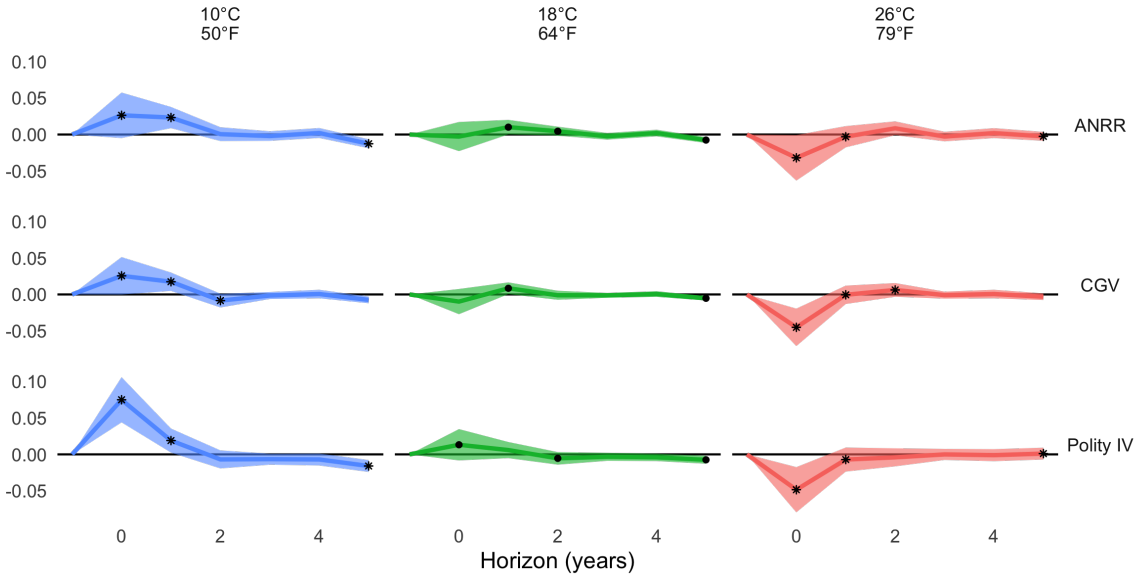


to infer the impact of a typical shock on the probability of democratic transition.

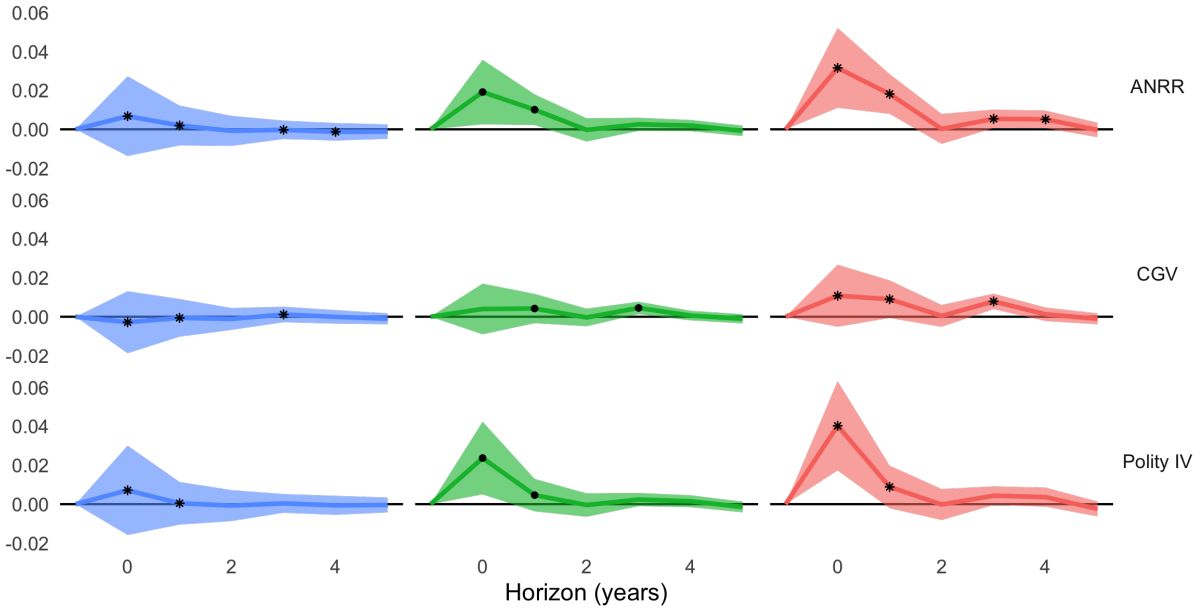
Figure 2.4b reports analogous results for the propensity of democracies to break down. These too exhibit large state dependencies though impacts to cool democracies are found to be null, suggesting they tend to be stable regimes. Impacts to hot countries in comparison are substantial, significant, and persistent, ascribing an increase of 0.032, 0.011 and 0.040 percentage points of democratic reversal in the year of the shock compared to unconditional transition probabilities of between 0.033-0.048. These correspond to proportional increases of 33.0%, 91.8%, and 91.0% for the ANRR, CGV, and Polity IV measures respectively.

Figure 2.4: Extensive margin: Effect of a 1°C shock on political transitions

(a) Change in propensity for democratic transition among autocracies



(b) Change in propensity for democratic reversal among democracies



2.5 Conclusion

In all, the three sets of complementary results presented throughout Section 2.4 provided internally consistent evidence supporting a "states of exception" theory in which adverse climate impacts both diminish latent public demand for democracy and the democratic character of its political institutions. This is contrasted against a competing "mercury uprising" theory with precedent outside climate contexts in which adversely impacted citizens recognize they have less to lose in contesting government suppression and act accordingly to manifest democratic change; to the extent that positive temperature shocks correspond to diminished opportunity costs of such political activity, the theory would imply exactly opposite institutional pressures to the ones recovered at each stage of our analysis.

Still, the present analysis is unsatisfying as a theory of environmental political economy absent consideration of the other side of the "market": the supply of democracy. Aidt and Franck (2015), Rahman et al. (2022), and Wood and Wright (2016) consider cases in which political leaders anticipate opportunities for political contestation or to consolidate emergency powers, acting to either pre-emptively suppress dissident activity or placate potential challengers. Indeed, our partial equilibrium framework is largely agnostic about the process by which the increased latent demand for democracy we attribute to shocks translates into macro-scale political transformations. Complementary analysis looking at the outcomes such as the incidence of social unrest, electoral participation, and state violence would enrich the work documented thus far.

Further, understanding inequality as both a consequence of climate change (Chapter 1) and a moderator of regime turnover (**democracy-inequality**) raises the question of how inequality in economic outcomes may relate to these inequalities in political representation. That climate change is expected to segment winners from losers within the same populace surely complicates equilibrium outcomes in contexts where the character of political institutions become malleable.

Chapter 3: Environmental disruption and the transmission of gendered norms

with Tanushree Goyal

> "Ammu said that human beings were creatures of habit and it was amazing the kind of things they could get used to."
>
> Arundhati Roy, *The God of Small Things*

3.1 Introduction

An active area of research intersecting the economics of gender and development asks how substantial differences in gendered outcomes may exist among countries at the same level of development. Though plenty of evidence corroborates the intuition that economic growth and women's empowerment positively impact one another, Duflo (2012) in a survey of this literature concludes that "the interrelationships are probably too weak to be self-sustaining."

Perhaps nowhere is this more stark than in South Asia where cultural conservativism regarding gender roles has displayed remarkable persistence despite rapid economic gains over recent decades. Indeed, compared to other countries with comparable levels of economic development, South Asian countries lag far behind in terms of female labor force participation, social integration, and health outcomes. Prior work has shown this disconnect between economic and gendered progress is reproduced within countries: the correlation between per capita income and many of these measures are weak and inconsistent in sign across Indian states (Jayachandran, 2021). And while policy intervention can overcome non-economic barriers to gender equality, cultural norms have proven remarkably resilient (Alesina et al., 2013; Ashraf et al., 2020).

In confronting this puzzle, we draw particular inspiration from a nascent political economy literature which uses empirical methods from modern quantitative social science to test long-standing theories of cultural transmission in evolutionary anthropology. Relevant work in this area, for example, has documented that matches in the marriage market can be either hastened or delayed by economic circumstance depending on the direction of marriage payments imposed by prevailing cultural norms (Corno et al., 2020). Operating through a consumption smoothing mechanism that imposes differences in marginal propensities to consume, income shocks are found to incentivize the recipients of marriage payments—the families of grooms in societies which practice dowry or those of brides where bride payments are the custom—to marry earlier. The gendered welfare implications are most salient when considering the resulting effect on the risk of exploitative child marriage. Interestingly, this implies that positive income shocks actually increase the incidence of

these arrangements in dowry societies which may contribute to the region's gender regressiveness amidst economic progress.

This chapters seeks to consider these issues within an emerging "cultural persistence" literature which investigates mechanisms explaining why some societies have historically exhibited a remarkable ability to transmit cultural traits, while others have undergone rapid cultural transformations. Giuliano and Nunn (2020) investigate one potential determinant: climatic instability experienced over ancestral generations spanning the years 500-1900. At a high level, the model's theory follow a Darwinian logic: ancestral lines that have been susceptible to highly variable climates would have found it optimal to adapt their cultural practices accordingly. The authors argue that the modern descendants of highly climatically unstable ancestral lines will have inherited this cultural adaptability.

We adapt this climatic-instability hypothesis, but propose instead that first-hand climatic instability should be sufficient to induce measurable within-lifetime cultural change. In particular, our theory suggests that random weather-induced shocks can impact intrahousehold bargaining power sufficiently that gendered outcomes can measurably change. This framing borrows from an established literature viewing family structures and spousal relations as domains for continual negotiation. When unexpected and severe droughts induce negative income shocks, the resultant shock to intra-household bargaining power should tend to manifest in gendered effects. To the extent that these environmental factors can affect the intergenerational persistence of gender norms, I refer to this proposed phenomenon as the environmental disruption of gender norms.

Gender norms being a broad term, I focus on a specific setting and consider a narrow subset of measurable "norms." A focus on rural South Asia allows us to hone in on a region where gender inequities are pronounced, where credit constraints are well known to exacerbate the effects of economic shocks and where environmental instability is ongoing and particularly fast-moving. India has been a principal site of investigation for such studies as it is notably marked my gendered cultural practices such as dowry payment and virilocality, for example.

Most importantly for my purposes, there is rich readily accessible secondary data collected from the region. Central to this paper is a unique module in the otherwise standard Indian implementation of the frequently used Demographic and Health Surveys program which surveys a representative subsample of female respondents about their attitudes and experiences with domestic violence. Domestic violence, as well as other exercises and measures in intrahousehold power, constitute the gender norms of study in this paper. In particular, I take care to consider different subclasses of gender norms: those related to respondent beliefs about the role of women, those related to specific actions and experiences with these gender roles, and gender norms they have adopted in their own household. I will argue that these subclasses of gender norms are differently vulnerable to disruption with beliefs likely to be relatively immutable while actions and the distribution of power within a household more likely to be responsive to external factors.

My empirical approach is three-pronged. As a motivating first step, I present correlational evidence illustrating the strong cultural persistence of gender norms. I find, unsurprisingly, that children of parents with a history of domestic violence are much more likely to themselves hold views that justify domestic violence and are also more likely to experience domestic violence themselves either as a perpetrator or victim. More interestingly, the persistence of these norms is higher for men than for women.

The second leg of my empirical approach seeks to identify the extent to which these same gender norms are susceptible to environmental disruption. Using plausibly exogenous precipitation shocks,

I find that random exposure to an additional drought between the ages of 11 and 15 results in a 0.60-month increase in a female respondent's age at first marriage, while the effects for men are null. Random exposure to a drought in the early years of a marriage also reduce reported controlling behavior of a respondent by 2.4 percentage points or 5%, but largely did not seem to affect the incidence of domestic violence over the course of the entire marriage. The implication is that for gender norms related to the parameters of control and bargaining power within a household, timing matters: norms are particularly malleable in the formative years of a marriage.

This issue of timing motivates the final set of regressions I use, which employ survival models to first estimate drought effects on the hazard into instances of child marriage—replicating the conclusion of a recent study by Corno et al. (2020) that droughts reduce these hazards—and then the drought effects on the hazard into a domestically violent marriage, which to my understanding is unprecedented in the economics literature since it is only made possible by a unique survey module. The central finding of the model is that droughts that occur in the early years of a marriage reduce the hazard of becoming domestically violent by between 1.8 and 2.0 percentage points in the year of the drought's occurrence. This amounts to a 48-53% reduction in the overall hazard of spousal violence. This lends credence to the notion that to the extent that domestic violence may be considered a repugnant exercise in intrahousehold bargaining power, the overall average effect of this reduction in bargaining power is a reduction or delay in physical violence rather than a "backlash" effect that is highlighted in work such as Sekhri and Storeygard (2014).

3.1.1 Gender norms: theories of transmission and change

There is a well established literature credibly documenting specific moments of disruption of gender norms. Boserup (1970) hypothesized that the role of women in society originated in different types of agricultural technology, which became the basis for widely cited work by Alesina et al. (2013). In fact, using the emergence of a specific technology as a source of gendered disruption has become its own literature with examples in Carranza (2014), Goldin and Katz (2002), and Greenwood et al. (2005).

Larcom et al. (2017) investigate the effects of strike-induced closures of a subset of stations of the London Underground. These closures forced Londoners to find new commuting routes, effectively forcing them to deviate from a habit and incur search costs to find alternative routes. That some of these newfound routes persisted past the duration of the strike is given a revealed-preference interpretation: the shock-induced costly search led to discovery of a superior alternative that had always been available.

Returning to gender norm applications, this evokes empirical work such as Luke and Munshi (2011). Studying female laborers at an Indian tea plantation, the authors find that "a relative increase in income weakens the family's ties to the ancestral community and the traditional economy" with effects entirely concentrated in laborers from historically disadvantaged castes. Their abstract concludes that "historically disadvantaged groups may, in fact, be especially responsive to new opportunities precisely because they have fewer ties to the traditional economy to hold them back." The suggestion is that the marginalized may be most responsive to external shocks with implications for the persistence of cultural norms. On the other hand, Dhar et al. (2022) finds that scheduled-caste children are more likely to adopt the attitudes of their parents than upper-caste children.

Two studies suggest that early-life exposure to differing family structures lead to measurable

changes in gender norms. Fernández et al. (2004) finds that war-induced shocks to female labor-force participation, by normalizing working mothers, led male children of such working mothers to seek out working partners in their adulthood. Unterhofer and Wrohlich (2004) find that the takeup of paternity leave induce positive effects on attitudes towards gender norms both in the offspring of the fathers and to the untreated network of these fathers.

3.1.2 The household as a forum for negotiation and bargaining

Luke and Munshi (2011), already mentioned above, is also a good reference for discussion of the household as a platform in which parties negotiate and bargain. In their framing, a random postitive shock to female income can induce breaks from tradition by decreasing dependence on the social safety net of ancestral communities. This effect is most pronounced on for those who come from disadvantaged ancestral communities as the safety net is less developed and thus is less costly to break from. The increase in female income also induced an increase in female bargaining power within the household as measured in influence over intrahousehold decision-making. Like Sekhri and Storeygard (2014), this study also warns that the empowered women's increased ability to act on their own true preferences presents a risk for "backlash" in the form of increased marital violence. This risk is not corroborated in my findings, at least as captured by average treatment effects.

3.1.3 Gendered impacts of weather shocks

Income mechanisms

In rural and credit-constrained settings where livelihoods depend more directly on weather exposure, drought or flooding events are commonly used as proxies for income. In the intersection with gender economics, such shocks are modeled as binding expenditure constraints such that in a gender-biased household, a reduction in available resources will lead parents to prioritize investments in male children to the detriment of female children. Empirical work has reported measurable impacts of weather shocks on female adult outcomes (Maccini & Yang, 2009), female educational attainment (Björkman-Nyqvist, 2013), relative mortality rates of female children (Rose, 1999), and the likelihoods of child marriage and early fertility (Corno et al., 2020).

Behavioral mechanisms

A second category of weather shocks looks at a more psychological mechanism. A burgeoning literature has found particularly hot days make people more stressed with measurable impacts on outcomes ranging from performance on exams (Park et al., 2020), economic productivity (M. Burke et al., 2015b), and aggression (Baylis et al., 2018), which naturally invite gendered impacts. Heightened aggression from heat exposure gives rise to an extensive literature on temperature-induced violence even aggregating to the level of the geopolitical (**violence**). In India, the frequency of farmer suicides (T. A. Carleton, 2017), dowry deaths, and reported domestic violence (Sekhri & Storeygard, 2014) have all been linked to extremes of weather.

3.2 Data

3.2.1 Demographic and Health Surveys

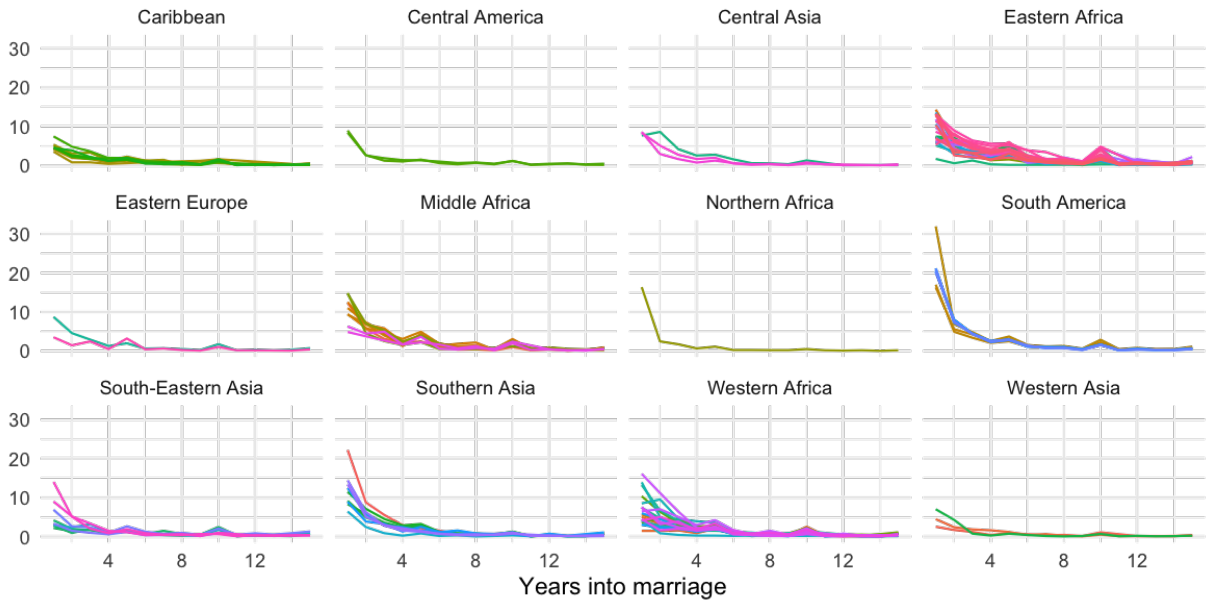
The Demographic and Health Surveys Program (DHS) are administered by the United States Agency for International Development (USAID), has conducted hundreds of representative household surveys collecting extensive information on health, nutrition, and population in over 90 mostly developing countries. These surveys are highly uniform in structure enabling useful scaling across vastly different settings. However, the Indian DHS is particularly useful for our purposes in two ways. First, it is one of the subset of DHS surveys which include village-level coordinate data, which allow the user to assign almost all of the 600,000 surveyed households to one of 28,526 unique coordinates. To preserve the anonymity of respondents, 99% of these coordinates are given a jitter of between 2-10 miles depending on the population density of the "true" region. Figure **??** depicts the geographic breadth of these cluster-level coordinates. This is useful for us as it allows us to integrate external datasets to match a given household to their history of local weather realizations.

Secondly, the Indian DHS is distinguished by its incorporation of the National Family Health Survey (NFHS), which supplements the traditional DHS survey with additional modules related to women's health, fertility, and births. Perhaps for this reason, these representative surveys are still largely imbalanced before post-stratification as the surveys entailed just under 699,686 female respondents compared to 112,122 male respondents. These female respondents are also asked much more extensive questions. Of these, the most relevant come from a sub-module on domestic violence, which are conducted with a representative subsample of 60,610 women. As enumerated in Section C.1.1, we split these survey questions into distinct categories pertaining to attitudes towards domestic violence, experiences with domestic violence, controlling behavior demonstrated by their spouse or cohabiting partner, attitudes towards spousal sexual consent, and intrahousehold bargaining power.

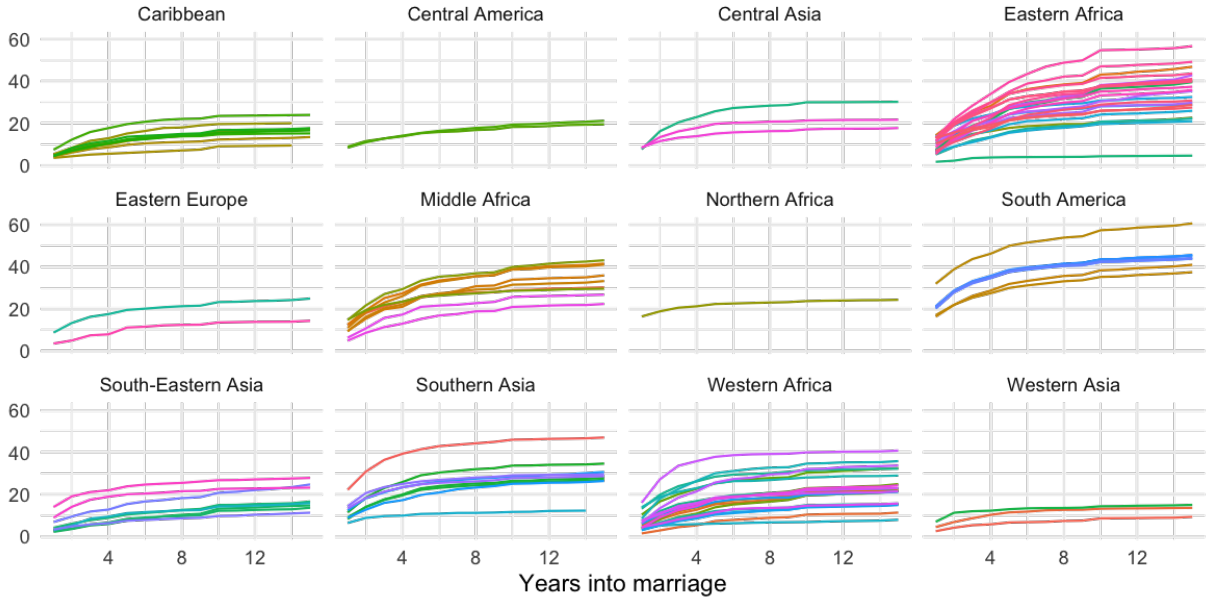
Within each module, I also add variables enumerating the number of affirmative answers given by the respondent and an indicator for whether any answers are affirmative. The DHS surveys ask more standard questions such as date of birth, family composition, date of marriage, religion, and caste which we make use of. It will also be useful to refer to descriptive statistics for these variables in order to assess the magnitude of impacts and so Tables **??** and **??** in the appendix give summary statistics for the variables we use as outcomes, subject to some subsetting for comparability. The women's sample reported here are a subset of the full women's sample selected to participate in the IHDS' domestic-violence module. Throughout our analysis, we use the full sample of over 600,000 female respondents whenever we are not dependent on this module. Note first that there are variables that overlap between the women's and men's recode but the set of questions asked of men are a subset of those asked women. It will also be important to note that marriage.age, which appears first in both summary tables, is given in units of months. I also illustrate the spatial heterogeneity in the region-level reported incidence of (*dv.any2*) and attitudes (*dv.justify.any*) towards domestic violence in Figure **??**.

Figure 3.1: Incidence of self-reported domestic violence by year of marriage

(a) Timing of first violent event



(b) Cumulative percentage of marriages ever violent



3.2.2 Weather data

Historical precipitation data come from the University of Delaware ("UDel") series maintained[1], which is constructed from interpolations of data from tens of thousands of weather stations worldwide. This is in contrast to reanalysis products, less frequently used in this literature, which combine real observations then subjects them to adjustment by subservience to localized climate models to avoid possible biases of interpolation. Our choice of the UDel series is in keeping with studies most recently and most frequently published in this subliterature but a comprehensive robustness exercise would invoke a wider range of both weather station and reanalysis products, such as the increasingly favored Berkeley Earth series.

The UDel data is provided at 0.5-degree monthly resolution, and spans the period 1900-2017. We make use of this extensive history of observations in our study when we construct our weather shocks, which we loosely refer to as droughts. Because a given level of precipitation would be differently remarkable across the extremely diverse terrains of the Indian subcontinent, we instead define droughts locally. For a given set of cluster coordinates, we select a set of gridcells located near the given coordinates and then extract every temperature and precipitation observation from every month at those gridcells from 1950 to 2017. We then annualize these monthly observations to construct annual precipitation measures for 67 years for that given location. Following M. Burke et al. (2014), we then fit a gamma distribution to these observations. Given a month and year, we then annualize the precipitation observed over the preceding 12 months and define that period as a drought if it falls beneath the 15th percentile of the fitted gamma distribution. In effect, we have made uniform the exposure to treatment across all locations and its random and relatively frequent occurrence should be a source of useful variation in treatment.

A critical additional advantage of defining droughts in this manner is that if precipitation realizations are sufficiently independent of one another, all locations are exactly as susceptible to experiencing a precipitation shock in a given year and that the local quantile of a given year of precipitation is intuitively unrelated to any local characteristics likely to influence our measures of cultural attitudes and gender norms.

3.3 Methods

I make use of three regression specifications to explore different facets of our research question. First, we look at simple correlations in gender norms between successive generations as a descriptive examination of patterns in intergenerational cultural persistence. Next, I estimate the impact of weather shocks on several sets of gendered outcomes to investigate the sensitivity of gendered behaviors and outcomes to economic disruption. The time dependence of these effects motivates the final approach, which makes use of survival models to measure the impacts of droughts on susceptibility to child marriage and falling victim to spousal domestic violence. All regressions are weighted by the included survey weights.

3.3.1 Intergenerational persistence of domestic violence

This first class of regressions relates parental domestic violence to a respondent's attitudes towards and experiences with domestic violence. Regressions of this variety are specified as follows:

[1] Willmott and Matsuura, 2022.

$$Y_{i,r,t} = \beta V_{i,t-1}^{\text{parent}} + \Pi X_{i,r,t} + \alpha_r + \epsilon_{i,r,t} \tag{3.1}$$

Here, outcomes $Y_{i,r,t}$ measure the response of the generation-t respondent from family i residing in region r to one of several survey questions. These questions can themselves be classified by modules such as the justifiability of domestic violence under hypothetical circumstances, experience with domestic violence, or intrahousehold decisions within the purview of the respondent. For example, a typical outcome may be a binary-coded response to the question of whether the respondent is empowered to make decisions about large household purchases. Most responses are coded in this binary fashion, but some modules also include a count of questions within the module to which the respondent responded affirmatively or a binary meta-variable indicating whether any of the questions in the module were answered affirmatively.

$V_{i,r,t-1}^{\text{parent}}$ is an indicator variable that takes on a value of 1 if the respondent reports their father—considered generation $t-1$ of family i—was ever domestically violent. Thus when regressing a binary outcome variable, our coefficient of interest, β, is bounded by -1 and 1 and is interpreted as a crude measure of the transmissivity of the particular outcome variable, enabling us to make interpretations along the lines of "having a spousally abusive father is associated with a β percentage-point decrease in the likelihood of having an affirmative response to outcome Y." When regressing a count outcome variable, β is interpreted as the increase in the number of affirmative answers in the module associated with having a domestically violent father. Importantly, while we include a sparse set of available cultural controls $X_{i,r,t}$ and include regional fixed effects α_r, these claims should not be interpreted causally. I will argue these correlations and their signs are nonetheless instructive and useful for motivating the main specifications that follow.

3.3.2 Effects of drought exposure on gendered attitudes and actions

Regressions of this variety are specified as follows:

$$Y_{i,r,y} = \sum_k \beta_k D_{r,y+k} + \alpha_r + \gamma_y + \varepsilon_{i,r,t} \tag{3.2}$$

The outcome $Y_{i,r,y}$ may entail the same binary responses as mentioned in Subsection 3.3.1 for respondent i living in region r born in year y. We will also first consider additional outcomes such as respondent's age at first marriage and educational attainment.

$D_{r,y+k}$ represent dummary variables indicating whether a locally-defined drought occurred during the period indexed by $y+k$, taking place k periods after some reference year y. In practice, we include droughts from two different period. First, we consider drought events in the respondent's first 16 years of life, including a "year zero" indicating whether there was a local drought in the 12 months preceding their month of birth. The next 15 years are binned into five-year intervals and are thus entered as count variables measuring how many drought events occurred between ages 1-5, 6-10, and 11-15. In a second specification, we instead consider drought shocks relative to the first year of cohabitation or marriage. This set of drought shocks also consider "year zero" events for the 12 months preceding the date of marriage which we take as the reference year y, and then the first five years of marriage also enter the regression as indicator variables, unbinned. The motivation here is to run separate regressions estimating impacts from early-life drought events at different life stages as well as early-marriage drought events when intrahousehold relationships are being

formed.

All regressions include year-of-birth or year-of-marriage fixed effects γ_y and regional fixed effects α_r to absorb the influence of time-invariant local characteristics. Standard errors are clustered at the regional level as drought events are likely to be spatially correlated within the same year. The identifying assumption for this estimation strategy is that within-location and within-birth-cohort variation in weather shocks is plausibly exogenous. Given that by construction, these weather shocks by their local long-run distribution (a process delineated in Subsection 3.2.2), this assumption is innocuous to the extent that there is insignificant autocorrelation in the local incidence of drought. If so, then β_k may then be causally interpreted as the change in the outcome variable associated with an additional drought experienced over the particular year or year-bin represented.

An advantage of this specification is its ability to distinguish the relative importance of the timing of shocks. As shocks experienced relative to some baseline event enter as separate dummies, the resulting estimates of β_k may be informative of when a drought event is most influential on the outcomes of interest even absent time-coded information about the process of forming gendered preferences, negotiating intrahousehold dynamics, and the incidence of domestic violence. In principal, the identifying assumption also implies non-correlation between these drought regressors enabling relatively efficient estimation.

3.3.3 Hazards into child marriage and domestic violence

Following Currie and Neidell (2005) and Corno et al. (2020), we use hazard models to estimate the impact of environmental factors on the hazard of some terminal event occurring, which we denote as a switch of a time t-indexed binary variable E_t from 0 to 1. These are panel data models specified as follows:

$$E_{i,r,k,t} = \beta D_{r,k,t} + \alpha_r + \lambda_k + \gamma_t + e_{i,r,k,t} \tag{3.3}$$

Currie and Neidell (2005) used variation in carbon monoxide to identify the hazard risk of infant mortality while Corno et al. (2020) used drought shocks to estimate the hazard of child marriage and early fertility. Here, we first attempt to replicate Corno et al. (2020) using this updated round of data and then perform a similar exercise to estimate the hazard of the first incidence of domestic violence in our data. In the first case the event outcome $E_{i,r,k,t}$ is an indicator of whether respondent i currently living in region r born in cohort k was married by the time they were age t. $D_{g,k,t}$ is an indicator of whether region r experienced a drought the year that someone born in cohort k would have been age t and so β measures the increase in the propensity to be married by the largest age $\bar{t}$ considered in the data.

This top-coding is important since it requires a reorganization of the data. Not only does the panel structure require one observation per individual-age combination, it also requires right-censorship of the data with respect to the outcome variable. Specifically, if we consider age 16 as the definitional cutoff for child marriage, then we must exclude all observations for individual-ages older than 16. Further, this being a survival model, we also exclude any observations after the first event, i.e., after the first positive occurrence, the individual is removed from the panel, which generally creates panel imbalance by design. Thus, a respondent who only gets married at age 20 would appear until age 16 and only with a sequence of zeroes for the outcome variable E. The inclusion of age fixed effects γ_t accounts for differential expected propensities to marry at different ages and

cohort fixed effects λ_k absorb cohort-specific characteristics. We also include specifications with fixed effects at the state level, a more aggregated geographic unit, and following Corno et al. (2020) a set of state-by-cohort interacted fixed effects.

As a central contribution of this paper, I adapt the approach to estimate the hazard of the incidence of domestic violence in the early years of a marriage. This exploits a survey question in the DHS' domestic violence module asking a representative subset of women whether they had experienced domestic violence and if so, in which year of marriage it first occurred. The surprisingly granular level of detail and wide distribution of responses enable me to investigate the hazard of the incidence of domestic violence using years since marriage as the running time variable in place of imputed age though age fixed effects are maintained in addition to age-of-marriage fixed effects as a direct analogy to the Corno et al. model.

3.4 Results

Our first set of results build on this work to study the hazard into marital violence using data from the Demographic and Health Surveys. Figure 3.2 and Table 3.1 summarize findings from Indian data. Here, we find that the incidence of a drought perhaps counterintuitively reduces the probability of first-time domestic violence by 6.8-7.4% in the year of the drought's occurrence. Intuitively, the more severe the definition of drought, the larger is the estimated effect. Ongoing work entails exploration of the underlying mechanisms for this effect and the degree to which this effect generalizes across other countries and over time. We are also interested in exploring the effect of environmental shocks on the quality of eventual marriage matches as measured by age gaps, political disagreement, intrahousehold bargaining power, and differences in gendered attitudes.

We also intend to exploit the cohort variation in DHS survey respondents to investigate whether these economically determined deviations from norms hold over time, even generationally transmitted, or transient. To investigate whether the cultural effects of economic shocks are distinctly gendered in nature, we have data on non-gendered (i.e., caste-based and religion-based) attitudes to use as placebo tests. To ascertain whether these effects can partially explain the observed cross-national differences, we are in the process of scaling our domestic violence analysis to the more than 60 other developing countries in the DHS database.

3.4.1 Intergenerational persistence of domestic violence norms

The results of our correlation exercise are summarized in Tables 1-3. I also repeat the same exercise for the variety of gendered norms available in the survey data and present those results in Tables 8-11, which appear in the Appendix. We find, perhaps unsurprisingly, that whether a respondent reported that their father had ever beat their mother is a very strong and significant predictor of essentially the full suite of gendered variables for both male and female respondents.

3.4.2 Effects of rainfall shocks on gendered outcomes

Age of marriage and educational attainment

Table 4 reports the results of a regression of respondent age of marriage on counts of droughts experienced in their first 15 years of life, binned into five-year bins, plus a dummy for whether a

Figure 3.2: Incidence of domestic violence in India

(a) Baseline annual hazard into first-time marital violence



(b) Effect of different definitions of drought on annual hazard

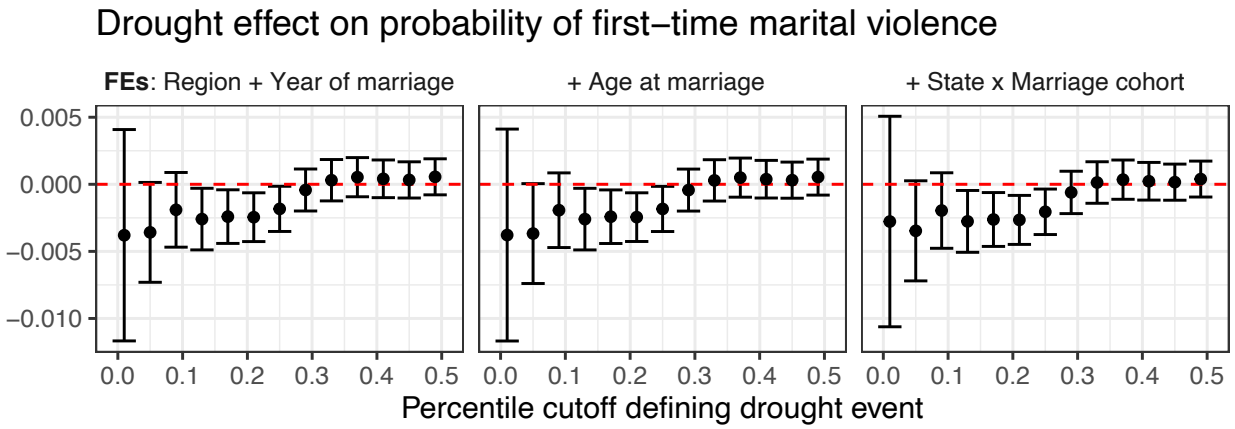


Table 3.1: Effect of droughts on first-time marital violence

Dependent Variable:	Hazard into first-time violent	
Model:	(1)	(2)
Drought	-0.002739***	-0.002978***
	(0.000979)	(0.000987)
Year of marriage	Yes	Yes
Age at marriage	Yes	Yes
Respondent age	Yes	Yes
Region	Yes	Yes
State × Marriage cohort		Yes
Observations	355,750	355,750
R^2	0.04374	0.04478
Within R^2	2.13×10^{-5}	2.47×10^{-5}

Clustered (region) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1* (Region clustered SEs)

drought occurred in the 12 months preceding their birth. While this specification does not specifically measure hazards into child marriage, the results are consistent with the findings of Corno et al. (2020) in that for women, only droughts experienced in the first years of bride eligibility (corresponding to ages 11 to 15) have a measurable effect on the age of marriage. It's important here to emphasize that units of *marriage.age* are given in terms of months so the results for the female regression would suggest that an additional drought experienced between the ages of 11 to 15 are associated with a 0.6-month increase in the expected age of marriage of a female respondent. In Corno et al. (2020), this is explained as a consumption-smoothing behavior specific to virilocal cultures that practice dowry. Given the negative income shock associated with a drought, the bride's family has two channels through which to smooth consumption: reduce the terms of the dowry price or delay marriage. This is corroborative evidence that the latter is at least one margin of adjustment.

In comparison, early-life droughts do not seem to have a comparable effect on the timing of male respondent's marriages. Corno et al. (2020) might argue that this is a privilege of virilocal societies where males are better able to smooth consumption without resorting to such channels of adjustment. While a negative income shock would theoretically induce males to hasten their marriages in order to use dowry payments as a cushioning advance, they also have the relative social safety net of labor force participation and the support of live-in parents who themselves may invest in their children as a tacit condition of the expectation that they are taken care of in their old age. Alternatively, this null result could be because men tend to marry later. Whereas women are seen as candidates for marriage as early as age 10, this is not the case for candidate grooms. Indeed, Tables **??** and **??** suggest the average age of marriage for women is 19 while for men it is over 23. For men, the relevant drought period may be somewhere between ages 16 and 24, which I do not currently have data on.

So for men, we instead only find a significant effect of pre-birth droughts: the age of marriage

Table 3.2: Effect of timing of drought occurrence on age of marriage

Dependent Variable:	marriage.age			
gender	female		male	
Model:	(1)	(2)	(3)	(4)
Variables				
drought.y0TRUE		0.0493		2.089**
		(0.3313)		(1.050)
droughts.5	-0.1550	-0.1542	0.0461	0.1132
	(0.1894)	(0.1905)	(0.4500)	(0.4538)
droughts.10	-0.2272	-0.2248	-0.7965*	-0.6972
	(0.1950)	(0.1939)	(0.4799)	(0.4897)
droughts.15	0.6145***	0.6145***	-0.2728	-0.2584
	(0.1846)	(0.1846)	(0.4462)	(0.4469)
Fixed-effects				
dob.year	Yes	Yes	Yes	Yes
region	Yes	Yes	Yes	Yes
Fit statistics				
Observations	473,078	473,078	65,789	65,789
R^2	0.12412	0.12412	0.20900	0.20913
Within R^2	0.00014	0.00014	0.00012	0.00029

One-way (region) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

is expected to increase by a full two months if the male's family experienced a drought in the 12 months prior to being born. This could be a spurious result, but one plausible interpretation is that these early-life conditions determine familial investment in children which in turn have persistent effects in adult outcomes, consistent findings by Maccini and Yang (2009) and Rose (1999).

In Table **??**, we find relatively muted effects in a similar regression using educational attainment as an outcome variable. While there are significant effects of droughts in a daughter's first five years of life on attainment of at least a primary education, they are counterintuitively negative in sign (suggesting the drought raised the likelihood of daughters attaining a primary education and small in size: just 0.3 percentage points for a subsample in which 68.1% of women receive at least a secondary education. There is a more intuitive negative effect of a drought experienced between ages 6 and 10, but again the effect size is very small despite its significance. No coefficients are significant for men, which again suggests investment in daughters is a more likely channel of adjustment for a male-biased family culture experiencing a negative income shock.

Attitudinal formation

For these outcomes considered so far, the timing of the drought to be considered is obvious: only droughts that occur in childhood can possibly be relevant to educational decisions and only droughts that occur in the years leading up to matching in the marriage market could be instructive. The timing is less obvious for other outcomes. It is less obvious during which years a person forms or is more impressionable about their attitudes towards domestic violence, for example. It is unclear whether the delegation of household responsibilities are a function of early-life upbringing or are determined after matrimony. This is why for these attitudinal outcomes, we consider the same binned set of drought counts from the first 15 years of life as well as indicators for droughts that occur in each of the first five years of marriage, as well as the year leading up to matrimony.

Results using measures of the controlling behavior of a husband are given in Tables **??** and **??** corresponding to estimates of the effects of binned early-life droughts and early-marriage droughts respectively. The aggregated outcomes in models (1) and (2) suggest null effects of early-childhood droughts. Scattered significant effects are small and likely spurious. In comparison, Table **??** estimates are suggestive of substantial early-marriage drought effects. A drought that occurs in the first year of a marriage is associated with a 2.4 percentage-point decrease in the likelihood of a wife reporting any controlling behavior by her husband. Given the sample average has 48.0% of women reporting at least one type of controlling behavior, this suggests a drought in the first year of marriage results in a 5% increase in the likelihood of the husband exhibiting controlling behavior. The theory here could be that the first year of a marriage is when intrahousehold relations and patterns are established. The disruption of this norm-setting period by a negative income shock could reduce the husband's bargaining power in the household, for example relative to a hypothetical outside option for the bride, and thus the husband is less able to act in a controlling manner.

Impacts of droughts on the formation of beliefs about the justifiability of domestic violence are reported in Tables **??**, **??**, and **??**. Similar results on the reported incidence of domestic violence are reported in Tables **??** and **??**. We also show the results for beliefs about the justifiability of the wife refusing sex in Tables **??** and **??**. In general, the pattern is of mostly insignificant results with no consistent pattern in signage among the few significant results.

Contrasting these inconclusive results with the more definitive results of the bargaining outcomes is perhaps instructive. Formation of attitudes are likely formed and refined continually over

time whereas intrahousehold bargaining decisions are the result of an implicit negotiation between two meeting parties with differential bargaining power. There is a specific time—marriage—at which these two parties meet and much of the matching process entails setting terms and evaluating relative positioning, for example through the practice of dowry payments as exploited and framed in analyses such as Corno et al. (2020). As such, it is then maybe not surprising that the terms of intrahousehold bargaining exhibit a sensitivity to income shocks, given they affect the parties' leverage, and that this should be measurably observed at the meeting of the two parties, which is to say the first year of marriage during which the terms of engagement are relatively fluid and not yet set in stone.

3.4.3 Hazards into early marriage and domestic violence

We now hone in on the importance of timing in our next set of results. As discussed in Subsection 3.3.3, the advantage of hazard models is that they estimate effects on the probability of a first occurrence of some event of interest. This is unprecedented in our analysis of domestic violence, which thus far has not been time-coded. The outcomes considered so far only entail indicators for whether the resopndent's father had ever beat their mother or whether they had ever experienced one or more specific types of domestic violence. We make use of a single question in the survey which asks how many years into marriage, if ever, the respondent first experienced domestic violence.

This draws inspiration from Corno et al. (2020), which estimates hazards into early marriage. This is also an exercise possible with our data so we begin first by replicating this analysis with results presented in Table **??**, which we present in the appendix. The results are entirely consistent with the original findings, although the effect size is substantially smaller. Whereas the original study ocncluded that a drought experienced between the ages of 10-16 reduced the likelihood of getting married in the same year by 0.46-0.47 percentage points, that range is 0.17-0.27 in my analysis. This amounts to a decrease in the hazard of early marriage of 1.2-1.9% compared to the original estimated magnitude of 2.8%.

The results for the effects of early-marriage droughts on the hazard of domestic violence are presented in Table 5. I find that across the range of specifications considered, the occurrence of a drought in the first five years of a marriage reduces the hazard of domestic violence by between 1.8 and 2.0 percentage points in the year of its occurrence. This would seem to suggest that a weather-induced negative income shock largely leads to a reduction, or at least a delay, in the first incidence of domestic violence. This lends credence to the notion that to the extent that domestic violence may be considered a repugnant exercise in intrahousehold bargaining power, the overall average effect of this reduction in bargaining power is a reduction or delay in physical violence rather than a "backlash" effect that is highlighted in work such as Sekhri and Storeygard (2014). Compared to a sample average hazard of 3.77% over the first five years of a marriage, this suggests the occurrence of a drought reduces the hazard of a first occurrence of a domestic violence episode by 48-53%.

Table 3.3: Hazard into domestic violence

Dependent Variable:		dv	
Model:	(1)	(2)	(3)
Variables			
droughtTRUE	-0.0184**	-0.0197***	-0.0179**
	(0.0073)	(0.0076)	(0.0077)
Fixed-effects			
year.of.marriage	Yes	Yes	Yes
age	Yes	Yes	Yes
state		Yes	Yes
state×marriage.cohort			Yes
Fit statistics			
Observations	47,035	47,035	47,035
R^2	0.03207	0.03810	0.04452
Within R^2	0.00021	0.00024	0.00020

One-way (region.name) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

3.5 Conclusion

Another adjustment margin: quality of couples match

A more complete analysis would be able to speak more directly to the intergenerational transmission of gender norms. This omission is an unfortunate consequence of the limitations of our dataset, which restricts the ability to conduct such an analysis in two ways. First, it mostly does not conduct surveys of multiple generations within the same household. Instead, it only surveys married women aged 15-49 and does not elicit much information about the respondent's relatives. A better implementation of our correlational exercise would be able to make use of equivalent questions for successive generations. The prevalence of parental domestic violence is very crude as a stand-in for generational gender attitudes.

Another refinefent to the above analysis would be able to exploit family structures. A unique and important feature of our rural Indian setting is the widespread practice of virilocality. This practice has substantial gendered effects in at least two important ways. First, it separates married women not just from their home location but from their support networks. Second, it introduces a unique family structure where the resident mother-in-law of the bride is able to impose substantial influence on the conduct, intrahousehold bargaining, and gendered attitudes of the married couple. Having cited work in Fernández et al. (2004) that also looks at the gendered effects of different family structures, exploring such dynamic family structures is not just promising, but necessary.

Just as Corno et al. (2020) compared findings in the Indian setting to findings in the African continent, so too could this study benefit from a cross-setting expansion. The power of the original paper came from the diretion of effects being so dramatically opposite in the two settings, clearly

attributable to differences in virilocality pratices and the diretion of marriage payments. This is not really a statement on the external validity of these findings considering the size of the Indian subcontinent, but the stark comparison afforded much more credibility to the authors' findings. The same can be done here.

Finally, the study could be extended to look at heterogeneous impacts. Luke and Munshi (2011) raised the provocative thesis that low-caste members are particularly likely to diverge from tradition because of their relatively less entrenched status in the economy. A differential analysis along caste lines could be similarly illuminating.

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Appendix A: Supplement to Chapter 1

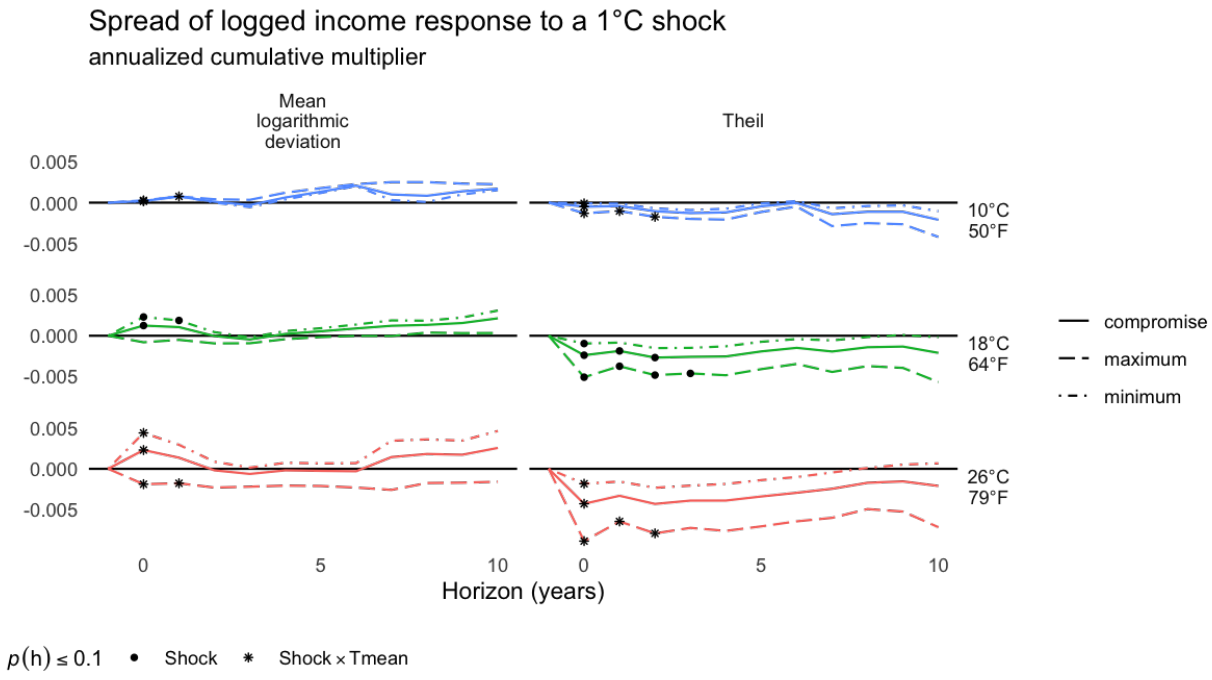
A.1 Data

A.1.1 Construction of inequality indices

Data from the World Inequality Database can be extracted at the country-percentile level with variables reporting averages, minimums, maximums, and shares of national income. As grouped data, its use for the construction of inequality indices still requires the modeler to make assumptions about the distribution of household incomes within these disaggregated groups. This becomes increasingly influential for the construction of entropy-based inequality measures such as the mean logarithmic deviation or the Theil index for higher levels of inequality across groups.

For a given inequality measure J, the lower bound J_L is calculated by assuming perfect equality within all groups such that all individuals are assigned their group mean income. The upper bound J_U is calculated by assigning a proportion of the individuals within a group the minimum income of the group and the remainder the maximum income of the group.

Figure A.1: Sensitivity of inequality index effects to assumptions about within-percentile inequality



Cowell, 2011 describes multiple 'compromise' assumptions between the two extremes, finding

that they each approximate an average where the lower bound is given twice the weight of the upper bound. Figure A.1 depicts IRFs of the two inequality measures under each of these three assumptions. For our purposes, we use the compromise construction whenever a choice is necessary.

A.2 Estimating persistent state-dependent responses to a temperature shock

Here, we provide additional exposition of our application of the method of local projections for inferring a dynamic social response to the temperature shock described in Section 1.2.3. The supplementary analysis here uses country-level GDP from the Penn World Tables (Feenstra et al., 2015) as the outcome variable of interest. The results documented here may also be compared to other estimates in the prolific literature on global climate impacts estimated by top-down methods, which most commonly use similar country-level aggregates as outcomes.

A.2.1 State-dependent impulse response of GDP

We first estimate Equation (1.3) on a country-year panel dataset using the h-period long difference in the logarithm of GDP as our outcome variable $\Delta_h y_{i,t+h}$. Figure A.2a illustrates the estimated impulse response of GDP up to a projection horizon of 6 years after a unit temperature shock. Colors correspond to three representative temperatures depicted in different colors. The plots in the second row depict the same functions annualized by simplying dividing estimates at each projection horizon h by $h + 1$.

The contemporaneous estimates in the IRFs in the first row of Figure A.2a depict a pattern now standard in the aggregate impacts literature featuring near-null temperature effects in cold climates, mild effects in mid-temperature climates, and the largest negative effects in hot climates. The evolution of these impacts over time is less commonly depicted and here produce a pattern qualitatively similar to those reported in NRK wherein per capita GDP appears to stay persistently depressed over the entire projection horizon. The second row depicts the same estimates but annualized by scaling down estimates by their corresponding horizon $h + 1$ to more clearly depict convergence behavior relative to the contemporaneous effect of the shock.

Global inequality effects of climate change thus are not just dependent on the shape of the impulse response responses across quantiles but also the distribution of warming over space and time as well as the actual distributions of incomes across quantiles. The simulation exercises to be described in Section 1.4 are valuable because they account for four relevant climate inequalities: inequality of temperature effects across an income distribution, inequality in the incidence of temperature effects across different climates, inequality in the global distribution of income, and spatial inequality in the incidence of absolute warming.

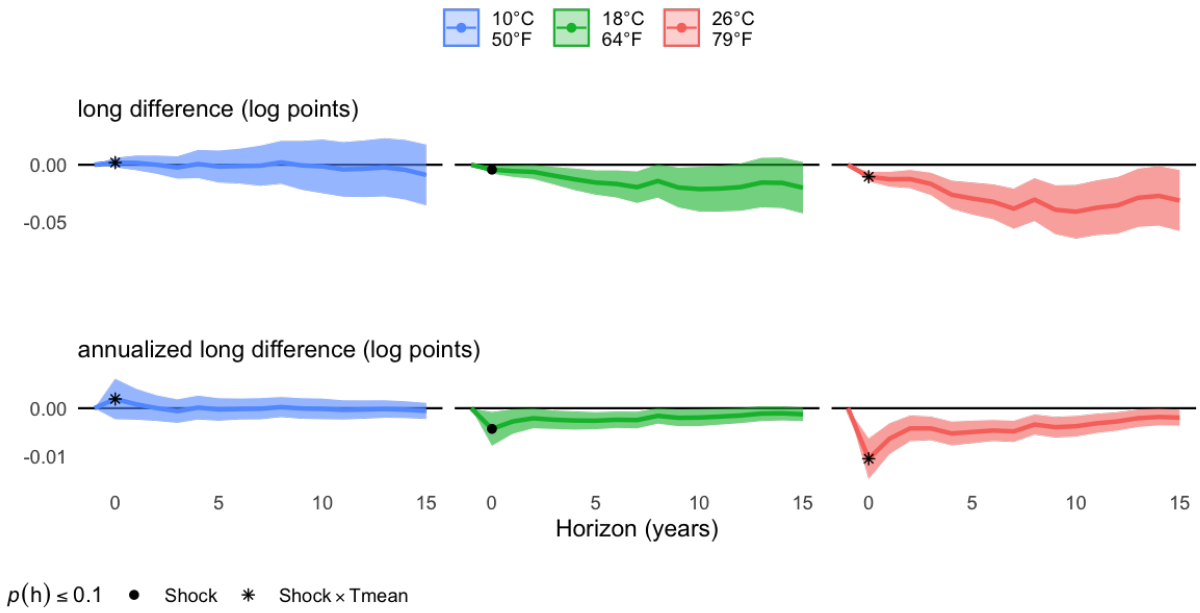
A.2.2 Inferring persistence from local projection estimates

The colored error bands depicted in Figure A.2a reflect pointwise estimation error. This is particularly important to include in this context since the volume of regressions underlying a single LP-IRF precludes convenient summary in a single regression table.[1] One may be tempted to

[1]For example, the results depicted in Figure A.2a derive from 16 different regressions. Later results derive from as many as 91. All graphically depicted errors and significance tests presented in this paper correspond to a significance level of $\alpha = 0.1$.

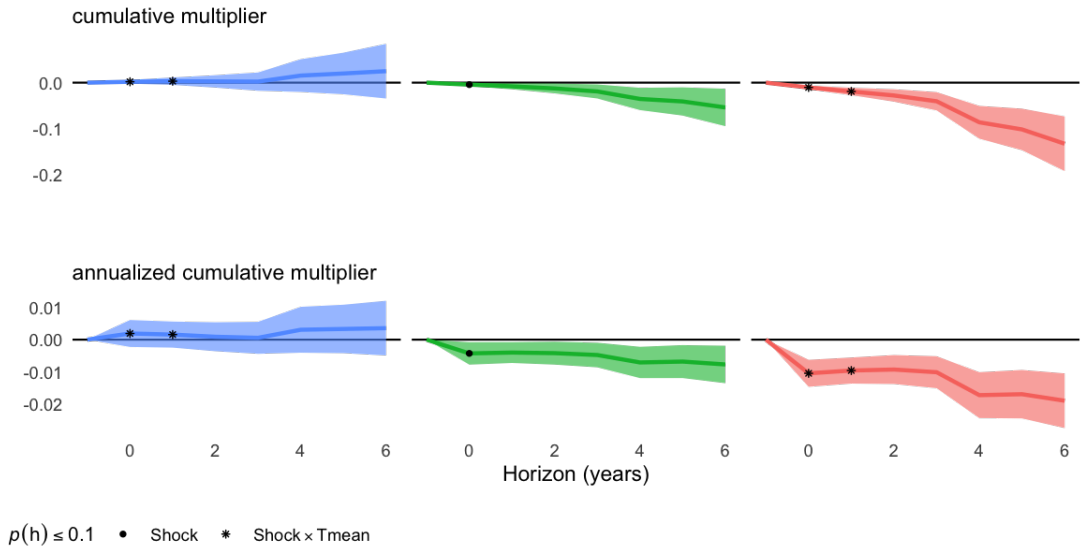
Figure A.2: State-dependent response of GDP to a 1°C shock

(a) State-dependent impulse response of GDP



The plots in the top row correspond to estimates of impulse responses of GDP estimated by the $H+1$ local projections regressions specified in Equation (1.3) for three representative temperatures depicted in different colors. The plots in the bottom row depict the same estimates and error bands annualized by dividing values at each horizon h by $h+1$

(b) State-dependent temperature multiplier of GDP



The cumulative multiplier is interpreted as the change in *cumulative* GDP over h periods attributable to a unit shock in $\widehat{\tau}$, net of its impact on future shocks $\widehat{\tau}$. Annualized multipliers represent the same estimates but divided by the projection horizon for visual convenience.

infer from the fact that the hot-country bands in red do not intersect the horizontal axis over the entire projection horizon that these estimates constitute significant evidence to reject a joint null hypothesis of level effects which persist for fewer than 15 years.

However, the uncertainty depicted by error bands corresponds only to individual hypothesis tests. Since estimates and residuals across an IRF are serially correlated—for example, the green bands straddle the horizontal axis but all estimates are negative— the relevant test of significance is a joint hypothesis test not represented by pointwise error. The significance of the main effect of the impulse response $\mathcal{R}_{\tau \to y}(h)$ amounts to testing the null hypothesis that all main effects up to h are 0:

$$H_0(h) : \beta_{1,0} = \beta_{1,1} = ... = \beta_{1,h} = 0$$

Since the number of coefficients being considered in the joint hypothesis increases in h, the coverage of the joint hypothesis test must be adjusted for the inclusion of each additional horizon to compensate for the problem of multiple hypothesis testing. Inverting this joint null hypothesis, Inoue et al. (2024) derives convenient estimators for the implied "significance band" defined to satisfy the following condition:

$$\mathbb{P}\left[\bigcap_{h=0}^{H}\left\{\zeta_{\frac{\alpha}{2(H+1)}}\frac{\sigma_h}{\sqrt{T-h}} < \widehat{\beta}_h < \zeta_{1-\frac{\alpha}{2(H+1)}}\frac{\sigma_h}{\sqrt{T-h}}\right\}\right] \geq 1-\alpha$$

The scaling of the significance level α by $2(H+1)$ is an implementation of the Bonferroni correction. In practice, rejecting the null for the main effect corresponding to horizon $h+1$ is sufficient to reject the null of non-persistence up to horizon h. Graphically, these bands are represented by a region straddling the horizontal axis such that estimates of β_{1h} contained within it are insufficiently large enough to reject the hypothesis of a null effect that the impulse response persists up to period h, analogous to the bands drawn to measure persistence in time series correlograms. The width of these bands increases with the projection horizon because the available estimation sample for a near-balanced panel is reduced by one period per unit for each additional projection.

These significance bands are specific to the main effect and are represented by the gray regions in the middle panels of Figure A.3a. Since ours is a state-dependent model, I derive equivalent significance bands specific to the interaction effects. These are defined relative to the point estimates of the main effects and are depicted by the gray regions in the panels on the left and right. Points outside these regions are considered sufficient to reject the null hypothesis that interaction effects do not persist up to h periods. By the linearity assumption, the null hypothesis of zero interaction effects for a 10° climate is rejected if and only if it is rejected for a 26° climate.

For comparison, Figure A.3b depicts a placebo test where I construct significance bands equivalent to those described above to test the persistence in the effect of temperature on another widely available and potentially non-stationary growth rate: that of population. As I am not aware of any theory or evidence to suggest temperature shocks would meaningfully impact country-level population growth[2], we should expect the tests to return precise null results. Indeed, we find with all point

[2]While there is an established literature on climate impacts on mortality rates (A. Barreca et al., 2016; A. I. Barreca & Shimshack, 2012; T. Carleton et al., 2022), absolute magnitudes are miniscule relative to total populations

estimates resulting from two separate population datasets easily contained by the corresponding significance bands. Notably, if we were to misuse the error bands as indicators of statistical significance, we would misinterpret the estimated IRFs as evidence to reject the null for large subintervals of the projection horizon for mid-temperature and hot climates. Instead, the entirety of these error bands are easily enveloped by the significance bands.

Since including significance bands, color-coded error bands, and point estimates for multiple IRFs can clutter space, I adopt a convention of omitting the significance bands entirely and instead coloring black the point estimates which are located outside the implicit error bands. In all IRFs beginning with those depicted in Figure A.2a, rounded points correspond to significant main effects while asterisks correspond to significant interaction effects. By this test, these results cannot reject the null hypothesis of one-period level effects.

A.2.3 The state-dependent temperature multiplier

Another reason to prefer temperature shocks to levels of temperature as treatment variables is that models which use the latter neglect to sufficiently account for autocorrelation in temperature itself when estimating dynamic effects (Bilal & Känzig, 2024; Nath et al., 2024). An analogy can be drawn to the estimation of multipliers in empirical macroeconomics: when estimating the impulse response of, for example, GDP to a fiscal policy shock, one must account for the tendency of stimulus programs to be implemented in stages or to be followed by additional stimulus. Otherwise, the effect sizes implied by simple impulse response functions misattribute cumulative changes in the outcome only to the initial unit shock; multipliers scale these effects at each horizon by the corresponding accumulation of shocks.

Figure 1.3 depicts the impulse response of temperature shocks to a pulse of itself for our three representative climates. Each exhibits very similar persistence patterns across two independent temperature dataset. An application of the persistence test covered in Section A.2.2 implies autocorrelations which persist for up to 10 years though only the first 1-2 periods are qualitatively large (approximately 21% and 9% respectively with insignificant heterogeneity across climates).[3] Cumulative multipliers are estimated in order to scale down the estimated impulse response of GDP $\mathcal{R}_{\tau\to y}(h)$ by the impulse response of the shock to itself $\mathcal{R}_{\tau\to\tau}(h)$ to account for this dynamic treatment schedule.

Cumulative responses are traditionally computed by first constructing *cumulative* versions of the two IRFs: $\mathcal{R}^c_{\tau\to y}(h) := \sum_{j=0}^{h} \mathcal{R}_{\tau\to y}(j)$ for the outcome of interest y and $\mathcal{R}^c_{\tau\to y}(h) := \sum_{j=0}^{h} \mathcal{R}_{\tau\to\tau}(j)$ for the shock $\widehat{\tau}$. Then the cumulative temperature multiplier $m_{\tau\to y}(h)$ would be calculated as the ratio

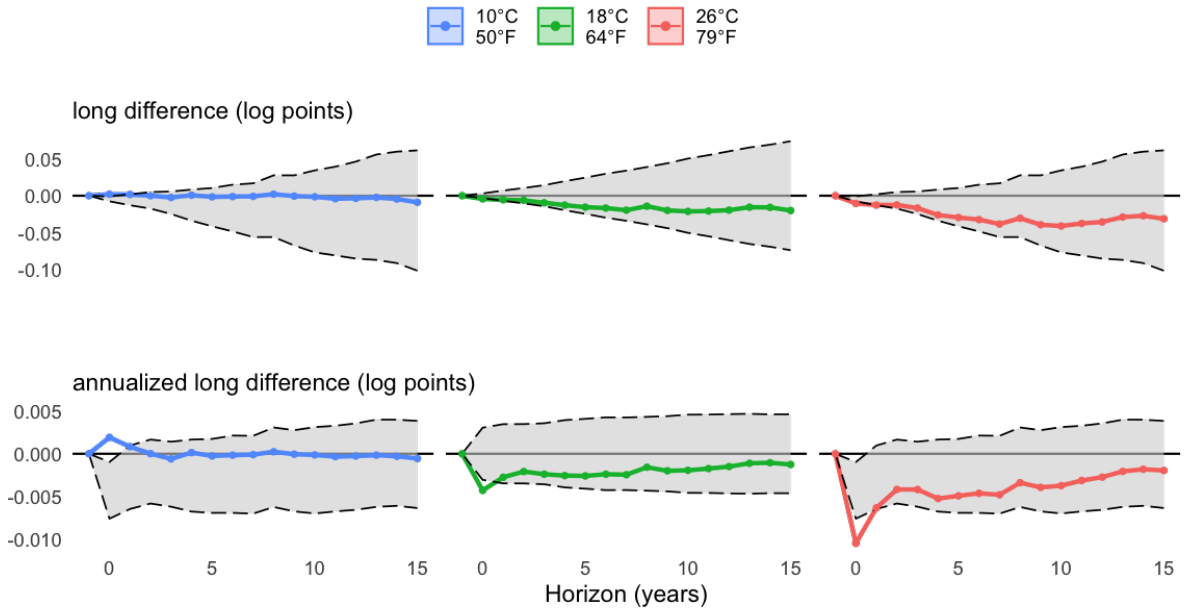
$$m_{\tau\to y}(h) := \frac{\mathcal{R}^c_{\tau\to y}(h)}{\mathcal{R}^c_{\tau\to\tau}(h)}$$

However, this approach has a couple of inconvenient downsides. First, as noted by Jordà (2024), computation of standard errors of a ratio of two random variables is complicated and non-standard. Second, it is computationally inefficient in the sense of requiring $2(h+1)$ (or $4(h+1)$ in our state-

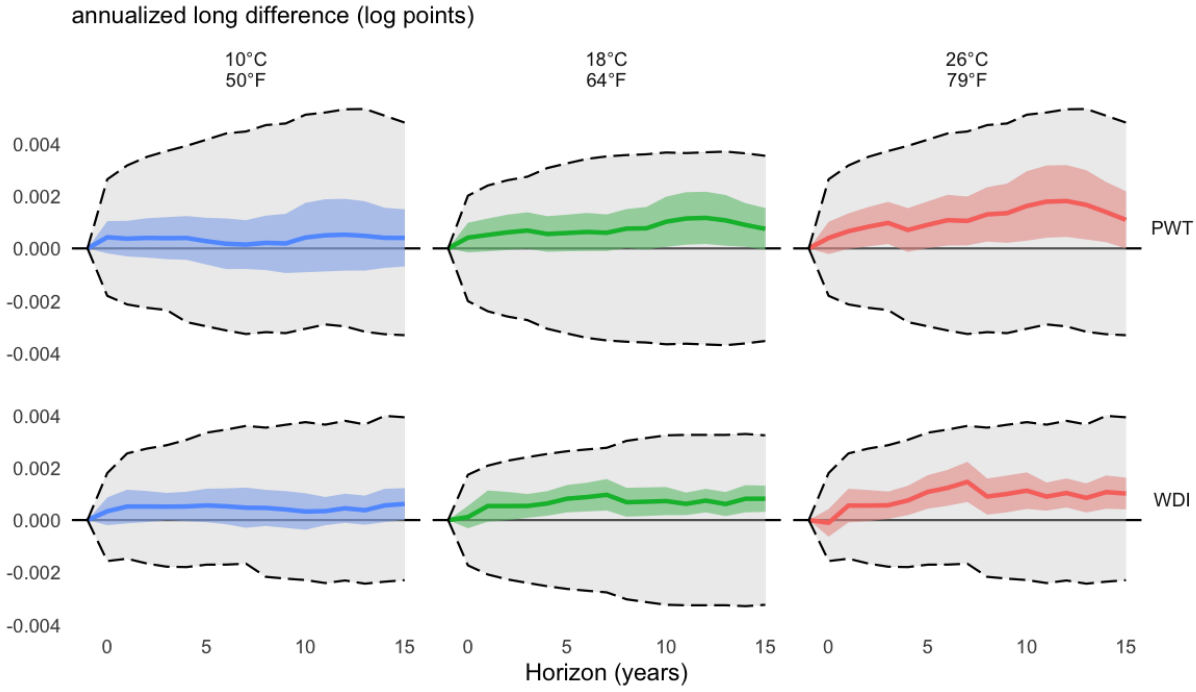
[3]This contrasts with the finding of NRK who measure shock persistence using levels of temperature to infer that temperature shocks remain 10-20% higher for at least nine years following a unit shock.

Figure A.3: Impulse responses to a 1°C shock, including significance bands

(a) GDP impulse response



(b) Placebo test: national population response



dependent case) separate local projections to obtain the corresponding multiplier for each projection horizon h.

Instead, we follow the recommendation to use the one-step method described in Ramey (2016) and applied in Ramey and Zubairy (2018). We adapt the method for our state-dependent setting through the procedure summarized in Algorithm 1.

We can use the resulting coefficients to trace a cumulative multiplier function $\mathcal{R}^c_{\tau\to y}(h;\overline{T})$ analogous to Equation (1.4) but which accounts for the effect of persistence in the shock. Since estimation of the coefficients follows directly from a system of local projection regressions, standard inference is preserved and the significance testing methods outlined in Section A.2.2 conveniently remain available. Figure A.2b demonstrates an application of the method to estimate temperature multipliers of GDP. The estimated multiplier over horizon h is interpreted as the difference in cumulative GDP over the h periods following an isolated 1°C shock relative to a counterfactual where the shock had never occurred.

Estimated cumulative multipliers are found to be substantial at larger horizons but the significance band test cannot reject a null of non-persistence beyond one period. 90% confidence intervals for the effect of an identified temperature shock on cumulative GDP are $[-0.49, 1.1]$ percent for 10°C climates and and $[-2.7, -1.1]$ percent for 26°C climates.

Algorithm 1: One-step state-dependent cumulative multiplier

for $h = 0$ **to** H **do**

Estimate the IRF $\mathcal{R}_{\tau\to\tau}(h)$ as defined in Equation (1.4), collecting state-dependent shock coefficients $\widehat{\alpha}_{1,h}, \widehat{\alpha}_{2,h}$;

Define $\Delta^c_h y_{i,t+h} := \sum_{j=0}^{h} \Delta_j y_{i,t+j}$;

Define $\widehat{\tau}^c_{i,t+h} := \widehat{\tau}_{it}\left[\sum_{j=0}^{h} \widehat{\alpha}_{1,j} + \widehat{\alpha}_{2,j}\overline{T}_{it}\right]$;

Estimate local projection:

$\Delta^c_h y_{i,t+h} = m_{1,h}\widehat{\tau}^c_{i,t+h} + m_{2,h}\widehat{\tau}^c_{i,t+h}\overline{T}_{it} + \lambda_h \overline{T}_{it} + \mathbf{Z}_{it}\boldsymbol{\gamma}_h + \mu_i + \eta_t + \varepsilon_{i,t+h}$;

end

Output: Multiplier coefficients $\widehat{m}_{1,h}, \widehat{m}_{2,h}, \widehat{\lambda}_h$ for $h \in \{0, ...H\}$

A.3 Supplementary results

Figure A.4: Cumulative state-dependent income responses to a 1°C shock within the top decile

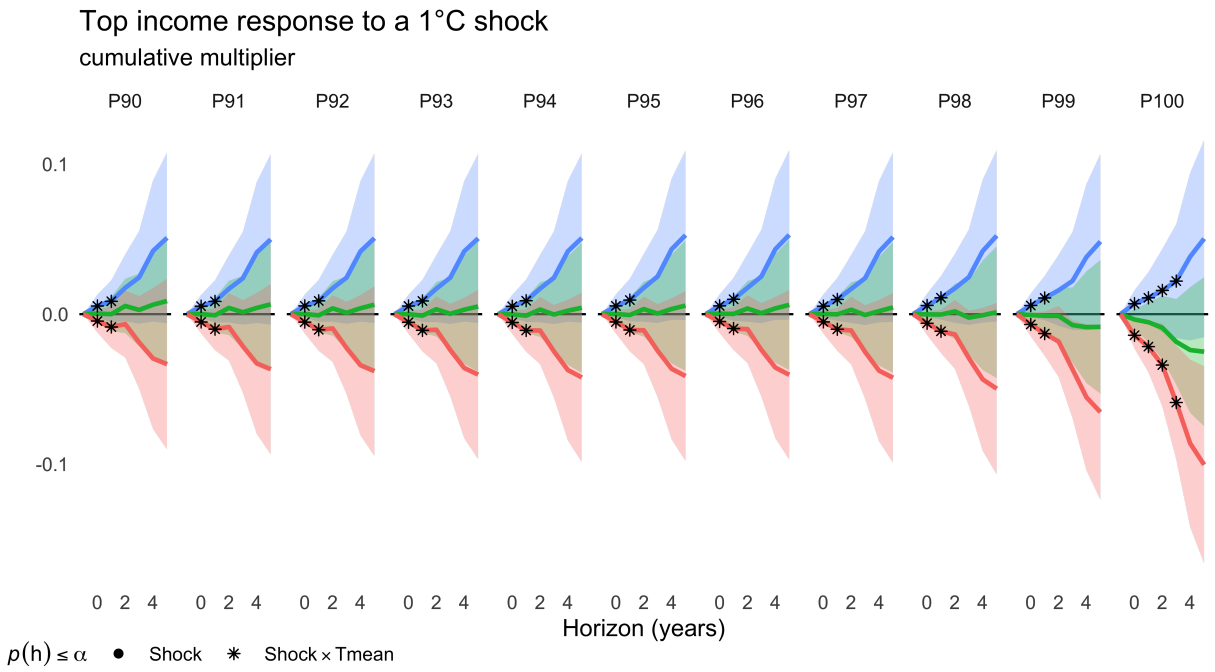
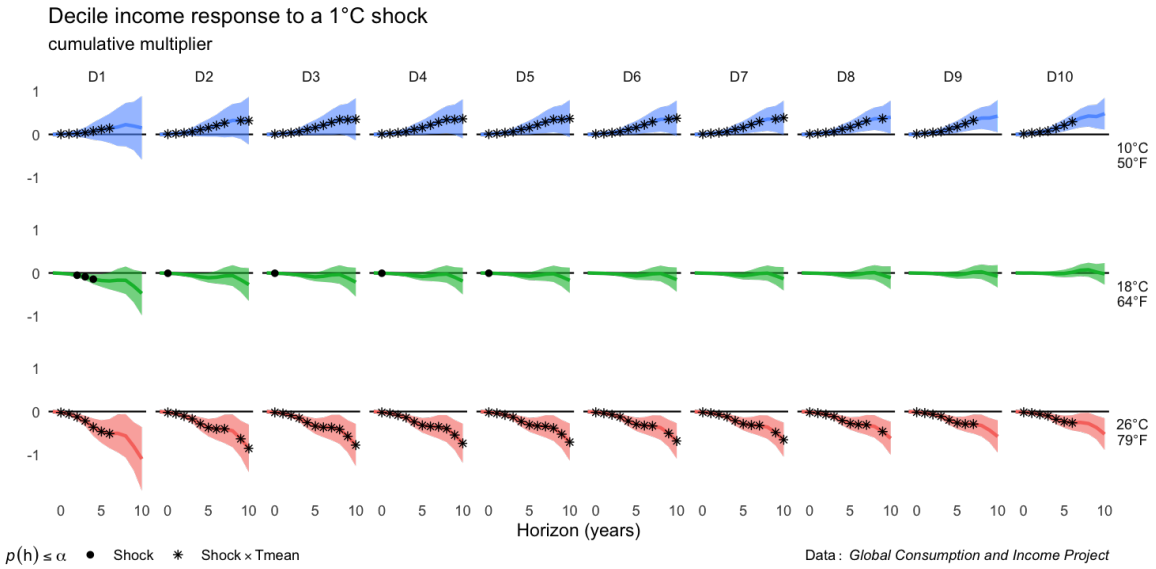
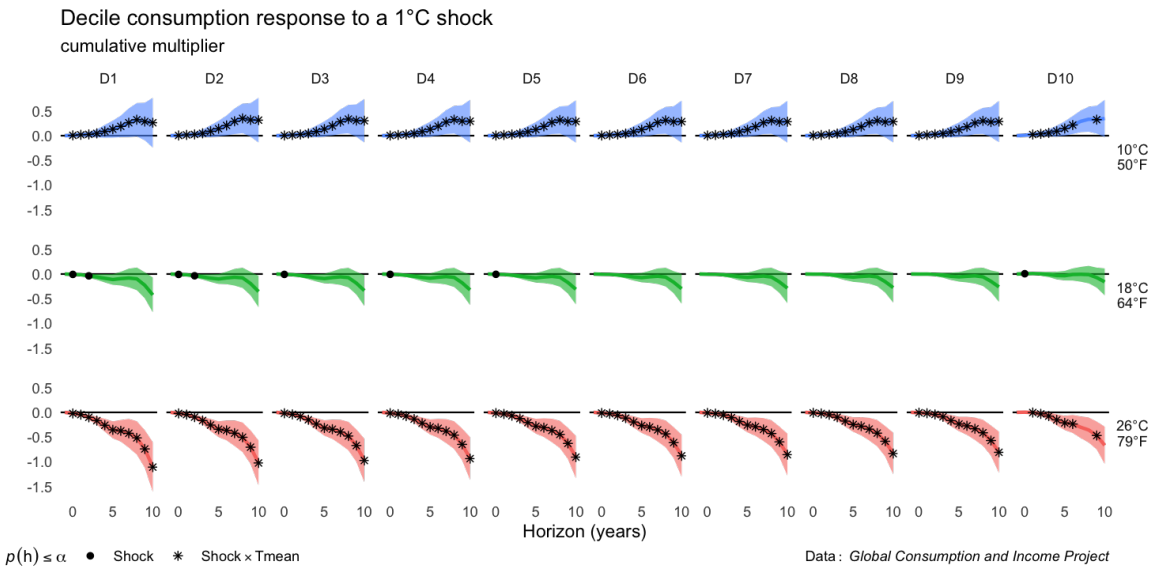


Figure A.5: Temperature multipliers estimated using survey microdata

(a) Impacts on income by decile



(b) Impacts on consumption by decile



A.4 Anthropogenic attribution analysis

To attribute a global inequality effect to historical anthropogenic climate change, we implement a counterfactual analysis which decomposes observed temperature effects into their natural and anthropogenic contributions, adapting methods commonly used in climate attribution science.

A.4.1 Climate simulation data

Climate simulation data comes from the contributions of climate modeling groups to the World Climate Research Program's Coupled Model Intercomparison Project (CMIP). The CMIP, initiated in 1995, is a framework to coordinate, standardize, and disseminate the results of state-of-the-art simulations of the global climate for the benefit of the international climate research community. Data used in this study comes from CMIP6, the sixth and newest[4] generation of models (Eyring et al., 2016). Data is provided as a collection of globally gridded time series summarizing the joint evolution of several hundred climate variables under various pre-specified calibrations. We will primarily be interested in the models' simulations of near-surface air temperature.

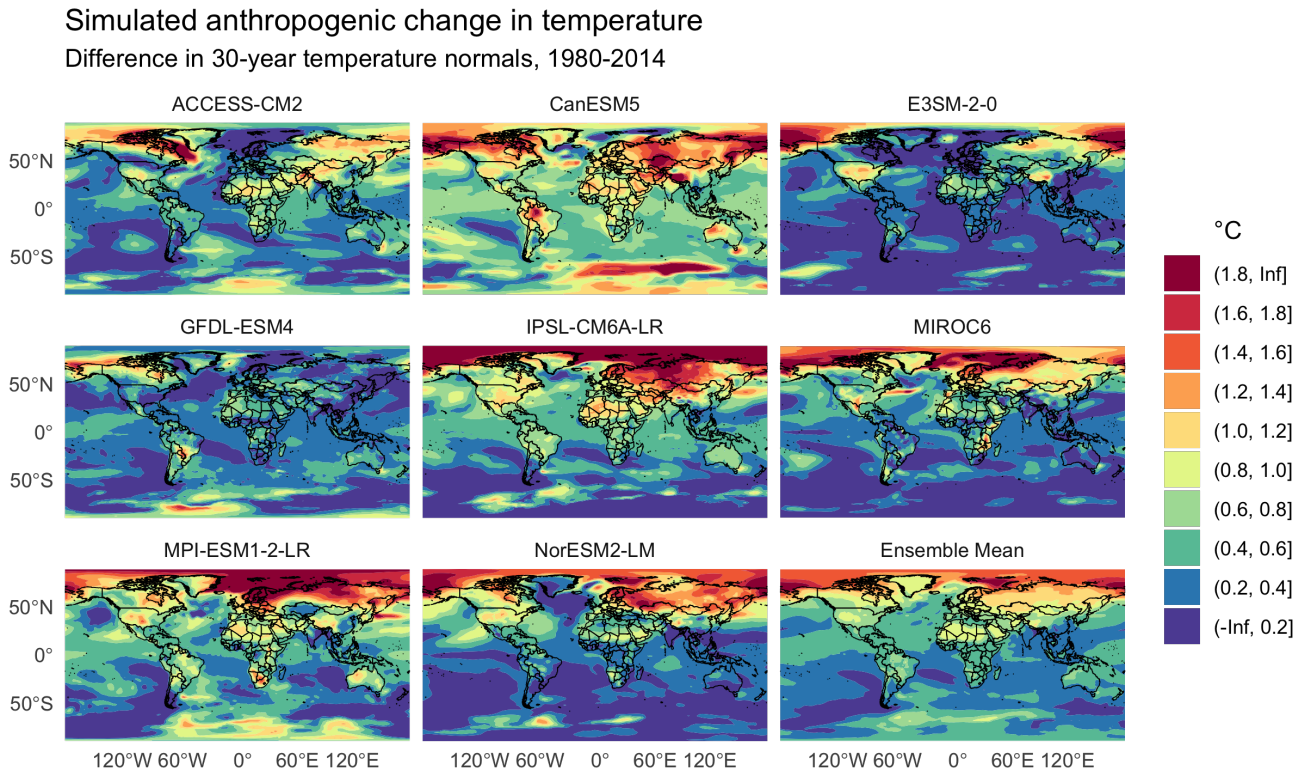
All participating models are "coupled", meaning that they explicitly account for interactions and feedback between distinct components of the global climate system such as the atmosphere, the cryosphere, land surface, and the ocean. Because the manner in which this integration is achieved is idiosyncratic to each model, computationally expensive, and highly sensitive to external calibrations, coordinating the set of "experiments" that participating models run is necessary for comparability and interpretability. For example, one of the four indexes which define an experiment is the realization index which corresponds to the set of geophysical initial conditions at the beginning of the simulation period. By holding the initial conditions fixed, differences in results across models within the same experiment can be more readily attributed to the distinguishing features of the model without being confounded by differences in implementation. Similarly, differences in the same climate model's results across different perturbations to the initial conditions provide a measure of the model's internal variability. The CMIP thus provides a systematic way of decomposing total variation across models and experiments both internal and external to the model. Coordination also enables identification of systematic and idiosyncratic biases in model designs that then inform the next generation of models.

Among the other three indexes which define an experiment, the forcing index is central to our counterfactual analysis. Forcings are factors which affect the net transfer of radiative energy in the climate system, essentially the energy the Earth receives from the sun minus that which it expels back into outer space. Anthropogenic forcings include, for example, industrial greenhouse gas emissions, deforestation, and the transformation of land for arable agriculture. Natural forcings include aerosol expulsions from major volcanic activity and variation in solar irradiance such as those associated with the 11-year solar cycle.[5] Comparing simulations which only differ in their forcings allow one to attribute differences in outcomes to the differences in forcings; we describe next how comparisons of this kind form the basis of both our historical and projection counterfactual analyses.

[4] After a delayed rollout, outputs from most CMIP6 models were made accessible by 2022

[5] Natural forcings are distinct from sources of natural variability such as the ENSO cycle because they are external to coupled climate models.

Figure A.6: Model-implied spatial distributions of global warming, 1980-2014



Simulated spatial distribution of anthropogenic warming for eight randomly selected climate models. The bottom-right plots the average distribution for all 13 CMIP6 models which ran both historical and historical-natural simulations.

A.4.2 Construction of historical counterfactuals

Our historical counterfactual analysis uses simulations from a category of experiments called "historical runs". For the CMIP6 generation of models, historical simulations begin at a pre-industrial baseline period in 1850 and run through to 2014. Among the forcing scenarios used in these historical experiments is one simply labeled "historical", which attempts to represent the impacts of influential forcings actually imposed on the climate system over this period, such as the deforestation of the Amazon rainforest and the eruptions of Krakatoa in 1883 and Mt. Pinatubo in 1991. Results from this historical run are required of all participating CMIP6 models since these "hindcasts" can be directly compared to observational data in order to evaluate model accuracy. Models collectively perform remarkably well at reproducing the historical record (Liu et al., 2021; Zelinka et al., 2020; Zhou et al., 2020). For this reason, studies which use CMIP data for the purpose of counterfactual analysis are advised to include the full available "multi-model ensemble" for experiments of interest (Knutti et al., 2010; Tebaldi & Knutti, 2007; van Vuuren et al., 2006).

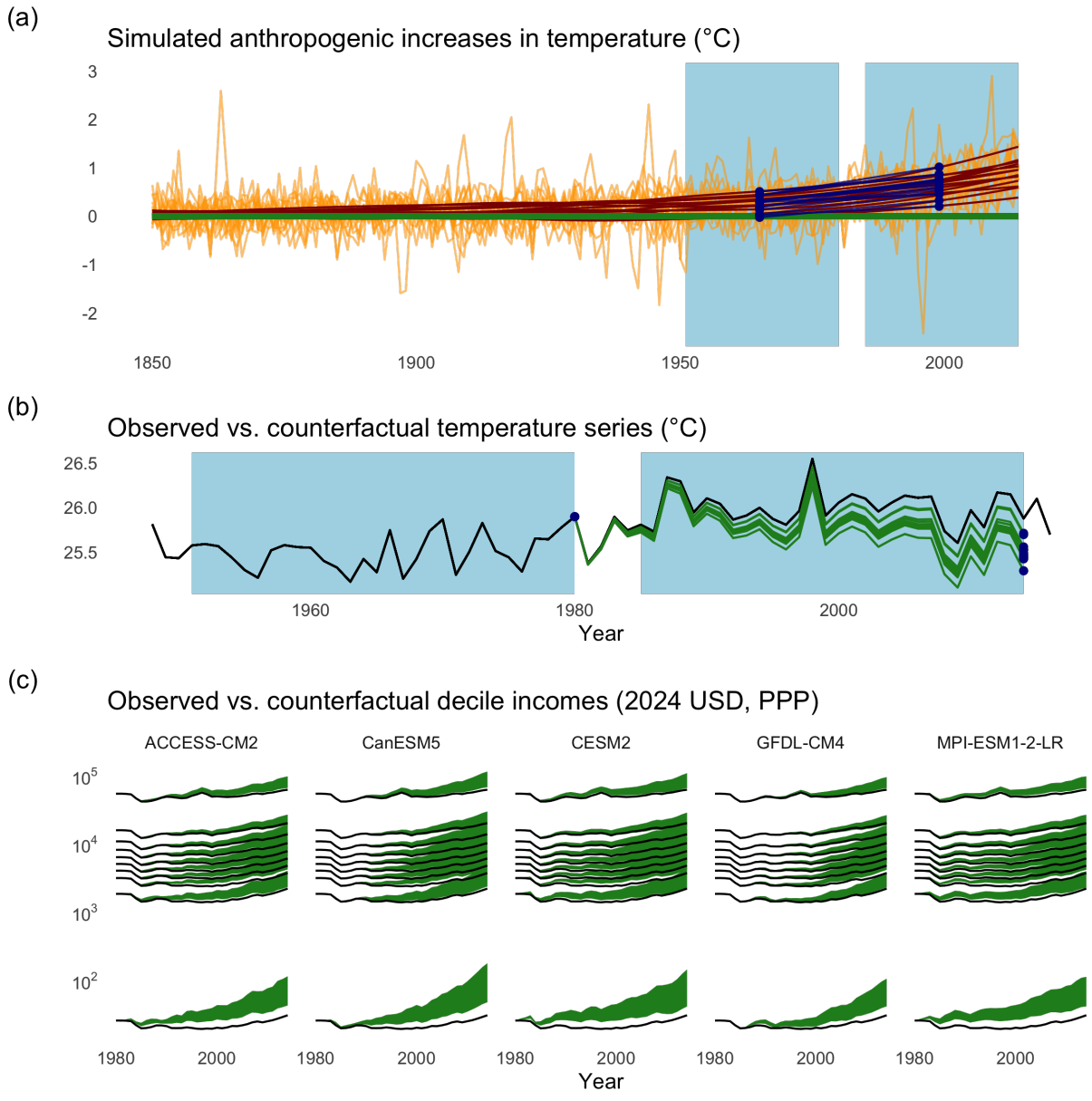
A subset of these models also simulate a set of optional 'historical-natural' experiments which are defined identically except with anthropogenic forcings held fixed at pre-industrial levels. The differences between the two forcings by the same model holding all other variables fixed is then interpreted as the model's simulation of anthropogenic contributions to climate change. Comparisons between historical and historical-natural experiments are commonly used in the subfield of attribution science concerned with quantitatively measuring anthropogenic contributions to the intensity, frequency, or probability of weather events or trends.

For each climate model m and country i, we use an observed series T and the two model-specific historical series $\mathcal{T}^{hist}$ and $\mathcal{T}^{nat}$ to construct a counterfactual series $\widetilde{T}$ where anthropogenic forcings have been held fixed since the first year of the simulation period. The iterative process is summarized in Algorithm 2 and illustrated in Figure A.7 using data from the Philippines as an example. In panel (a), differences between model-specific simulations of national temperature when including all forcings (orange) and when excluding anthropogenic forcings (green horizontal axis). Local regression fits are depicted in red. Light-blue shaded regions correspond to 30-year reference periods used to define temperature normals (dark blue dots) for the start and end of the simulation period 1980-2014.

In panel (b), The linearized difference between normals depicted as dark blue lines in (a) are subtracted from the observed temperature series (black) starting in 1980 to produce counterfactual temperature series (green). Each of these is interpreted as the temperature history that would have been realized had anthropogenic forcings been held fixed at 1980 levels according to a specific model of the climate system.

In panel (c), linearizing the difference between the climate normals defined at either end of the simulation period and subtracting the result from an observed temperature series implements the standard "delta method" for constructing bias-corrected climate counterfactuals. Counterfactual economic series for a random selection of five climate models are depicted in green and are constructed by iteratively applying the decile-level dynamic income responses depicted in Figure 1.4 to the observed and counterfactual temperatures depicted in (b). Their difference is interpreted as the change in income attributable to anthropogenic forcings since 1980. Green bands correspond to 90% confidence intervals generated from 500 bootstrap estimates of the response functions.

Figure A.7: Constructing historical counterfactuals holding anthropogenic forcings fixed at 1980 levels



Demonstration of the delta method for constructing bias-corrected counterfactuals using data for the Philippines as an example.

Algorithm 2: Constructing temperature counterfactuals holding anthropogenic forcings fixed at 1980 levels

Input: $\{\mathcal{T}^{hist}_{m,i,t}\}$,$\{\mathcal{T}^{nat}_{m,i,t}\}$,$\{T_{it}\}$

for $t^* \in \{1980, 2014\}$ **do**

$\overline{\mathcal{T}}^{hist}_{m,i,t^*} \leftarrow \frac{1}{30}\sum_{j=1}^{30} \mathcal{T}^{hist}_{m,i,t^*-j}$;

$\overline{\mathcal{T}}^{nat}_{m,i,t^*} \leftarrow \frac{1}{30}\sum_{j=1}^{30} \mathcal{T}^{nat}_{m,i,t^*-j}$;

$\overline{\delta}_{m,i,t^*} \leftarrow \overline{\mathcal{T}}^{hist}_{m,i,t^*} - \overline{\mathcal{T}}^{nat}_{m,i,t^*}$;

end

for $t = 1980$ **to** 2014 **do**

$\overline{\delta}_{m,i,t} \leftarrow (t - t_0)\frac{\overline{\delta}_{m,i,2014} - \overline{\delta}_{m,i,1980}}{2014-1980}$;

$\widetilde{T}^{nat}_{m,i,t} \leftarrow T_{it} - \overline{\delta}_{m,i,t}$;

end

Construct states $\{\overline{T}^{nat}_{m,i,t}\}$ by applying Equation (1.1a) to $\{\widetilde{T}^{nat}_{m,i,t}\}$ then subtracting by $\widetilde{T}^{nat}_{m,i,1980}$;

Construct shocks $\{\widehat{\tau}^{nat}_{m,i,t}\}$ by applying Equation (1.1a) and Equation (1.1b) to $\{\mathcal{T}^{nat}_{m,i,t}\}$;

Output: $\{\widetilde{T}^{nat}_{m,i,t}\}$,$\{\overline{T}^{nat}_{m,i,t}\}$,$\{\widehat{\tau}^{nat}_{m,i,t}\}$

Algorithm 3: Constructing economic counterfactuals holding anthropogenic forcings fixed at 1980 levels

Input: $\{\widetilde{T}^{nat}_{m,i,t}\}$,$\{\overline{T}^{nat}_{m,i,t}\}$,$\{\widehat{\tau}^{nat}_{m,i,t}\}$, B bootstrap samples indexed b

Initializing series:

for $b = 1$ **to** B **do**

Specify desired simulation interval: $(t_0, t_1) \leftarrow (1980, 2014)$;

Estimate $\mathcal{R}^{c}_{\tau \rightarrow Y}(h; \overline{T}, b)$ from Equation (1.5);

Specify maximum projection horizon H using persistence test from Section A.2.2;

Define IRFs $\mathcal{R}_{\tau \rightarrow Y}(h; \overline{T}, b) := \mathcal{R}^{c}_{\tau \rightarrow Y}(h; \overline{T}, b) - \mathcal{R}^{c}_{\tau \rightarrow Y}(h-1; \overline{T}, b)$;

Collect IRF coefficients;

$\{\widehat{\beta}_{1,h,b}, \widehat{\beta}_{2,h,b}, \widehat{\lambda}_{h,b}\}_{h=0}^{H}$;

Define $\delta^{\tau}_{m,b,i,t} := \widehat{\tau}^{nat}_{m,i,t} - \widehat{\tau}_{it}$;

Define $\delta^{\tau T}_{m,b,i,t} := \widehat{\tau}^{nat}_{m,i,t} \cdot \overline{T}^{nat} - \widehat{\tau}_{it} \cdot \overline{T}_{it}$;

Define $\delta^{T}_{m,b,i,t} := \overline{T}^{nat}_{m,i,t} - \overline{T}_{it}$;

Initialize $\delta^{Y}_{m,b,i,t} \leftarrow 0$;

Initialize $\widetilde{Y}^{nat}_{m,b,i,t_0} := Y_{i,t_0}$

end

Iteratively constructing series:

for $t = t_0 + 1$ **to** t_1 **do**

for $b = 1$ **to** B **do**

for $h = 0$ **to** H **do**

$\delta^{Y}_{m,b,i,t+h} \leftarrow \delta^{Y}_{m,b,i,t+h} + \widehat{\beta}_{1,h,b} \cdot \delta^{\tau}_{m,b,i,t} + \widehat{\beta}_{2,h,b} \cdot \delta^{\tau T}_{m,b,i,t} + \widehat{\lambda}_{h,b} \cdot \delta^{T}_{m,b,i,t}$;

end

$\widetilde{Y}^{nat}_{m,b,i,t} \leftarrow \left(1 + \log(\frac{Y_{it}}{Y_{i,t-1}}) + \delta^{Y}_{m,b,i,t}\right) \widetilde{Y}^{nat}_{m,b,i,t-1}$

end

end

Output: $\widetilde{N}^{nat}_{m,b,i,t}$

Figure A.8: Anthropogenic growth incidence curves by climate model, 1980-2014

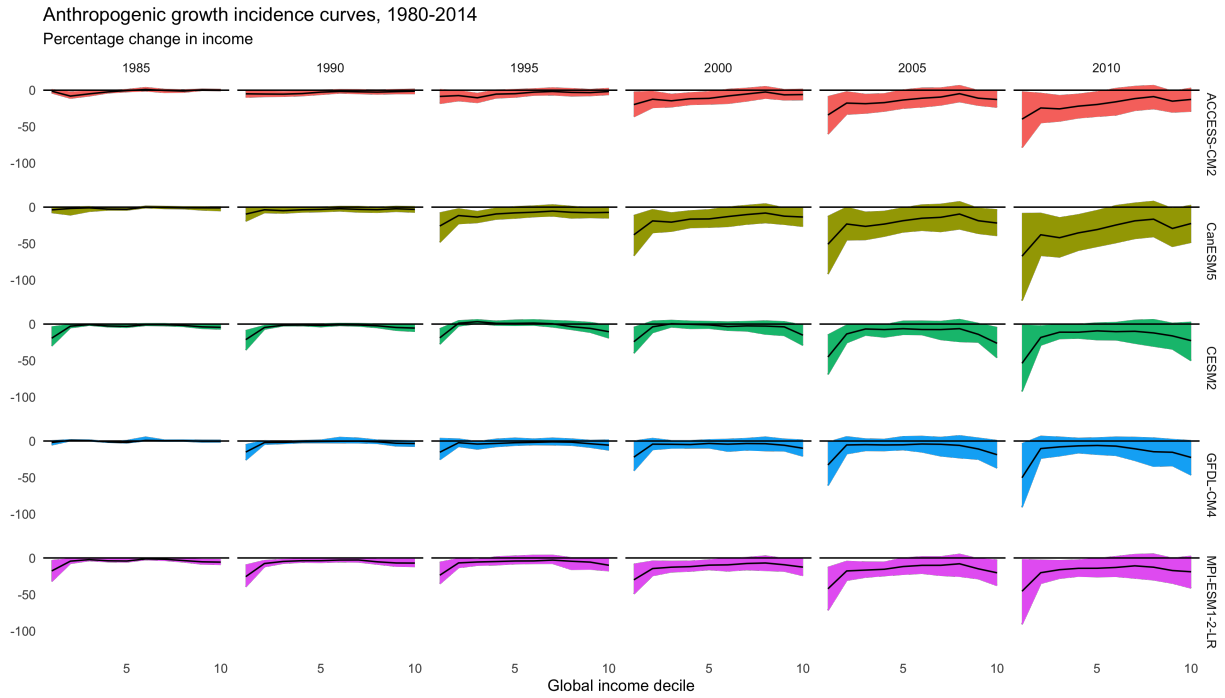
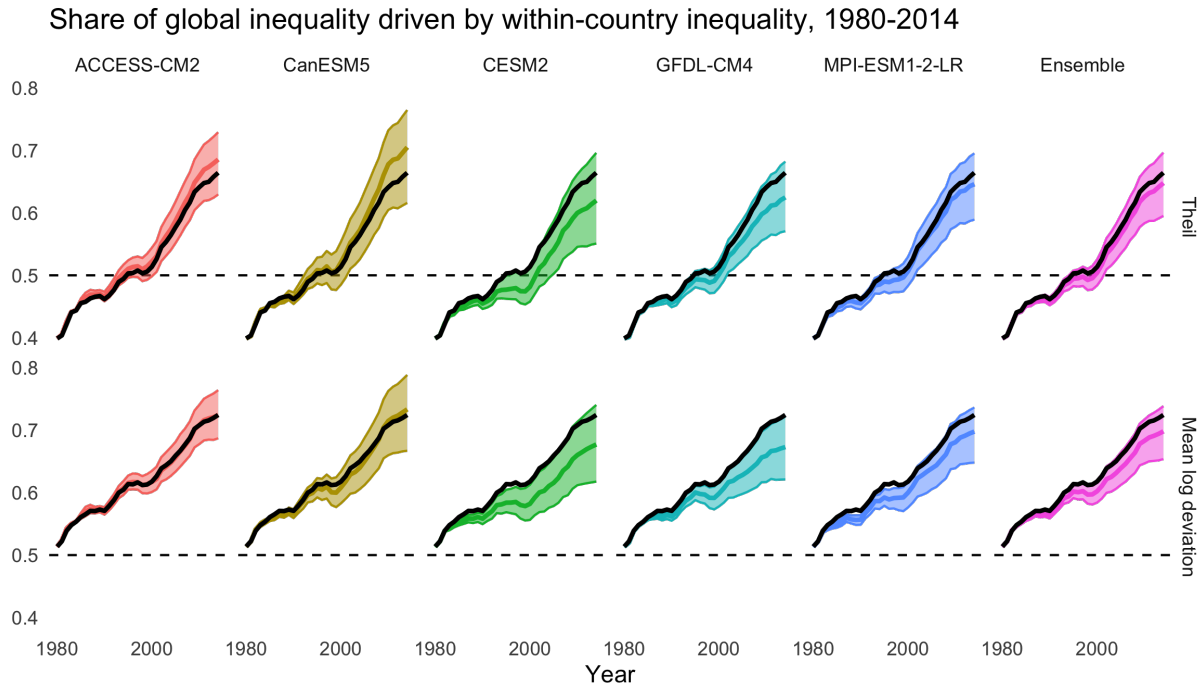


Figure A.9: Anthropogenic influence on within- vs. between-country inequality by climate model



Appendix B: Supplement to Chapter 2

B.1 Data descriptives

Figure B.1: Distribution of democracy scores

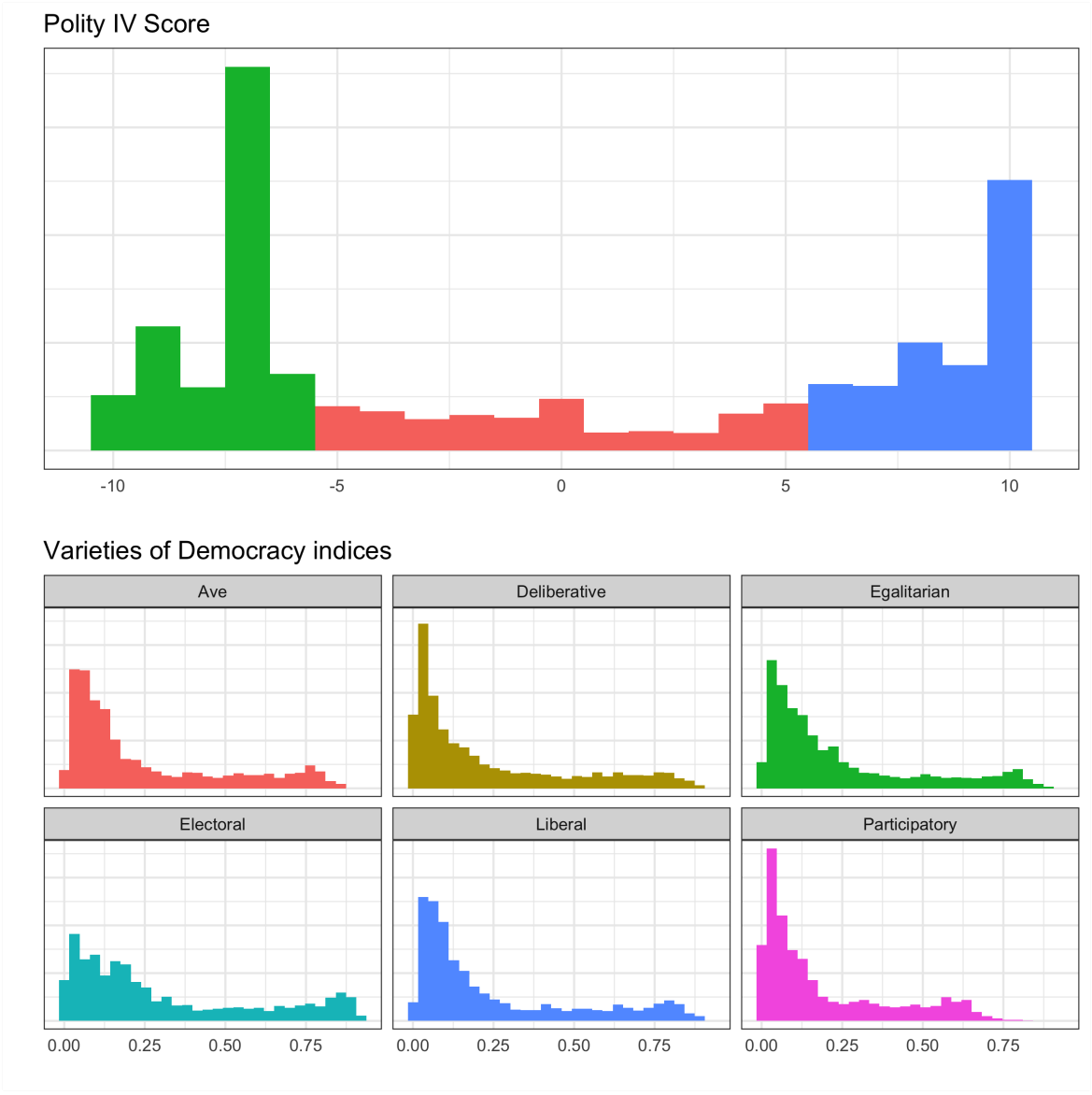


Table B.1: Correlations between democracy indices after de-meaning by country

	Polity	V-Ave	V-Elec	V-Lib	V-Part	V-Delib	V-Egal
Polity	1.000	0.707	0.679	0.706	0.698	0.699	0.697
VDem-Average	0.707	1.000	0.987	0.990	0.988	0.990	0.979
VDem-Electoral	0.679	0.987	1.000	0.973	0.971	0.968	0.952
VDem-Liberal	0.706	0.990	0.973	1.000	0.973	0.979	0.959
VDem-Participatory	0.698	0.988	0.971	0.973	1.000	0.972	0.964
VDem-Deliberative	0.699	0.990	0.968	0.979	0.972	1.000	0.962
VDemx -Egalitarian	0.697	0.979	0.952	0.959	0.964	0.962	1.000

Table B.2: Summary of different binary democracy indicators

Marshall and Gurr, 2020 (Polity IV)

- Country-year regimes scored between -10 and +10 based on competitiveness of political participation and constraints on executive authority
- A country-year is labeled a democracy if scoring at least +5, autocracy if scoring at most -5, anocracy otherwise

Freedom House, 2022 (FH)

- Country-year regimes classified as 'free', 'partially free', or 'not free' based on assessment of political rights and civil liberties
- A country-year is labeled a democracy if free or partially free

---

Cheibub et al., 2010 (CGV)

- Country-year regimes deemed 'democratic' if determined to satisfy four conditions for competitive elections
    1. Chief executive chosen by popular election or elected body
    2. Legislature is popularly elected
    3. More than one party competes in elections
    4. Alternation in power under identical electoral rules

Boix et al., 2012 (BMR)

- Country-year regimes are deemed democratic if it is determined that
    1. Political leaders are chosen through free and fair elections
    2. The electorate satisfies a threshold value of suffrage
- Missing values are assigned to observations during international war or state collapse, unlike other measures that retain these cases
- Long time frame (1800-2007) but in the dataset, only 48 years of data used, the second lowest after FH

Papaioannou and Siourounis, 2008 (PS)

- Focuses only on permanent transitions to democracy, no democratic breakdowns or transitory transitions
- ANRR suggests susceptibility to endogeneity in growth regressions

Acemoglu et al., 2019 (ANRR)

- A country-year regime is deemed 'free' if either:
    1. FH labels it free or partially free and Polity IV assigns a positive score
    2. If FH or Polity IV data is unavailable, CGV or BMR label it as democratic

B.2 Supplementary results

Table B.3: Impact of additional 1°C shock on support for democracy by survey question

Outcome:	Important to live in	Better than alternatives	Index
Model:	(1)	(2)	(3)
Variables			
$\text{Shock}^{(3)}$	-0.0511	-0.0935	-0.1152·
	(0.1639)	(0.0913)	(0.0585)
$\text{Shock}^{(3)} \times \overline{T}$	-0.0249	-0.0091	-0.0168·
	(0.0228)	(0.0080)	(0.0090)
$\overline{T}$	-2.070·	0.2943	-0.6651·
	(0.9996)	(0.5145)	(0.3382)
Gender	Yes	Yes	Yes
Precip	Yes	Yes	Yes
Fixed effects, # unique values			
Country	76	76	75
Year	11	13	11
Age	85	87	84
Birth year	92	93	92
Survey wave	4	4	4
Observations	160,486	189,456	150,292
Outcome mean	8.4417	3.2526	0.02250
R^2	0.06307	0.09477	0.08924

Clustered (Country & Year) standard-errors in parentheses
*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, .: 0.1*

Table B.4: Impact of additional 1°C shock on skepticism of democracy by survey question

Outcome:	Causes disorder	Too bureaucratic	Bad for economy	Index
Model:	(1)	(2)	(3)	(4)
Variables				
$\text{Shock}^{(3)}$	0.0803	0.2320***	0.0030	0.1235
	(0.0801)	(0.0423)	(0.0040)	(0.0815)
$\text{Shock}^{(3)} \times \overline{T}$	0.0107	0.0268***	0.0059**	0.0169*
	(0.0071)	(0.0027)	(0.0016)	(0.0072)
$\overline{T}$	-0.1319	-0.3056	-0.1672	-0.2732
	(0.3052)	(0.4138)	(0.3569)	(0.3732)
Gender	Yes	Yes	Yes	Yes
Precip	Yes	Yes	Yes	Yes
Fixed effects, # unique values				
Country	77	76	76	76
Year	13	13	13	13
Age	87	87	87	87
Birth year	95	95	95	93
Survey wave	4	4	4	4
Observations	185,838	185,402	181,414	170,798
Outcome mean	2.2753	2.5324	2.2669	-0.00387
R^2	0.08693	0.08300	0.07527	0.10248

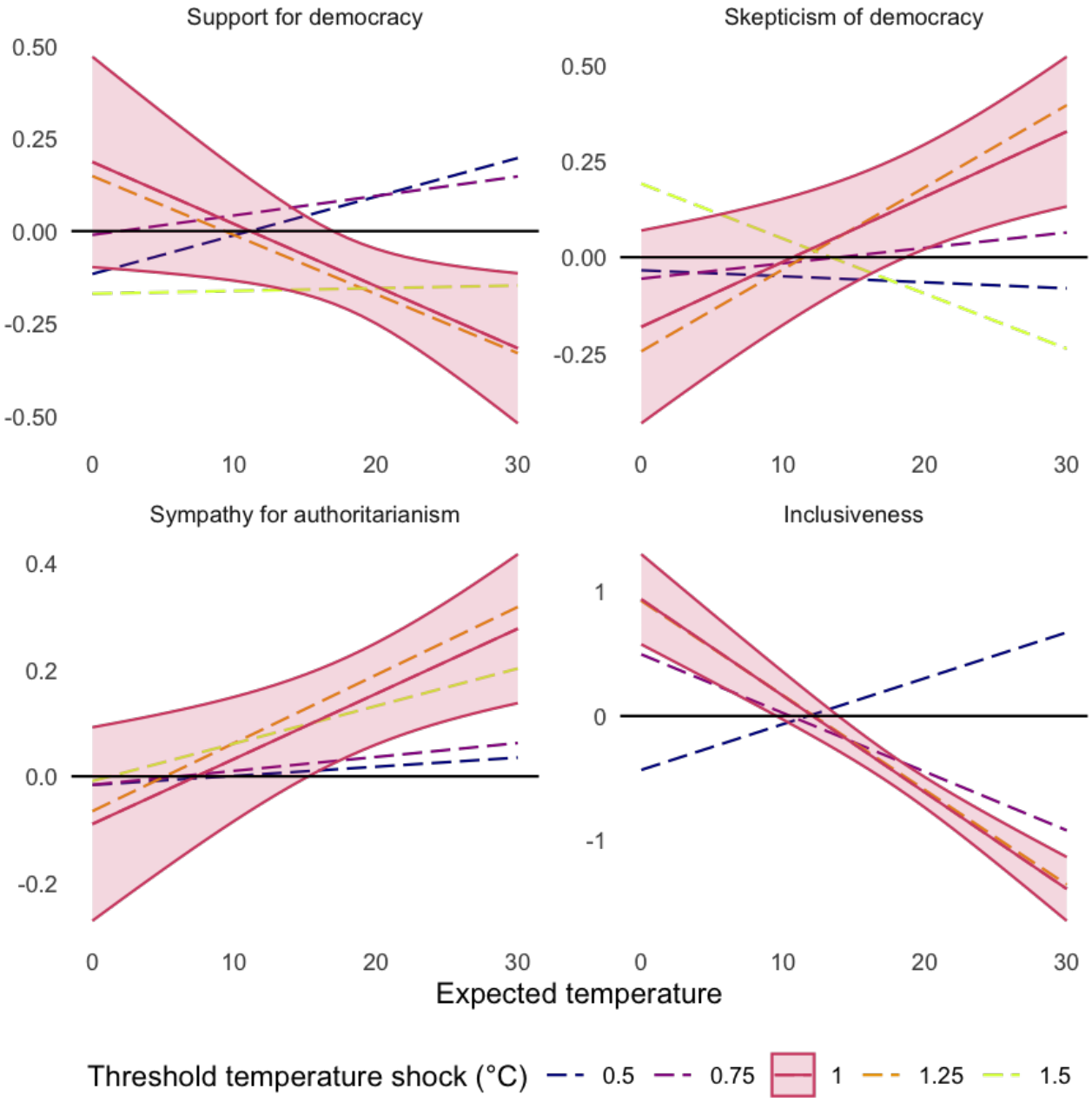
Clustered (Country & Year) standard-errors in parentheses
*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, .: 0.1*

Table B.5: Impact of additional 1°C shock on preference for authoritarian features by survey question

Outcome:	Strongman	Army rule	Experts over government	Index
Model:	(1)	(2)	(3)	(4)
Variables				
$\text{Shock}^{(3)}$	0.4612*	0.2957*	0.1370	0.1303*
	(0.1628)	(0.1246)	(0.0831)	(0.0568)
$\text{Shock}^{(3)} \times \overline{T}$	0.0462**	0.0272*	0.0146·	0.0122*
	(0.0157)	(0.0111)	(0.0082)	(0.0052)
$\overline{T}$	0.4493*	-0.1414	0.1461	0.0832
	(0.1754)	(0.1650)	(0.1719)	(0.1460)
Gender	Yes	Yes	Yes	Yes
Precip	Yes	Yes	Yes	Yes
Fixed effects, # unique values				
Country	98	98	98	98
Year	21	21	21	21
Age	87	87	87	87
Birth year	101	101	101	101
Survey wave	8	8	8	8
Observations	351,770	349,842	342,351	317,655
Outcome mean	2.2227	1.7531	2.6405	-0.03017
R^2	0.16859	0.21701	0.10108	0.18527

Clustered (Country & Year) standard-errors in parentheses
*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, .: 0.1*

Figure B.2: Effect on political preferences of an additional temperature shock in last three years



Analogous to Figure 2.2 but instead of varying the number of years over which shocks are enumerated, we vary the minimum magnitude to qualify as a temperature shock.

Table B.6: Impact of additional 1°C shock on inclusiveness

Outcome:	Free elections	Civil rights	Gender equality	Tax the rich	Income equality	Unemployment aid	Index
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Variables							
$Shock^{(3)}$	-0.0498	-0.4239$^{\cdot}$	0.2250	-0.1899	19.38*	-0.0151	-0.4582**
	(0.1486)	(0.2244)	(0.1389)	(0.2499)	(5.119)	(0.1883)	(0.0704)
$Shock^{(3)} \times \overline{T}$	-0.0188	-0.0641**	0.0085	-0.0657$^{\cdot}$	2.885*	-0.0400$^{\cdot}$	-0.0775**
	(0.0215)	(0.0188)	(0.0160)	(0.0295)	(0.8057)	(0.0214)	(0.0116)
$\overline{T}$	-2.325$^{\cdot}$	-2.041	-0.1941	-1.115	0.7127	-1.702	-0.3258*
	(1.190)	(1.215)	(0.8698)	(1.078)	(1.789)	(1.323)	(0.0970)
Gender	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Precip	Yes	Yes	Yes	Yes	Yes	Yes	Yes
effects Country	75	75	75	75	59	75	58
Year	11	11	11	11	6	11	6
Age	85	85	85	85	83	85	83
Birth year	92	92	92	92	87	92	86
Survey wave	4	4	4	4	3	4	3
Observations	157,769	153,650	158,623	156,230	85,972	157,150	77,392
Outcome mean	8.2808	7.6900	8.2046	6.4546	5.9951	7.1533	-0.05107
R^2	0.07563	0.08909	0.10281	0.10524	0.15725	0.10367	0.17156

Clustered (Country & Year) standard-errors in parentheses
*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, .: 0.1*

Appendix C: Supplement to Chapter 3

C.1 Data

C.1.1 DHS survey questions related to intimate-partner violence

1. **Attitudes towards domestic violence**: under the following hypothetical situations, the respondents (both the subsample of women and the whole sample of men) are asked whether a husband would be justified in responding violently to his wife

    - *dv.justify.wifegoesout*: If the wife goes out without permission
    - *dv.justify.wifeneglectchildren*: If the wife neglects their children
    - *dv.justify.wifeargues*: If the wife argues with her husband
    - *dv.justify.wifenocook*: If the wife has refused sex
    - *dv.justify.cheat*: If the wife has not cooked
    - *dv.justify.cheat*: If the wife was unfaithful
    - *dv.justify.disrespect*: If the wife was disrespectful
    - *dv.father.beat.mother*: Did your father ever beat your mother?

2. **Incidence of domestic violence**: has your husband ever performed the following actions to you (asked only of the domestic-violence subset of female respondents)

    - *dv.pushed*: pushed
    - *dv.slapped*: slapped
    - *dv.punched*: punched
    - *dv.kicked*: kicked
    - *dv.strangled*: strangled
    - *dv.weapon*: used a weapon
    - *dv.less*: performed any of the "less" severe actions enumerated above
    - *dv.severe*: performed any of the "more" severe actions enumerated above
    - *dv.rape*: forced you to perform an unwanted sexual act
    - *dv.rape*: how many years into your marriage did your husband first beat you, if ever

3. **Spousal controlling behavior**: does your husband exhibit the following behavior

    - *control.jealous*: jealousy when you talk to other men
    - *control.accuse*: accuse you of unfaithfulness
    - *control.meetfriends*: require his permission for you to meet your friends

- *control.nofamily*: limit your contact with your family
- *control.where*: demand to know where you are at all times
- *control.nomoney*: not trust you with household spending

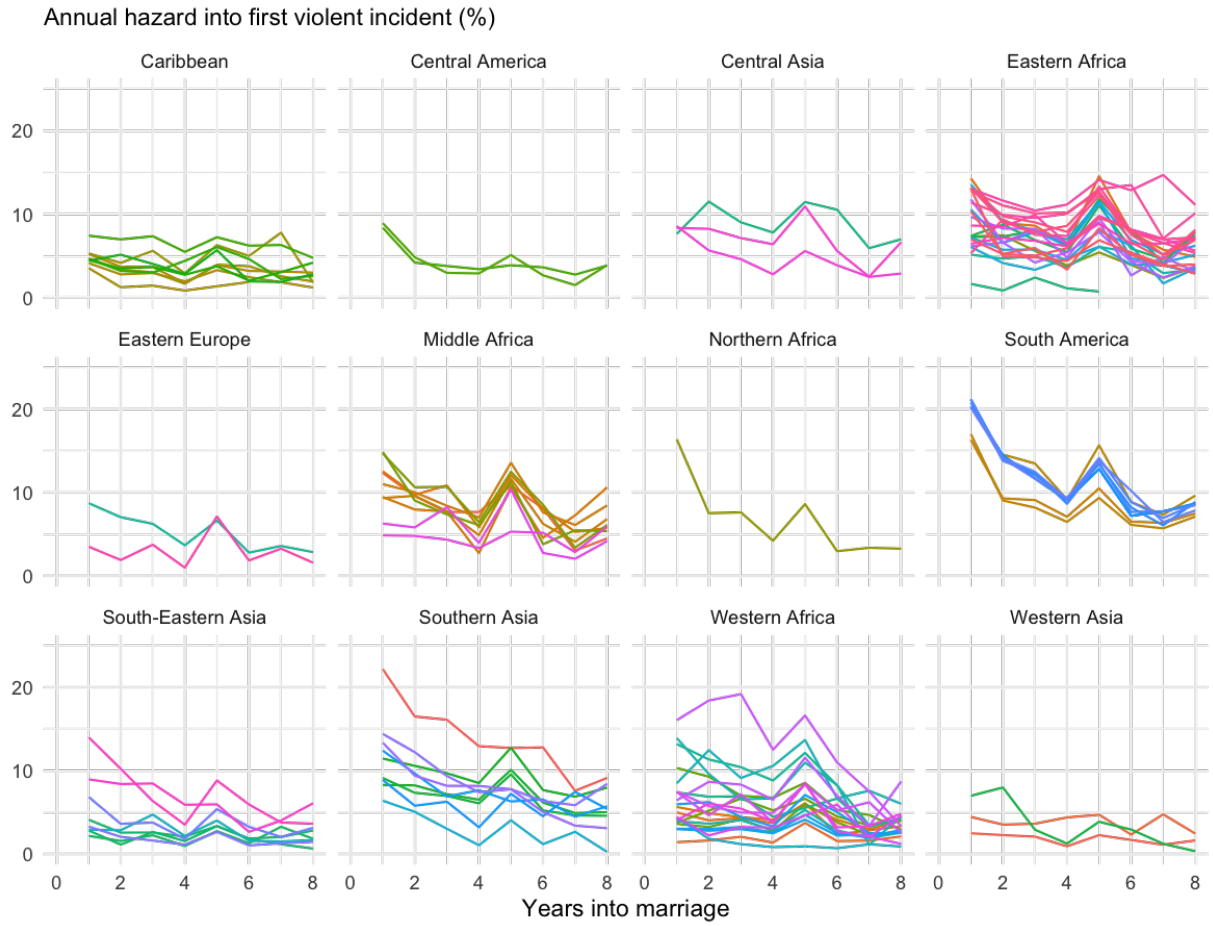
4. **Refusal to have sex**: under the following hypothetical situations, is the wife justified in refusing sex with the husband (asked of both female and male respondents)

    - *attitude.wiferefusesex.sti*: if it is known the husband has a sexually transmitted infection
    - *attitude.wiferefusesex.cheat*: if it is known the husband has been unfaithful
    - *attitude.wiferefusesex.mood*: if the wife is not in the mood

5. **Intrahousehold bargaining**: does the wife have influence in the following household decisions? (asked of both female and male respondents)

    - *bargaining.healthcare*: healthcare spending
    - *bargaining.largepurchase*: large household purchases
    - *bargaining.visitrelatives*: whether the household can visit her family
    - *bargaining.money*: to spend the household's earned income

Figure C.1: Annual hazard into first-time violence



C.2 Correlations between parental violence and attitudes towards domestic violence

While again cautioning that these correlations should not be interpreted causally, we find unsurprising but instructive positive associations between this parental history of domestic violence and predispositions to justify spousal violence under hypothetical scenarios (Tables 1 and 2). Female respondents who reported a history of parental spousal violence were around 10 percentage points more likely to say domestic violence was justifiable under at least one of the presented scenarios, while this figure is substantially larger—17-18 percentage points—for men. Compared to sample averages, these represent a 20% increase for women and a 44-47% increase for men.[1] This is suggestive of the possibility that cultural norms are differently persistent for the different genders and often substantially so.

These gendered differences in persistence also emerge when looking at attitudes towards intrahousehold bargaining power as seen by comparing results in Tables **??** and **??**. It is a feature of both the female and male respondents that those whose fathers beat their mothers are less likely to report wives having influence over healthcare spending and large purchases, but the healthcare difference is insignificant for women and relatively muted (-1.8 percentage points) for women when it comes to large purchases compared to -6.0 percentage points for men.

Across other outcomes, we find in Table **??** that attitudes towards spousal control are also strongly predicted by whether a respondent's father was spousally violently abusive. This module was only asked of female respondents so no gendered differences can be measured. Table **??** and **??** find null relationships between acceptability of reasons for a wife to refuse sex and reported parental violence except in counts of questions answered affirmatively. This could be indicative of poor persistence of this norm, but I think what is more likely to be the case, incidence of domestic violence is an imperfect proxy for justifying refusal of sex. An ideal intergenerational persistence will compare much more like events to one another and the sparsity of parental gender roles is a limitation of the DHS dataset compared to, say the Rural Economic & Demographic Surveys, a point we return to in Section 1.5.

We can however relate incidence of parental spousal violence to respondent experience with spousal violence as we do in Table 3. These comparisons of means finds that whether a respondent's father had ever beat their mothers is highly predictive of whether they had themselves experienced any of the suggested acts of violence, from being pushed, slapped, or punched, to being threatened with a weapon or having unwanted sex with percentage-point increases ranging between 7.8 to 27.0. Given the already low prevalence of these acts reported on average in the summary statistics in Table **??**, these represent percentage differences of over 100% dependng on the act. The suggestion here is unsurprising but stark: proclivity for spousal violence is highly persistent across generations.

I here flag, however, that this remains somewhat imperfect as a measure of cultural persistence for a couple of reasons. First, the questions asked of the respondents' first-hand experience with spousal violence is not the simple binary question asked of their parents. Beyond the fact respondents are likely differently willing to admit to parental spousal abuse versus their own experience with spousal abuse, the parent question asks unspecifically whether the respondent's father had ever beat his wife or partner while the respondent questions are more specific, asking about whether they

[1] It is separately interesting to note that baseline proclivities to justify domestic violence are higher on average for women (0.499) than men (0.384). In fact, women are more likely than men to argue that domestic violence is a justified response under every given scenarios, even when applying survey weights to take weighted averages. I don't yet have a working explanation for this other than speculative amateur psychological ones.

Table C.1: Attitudes towards domestic violence: women

Dependent Variables:	dv.justify.any		dv.justify.count		dv.justify.wifegoesout		dv.justify.wifeneglectchildren	
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Variables								
dv.father.beat.motherTRUE	0.0979***	0.1043***	0.3899***	0.4079***	0.0628***	0.0628***	0.0741***	0.0783***
	(0.0098)	(0.0104)	(0.0464)	(0.0481)	(0.0094)	(0.0096)	(0.0107)	(0.0103)
low.casteTRUE		0.0600***		0.3241***		0.0434***		0.0506***
		(0.0105)		(0.0453)		(0.0082)		(0.0098)
hinduTRUE		0.0016		-0.0287		-0.0117		-0.0040
		(0.0116)		(0.0522)		(0.0093)		(0.0097)
Fixed-effects								
region	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics								
Observations	60,610	57,920	60,610	57,920	60,610	57,920	60,610	57,920
R^2	0.14672	0.14174	0.14021	0.13422	0.12922	0.12039	0.16983	0.16475
Within R^2	0.00679	0.00822	0.00516	0.00768	0.00343	0.00501	0.00442	0.00569

Dependent Variables:	dv.justify.wifeargues		dv.justify.wiferefusesex		dv.justify.wifenocook		dv.justify.cheat		dv.justify.disrespect	
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Variables										
dv.father.beat.motherTRUE	0.0650***	0.0710***	0.0197***	0.0241***	0.0356***	0.0421***	0.0496***	0.0451***	0.0831***	0.0844***
	(0.0096)	(0.0100)	(0.0072)	(0.0079)	(0.0078)	(0.0083)	(0.0083)	(0.0091)	(0.0094)	(0.0098)
low.casteTRUE		0.0584***		0.0301***		0.0439***		0.0491***		0.0486***
		(0.0098)		(0.0067)		(0.0077)		(0.0086)		(0.0093)
hinduTRUE		-0.0120		-0.0133		0.0035		0.0057		0.0031
		(0.0095)		(0.0084)		(0.0083)		(0.0113)		(0.0127)
Fixed-effects										
region	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics										
Observations	60,610	57,920	60,610	57,920	60,610	57,920	60,610	57,920	60,610	57,920
R^2	0.10199	0.09690	0.05556	0.04833	0.07108	0.06655	0.07488	0.06726	0.11131	0.10480
Within R^2	0.00333	0.00502	0.00051	0.00132	0.00130	0.00287	0.00216	0.00361	0.00493	0.00625

One-way (region) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table C.2: Attitudes towards domestic violence: men

Dependent Variables:	dv.justify.any		dv.justify.count		dv.justify.wifegoesout		dv.justify.wifeneglectchildren	
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Variables								
dv.father.beat.motherTRUE	0.1691***	0.1818***	0.6757***	0.7086***	0.0888***	0.1025***	0.1103***	0.1184***
	(0.0097)	(0.0119)	(0.0411)	(0.0467)	(0.0086)	(0.0098)	(0.0093)	(0.0110)
low.casteTRUE		0.0478***		0.1815***		0.0224***		0.0317***
		(0.0107)		(0.0383)		(0.0073)		(0.0082)
hinduTRUE		-0.0097		-0.0520		-0.0152*		-0.0180*
		(0.0142)		(0.0451)		(0.0088)		(0.0107)
Fixed-effects								
region	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics								
Observations	66,445	63,102	66,445	63,102	66,445	63,102	66,445	63,102
R^2	0.16082	0.15079	0.15977	0.15446	0.10051	0.09475	0.14329	0.13563
Within R^2	0.01918	0.02182	0.02128	0.02472	0.00953	0.01054	0.01276	0.01505

Dependent Variables:	dv.justify.wifeargues		dv.justify.wiferefusesex		dv.justify.wifenocook		dv.justify.wifecheat		dv.justify.wifedisresp	
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Variables										
dv.father.beat.motherTRUE	0.1189***	0.1219***	0.0562***	0.0518***	0.0649***	0.0663***	0.1118***	0.1119***	0.1248***	0.1358*
	(0.0085)	(0.0103)	(0.0075)	(0.0084)	(0.0070)	(0.0083)	(0.0092)	(0.0104)	(0.0095)	(0.0108
low.casteTRUE		0.0374***		0.0097*		0.0095		0.0358***		0.0350*
		(0.0080)		(0.0050)		(0.0058)		(0.0087)		(0.0093
hinduTRUE		-0.0147		-0.0020		0.0069		-0.0078		-0.001
		(0.0098)		(0.0066)		(0.0070)		(0.0119)		(0.0106
Fixed-effects										
region	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics										
Observations	66,445	63,102	66,445	63,102	66,445	63,102	66,445	63,102	66,445	63,102
R^2	0.09459	0.08842	0.07699	0.06950	0.07654	0.06527	0.12279	0.11792	0.11707	0.1097
Within R^2	0.01374	0.01578	0.00646	0.00764	0.00728	0.00868	0.01147	0.01281	0.01212	0.0142

One-way (region) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table C.3: Incidence of domestic violence

Dependent Variables:	dv.pushed		dv.slapped		dv.punched		dv.kicked		dv.strangled	
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Variables										
dv.father.beat.motherTRUE	0.1415***	0.1393***	0.2560***	0.2432***	0.0917***	0.0898***	0.0965***	0.0885***	0.0119***	0.0143***
	(0.0075)	(0.0082)	(0.0088)	(0.0095)	(0.0060)	(0.0066)	(0.0072)	(0.0078)	(0.0023)	(0.0025)
low.casteTRUE		0.0310***		0.0612***		0.0187***		0.0231***		0.0024
		(0.0060)		(0.0096)		(0.0051)		(0.0061)		(0.0020)
hinduTRUE		-0.0055		-0.0083		-0.0052		-0.0067		-0.0007
		(0.0074)		(0.0100)		(0.0050)		(0.0054)		(0.0021)
Fixed-effects										
region	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics										
Observations	60,610	57,920	60,610	57,920	60,610	57,920	60,610	57,920	60,610	57,920
R^2	0.07979	0.07630	0.12803	0.12345	0.06624	0.06276	0.06821	0.06422	0.02205	0.01612
Within R^2	0.02838	0.03053	0.05498	0.05691	0.01843	0.01869	0.01947	0.02017	0.00159	0.00162

Dependent Variables:	dv.weapon		dv.less		dv.severe		dv.rape		dv.any2	
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Variables										
dv.father.beat.motherTRUE	0.0048***	0.0071***	0.2695***	0.2587***	0.0998***	0.0921***	0.0846***	0.0778***	0.2018***	0.1917***
	(0.0018)	(0.0019)	(0.0089)	(0.0097)	(0.0072)	(0.0077)	(0.0073)	(0.0076)	(0.0085)	(0.0094)
low.casteTRUE		0.0007		0.0653***		0.0224***		0.0158***		0.0439***
		(0.0013)		(0.0098)		(0.0063)		(0.0053)		(0.0081)
hinduTRUE		-0.0002		-0.0040		-0.0055		-0.0026		-0.0005
		(0.0017)		(0.0103)		(0.0054)		(0.0058)		(0.0089)
Fixed-effects										
region	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics										
Observations	60,610	57,920	60,610	57,920	60,610	57,920	60,610	57,920	60,610	57,920
R^2	0.02063	0.01478	0.13248	0.12842	0.06839	0.06431	0.06092	0.05561	0.10746	0.10358
Within R^2	0.00050	0.00015	0.05763	0.06002	0.01988	0.02030	0.01829	0.01784	0.04280	0.04403

One-way (region) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

had experienced a specific type of violence, such as slapping or punching. For example, on average the data suggests respondents more frequently report having experienced "less severe" spousal violence than they do that their father beat their mother. But the opposite is the case for whether the respondent reported having experienced "severe" spousal violence. It's unclear how to map the different acts of violence into what the respondent understand by "beating" so without a more comparable question, we cannot even necessarily determine in the aggregate whether spousal violence has gone up or down

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